

MDES

Technical Specifications for Dual and Single Message
Systems

28 October 2025



Contents

MDES Technical Specifications for Dual and Single Message Systems.....	6
Chapter 1: Using this guide.....	9
Scope.....	10
How to use this guide.....	10
Related publications.....	11
Chapter 2: MDES overview.....	12
Digitization and tokenization.....	13
Network Messages.....	13
Account Status Inquiry Messages.....	14
IPM MPE and 80-Byte Financial Institution Table (FIT) identification of MDES token	
BINs/account ranges.....	15
Token file maintenance.....	15
Identifying token transactions.....	16
MDES transaction validation and decision matrix.....	18
Transaction analysis service and user experiences for MCBP tokens.....	19
Token transactions.....	20
Process flows.....	20
Standard authorization message process flow.....	20
Token to PAN mapping failure process flow.....	21
Chip pre-validation or dynamic CVC 3 pre-validation or DSRP validation failure	
process flow.....	22
Clearing (Dual Message System) process flow.....	24
Suspended or deactivated tokens (Clearing).....	25
Contactless transactions.....	25
Online PIN support.....	25
Mastercard Consumer-Presented QR Transactions.....	26
E-Commerce Transactions with Digital Secure Remote Payment (DSRP) Cryptograms..	26
E-commerce Transactions with Dynamic Token Verification Code (DTVC).....	27
Static tokens.....	27
Dynamic magnetic stripe transactions.....	27
Merchant-initiated Transactions (MITs).....	28
Account status inquiries, refunds and incremental authorizations.....	28
Mastercard MoneySend transactions	29
Payment Account Reference (PAR).....	29

Chapter 3: Acquirer-specific information.....	33
IPM MPE and 80-byte Financial Institution Table (FIT) File.....	35
Contactless transactions.....	35
Dual Message System (Authorization) and Single Message System.....	35
250-byte batch data file.....	39
Single Message Transaction Manager.....	40
Online PIN.....	40
Dual Message System (Clearing).....	41
Mastercard Consumer-Presented QR Transactions.....	42
Dual Message System (Authorization) and Single Message System.....	43
250-byte batch data file.....	47
Single Message Transaction Manager.....	47
Dual Message System (Clearing).....	47
E-commerce transactions with Digital Secure Remote Payment (DSRP) Crypto.....	49
Dual Message System (Authorization) and Single Message System - EMV Data.....	49
Dual Message System (Clearing) - EMV Data.....	55
Dual Message System (Authorization) and Single Message System - UCAF data.....	56
Dual Message System (Clearing) - UCAF data.....	65
E-commerce transactions with Dynamic Token Verification Code (DTVC).....	69
Dual Message System (Authorization) and Single Message System.....	69
Dual Message System (Clearing).....	75
Dynamic magnetic stripe transactions.....	76
Merchant-initiated Transactions (MITs).....	80
MoneySend Payment Transactions.....	82
Dual Message System (Authorization) and Single Message System - e-Commerce	
MoneySend Funding Transactions.....	83
Dual Message System (Clearing) - E-Commerce MoneySend Funding Transactions.....	89
Dual Message System (Authorization) and Single Message System - e-Commerce	
MoneySend Payment Transactions.....	90
Dual Message System (Clearing) - E-Commerce MoneySend Payment Transactions.....	96
Dual Message System (Authorization) and Single Message System - Contactless	
MoneySend Funding Transactions.....	97
Dual Message System (Clearing) - Contactless MoneySend Funding Transactions.....	102
Dual Message System (Authorization) and Single Message System - Contactless	
MoneySend Payment Transactions.....	103
Dual Message System (Clearing) - Contactless MoneySend Payment Transactions.....	108
Purchase mail order/telephone order transactions.....	109
 Chapter 4: Issuer-specific information.....	 112
Pre-digitization and post-digitization network messages.....	114
Single Message System.....	118

IPM MPE and 80-byte Financial Institution Table (FIT) File.....	118
Funding PAN and funding PAN expiry through ABU.....	118
Bulk files.....	118
Account Management System File updates (bulk file R311).....	118
Contactless transactions.....	119
Dual Message System (Authorization) and Single Message System.....	119
Token Mapping Failures, Suspended/Deactivated Tokens, and Pre-validation	
Failures.....	124
Reports.....	129
Online PIN.....	130
Dual Message System (Clearing).....	131
Mastercard Consumer-Presented QR Transactions.....	132
Dual Message System (Authorization) and Single Message System.....	133
Token Mapping Failures, Suspended/Deactivated Tokens, and Pre-Validation	
Failures.....	138
Reports.....	141
Dual Message System (Clearing).....	141
E-commerce transactions with Digital Secure Remote Payment (DSRP) Crypto.....	143
Dual Message System (Authorization) and Single Message System EMV Data.....	143
250-byte batch data file.....	147
Single Message Transaction Manager.....	148
Dual Message System (Clearing) - EMV Data.....	148
Dual Message System (Authorization) and Single Message System - UCAF data.....	150
Dual Message System (Clearing) - UCAF data.....	160
Token Mapping Failures, Suspended/Deactivated Tokens, and Pre-validation Failures...	163
E-Commerce Transactions with Dynamic Token Verification Code (DTVC).....	169
Dual Message System (Authorization) and Single Message System.....	170
Token Mapping Failures, Suspended/Deactivated Tokens, and Pre-Validation	
Failures.....	175
Dual Message System (Clearing).....	176
Dynamic magnetic stripe transactions.....	178
Merchant-initiated Transactions (MITs).....	183
Stand-In authorization.....	189
MoneySend Transactions.....	190
Dual Message System (Authorization) and Single Message System - e-Commerce	
MoneySend Funding Transactions.....	190
Dual Message System (Clearing) - e-Commerce MoneySend Funding Transactions.....	196
Dual Message System (Authorization) and Single Message System - e-Commerce	
MoneySend Payment Transactions.....	198
Dual Message System (Clearing) - e-Commerce MoneySend Payment Transactions.....	202
Dual Message System (Authorization) and Single Message System - Contactless	
MoneySend Funding Transactions.....	204
Dual Message System (Clearing) - Contactless MoneySend Funding Transactions.....	209

Dual Message System (Authorization) and Single Message System - Contactless
MoneySend Payment Transactions..... 211

Dual Message System (Clearing) - Contactless MoneySend Payment Transactions..... 216

Purchase mail order/telephone order transactions..... 218

Appendix A: Guidance for acquirers on typical e-Commerce purchase
submissions..... 221

Appendix B: Guidance for acquirers on typical e-Commerce MITs and
CITs transactions..... 227

Appendix C: MDES service indicators..... 241

Notices..... 242

MDES Technical Specifications for Dual and Single Message Systems

Overview

Enhancements to the Mastercard Dual Message System and Single Message System to support the Mastercard Digital Enablement Service (MDES) are published quarterly. This document consolidates all MDES release articles related to MDES published since that time, into a single guide for issuers, acquirers, and processors.

For information about the MDES release articles, refer to the [Related publications](#) chapter.

To understand all the implementation steps required for an issuer to enroll in and use MDES, refer to the *MDES Issuer Implementation Guide*, available on the Technical Resource Center (TRC) on Mastercard Connect™, www.mastercardconnect.com.

Audience

Issuer, acquirer, and processor operations staff that need to understand the impact of MDES on their operational activities.

Issuer, acquirer, and processor staff responsible for implementing system and process changes to support MDES.

Questions

For more information, questions or feedback about this document, click [Contact Customer Support](#).

Details

Table 1: Publication information

Metadata	Value
Audience	Acquirer, Branded Processor, Issuer, Processor
Product	Mastercard Digital Enablement Service (MDES)
Region	Global
Type	Guide
Publish Date	28 October 2025

Table 2: Summary of changes

Description of change	Where to look
Added details on the Digital Commerce Solution Indicator in DE 48 (Additional Data: Private Use), subelement 48 (Digital Commerce Solutions Indicators).	• Dual Message System (Authorization) and Single Message System - UCAF data • Dual Message System (Clearing) - UCAF data • Dual Message System (Authorization) and Single Message System • Dual Message System (Clearing) • Dual Message System (Authorization) and Single Message System - UCAF data • Dual Message System (Clearing) - UCAF data • Dual Message System (Authorization) and Single Message System • Dual Message System (Clearing)
Added DE 104 (Digital Payment Data).	• Dual Message System (Authorization) and Single Message System • Dual Message System (Authorization) and Single Message System EMV Data
Added PDS 0016 and PDS 0218.	Dual Message System (Clearing) - UCAF data
Added subelements 87 Card Validation Code Result and 92 CVC 2 to DE 48 (Additional Data: Private Use). Added subelements 004 Digital Service Provider and subelement 005 Digital Authentication Data to DE 104 (Digital Payment Data).	• Dual Message System (Authorization) and Single Message System - UCAF data • Dual Message System (Authorization) and Single Message System • Merchant-initiated Transactions (MITs)
Added an appendix which provides guidance for acquirers on typical e-Commerce purchase submissions.	Guidance for acquirers on typical e-Commerce purchase submissions
Added an appendix which outlines the transaction coding for MITs and CITs.	Guidance for acquirers on typical e-Commerce MITs and CITs transactions
Added an appendix for the MDES Service Indicators. Added DE 119 MDES Service Indicators to the: • 0100 Authorization Request and 0200 Financial Transaction Request messages. • 0110 Authorization Request Response and 0210 Financial Transaction Request Response messages.	MDES service indicators and throughout the guide

Description of change	Where to look
Added the values for DE 48, subelement 48, subfields 1, 2, 3, and 4.	• Dual Message System (Clearing) - UCAF data • Dual Message System (Authorization) and Single Message System - UCAF data • Dual Message System (Authorization) and Single Message System - UCAF data • Dual Message System (Clearing) - UCAF data
Changed: • <i>Customer Interface Specification to Mastercard Network Processing Dual Message Authorization System Guide</i> • <i>Single Message System Specification to Mastercard Network Processing Dual Message Authorization System Guide</i> • <i>Clearing Manual to Mastercard Network Processing Dual Message Clearing System Guide</i>	Throughout the guide

Chapter 1 Using this guide

This section provides information regarding the purpose of this guide and related publications.

Scope.....	10
How to use this guide.....	10
Related publications.....	11

Scope

This guide includes:

- Information about the types of pre-digitization and post-digitization network messages that issuers can receive when implementing MDES.
- The different token transaction types supported by MDES.
- Summaries of the general message flows and processing for token transactions.
- Overview of MDES transaction validation and decision processing, including the Transaction Analysis service.
- Information about the Payment Account Reference (PAR) and when it can be added to messages.
- For each token transaction type, the data elements and values that:
 - Acquirers must be prepared to send to, and receive from, Mastercard when supporting token transactions.
 - Issuers must be prepared to receive from, and send to, Mastercard when supporting token transactions processed by MDES.

NOTE: Subsequent functionality, changes, and enhancements for future releases will be incorporated into this document during bi-annual updates of the Dual Message System and Single Message System core manuals.

How to use this guide

The Token transaction message information in the [Acquirer-specific information](#) and [Issuer-specific information](#) sections has been converted to message tables to present the information more clearly.

Many of the message tables have 'Req' and 'Res' columns to indicate whether the elements can be present in the request and response messages (respectively), for example:

Table 3: Message

DE	Name/Subelement	Comments/ Subfield	Req	Res
2	PAN	Cardholder's token	√	√
14	Date, Expiration	PAN expiration date	√	•

The tables are meant to assist with implementations. For more detailed message layout information, refer to the relevant guide:

- *Mastercard Network Processing Dual Message Authorization System Guide*
- *Mastercard Network Processing Single Message System Guide*
- *Mastercard Network Processing Dual Message Clearing System Guide*

Related publications

The following documents and resources provide information directly related to the subjects discussed in this Technical Specification.

MDES Manuals

- *MDES Issuer Implementation Guide*
This guide describes the services provided by MDES, how they work, and introduces what issuers need to do to use those services
- *MDES Digital Secure Remote Payment Acquirer Implementation Guide*
This guide includes all implementation steps required for an acquirer to support MDES Digital Secure Remote Payment (DSRP) transactions.

Chapter 2 MDES overview

MDES supports tokenization and card digitization for use during contactless e-commerce and Dynamic Magnetic Stripe Data transactions.

Digitization and tokenization.....	13
Network Messages.....	13
Account Status Inquiry Messages.....	14
IPM MPE and 80-Byte Financial Institution Table (FIT) identification of MDES token BINs/ account ranges.....	15
Token file maintenance.....	15
Identifying token transactions.....	16
MDES transaction validation and decision matrix.....	18
Transaction analysis service and user experiences for MCBP tokens.....	19
Token transactions.....	20
Process flows.....	20
Standard authorization message process flow.....	20
Token to PAN mapping failure process flow.....	21
Chip pre-validation or dynamic CVC 3 pre-validation or DSRP validation failure process flow.....	22
Clearing (Dual Message System) process flow.....	24
Suspended or deactivated tokens (Clearing).....	25
Contactless transactions.....	25
Online PIN support.....	25
Mastercard Consumer-Presented QR Transactions.....	26
E-Commerce Transactions with Digital Secure Remote Payment (DSRP) Cryptograms.....	26
E-commerce Transactions with Dynamic Token Verification Code (DTVC).....	27
Static tokens.....	27
Dynamic magnetic stripe transactions.....	27
Merchant-initiated Transactions (MITs).....	28
Account status inquiries, refunds and incremental authorizations.....	28
Mastercard MoneySend transactions	29
Payment Account Reference (PAR).....	29

Digitization and tokenization

Digitization is the personalizing and provisioning of account credentials, cryptographic keys (EMV/Chip), and associated data into mobile devices or secure cloud-based servers.

Tokenization is the replacement of the cardholder's primary account number (PAN) with an alternative number, referred to as a token.

Network messages support issuers in the digitization and tokenization of a card into a mobile device or cloud-based server. Support of most of these messages is optional. Refer to subsequent sections of this manual for an overview on digitization network messages.

For full details on network messages, refer to the *Mastercard Network Processing Dual Message Authorization System Guide* and *Mastercard Network Processing Single Message System Guide*.

Network Messages

Issuers participating in MDES may choose to receive digitization requests or notifications during the digitization process and token life cycle, either in the ISO 8583-1987 message format or in the format required by the equivalent Web Service (API call).

The formats for each API call are specified in the MDES Pre-Digitization Application Programming Interface (API) documentation available on the Mastercard Developers web site (<https://developer.mastercard.com>). For further information on the MDES Pre-Digitization API, contact your Mastercard representative.

Pre-digitization and post-digitization messages perform the following:

- Tokenization Eligibility Request (TER) and Tokenization Authorization Request (TAR) messages enable issuers to provide a real-time decision as to whether Mastercard can digitize a card and designate a token on their behalf.
- The Activation Code Notification (ACN) message provides issuers with an activation code that they will deliver to the cardholder as a second means of authentication.
- The Tokenization Complete Notification (TCN) message informs issuers of the token assigned to a PAN.
- The Tokenization Event Notification (TVN) message informs issuers of unsuccessful activation code entry attempts and subsequent invalidation of an activation code, or when a token is suspended, resumed, or deactivated.

NOTE: When Mastercard is BIN Controller, TCN may include DE56 (Payment Account Data), subelement 01, subfield 01 (Payment Account Reference).

Issuers that choose to receive pre- and post-digitization network messages in the ISO 8583-1987 format may choose to receive the messages as:

- TER and TAR: Authorization Request/0100 or Financial Transaction Request/0200 messages
- ACN, TCN, and TVN: Authorization Request Request/0100, Financial Transaction Request/0200, or Administrative Advice/0620 messages

Issuers may optionally support the use of the Issuer File Update Request/0302 message layout for issuer token maintenance. Issuers may also use the bulk method for token file maintenance.

NOTE: Issuer receipt of any of these digitization messages is optional. When issuers choose not to receive a digitization message, they will receive an Account Status Inquiry (ASI) network message to authorize digitization unless they have opted to receive Authorize Service API messages from the MDES Pre-Digitization API.

NOTE: The ASI messages and pre-digitization network messages sent by MDES contain the indication that they originate from the United States. For Local Use Only account ranges, MDES adjusts the country indicators (country code and currency code) to make the corresponding ASI or pre-digitization network messages look like they originate from the domestic network. This way, the ASI and pre-digitization messages will not be systematically declined for account ranges marked as Local Use Only (see Local Use Only Account Ranges).

Account Status Inquiry Messages

Account Status Inquiry (ASI) messages allow acquirers to send Account Status Inquiry transactions to validate aspects of a cardholder account.

ASI messages can be submitted with MDES tokens, therefore applicable MDES On-Behalf services will be performed on these messages, such as PAN mapping service and Cryptographic validation service (cryptography data is not required for ASI messages, but will be validated if present).

As the PAN mapping service (On-behalf service 50) will be performed on ASI messages, the issuer will receive the PAN (DE 2) and the Token related data (DE 48, subelement 33) in Authorization request/0100 or Financial transaction request/0200 messages.

NOTE: If cryptography data was provided (not required), then the issuer will also receive the applicable On-behalf cryptographic validation service and results, that is OBS 51 or 61.

Then the issuer, at its discretion, sends the acquirer an Authorization Request Response/0110 or Financial Transaction request response/0210 messages where DE 39 (Response Code) may contain either value 00 (Approved or completed successfully), 05 (Do not honor), 85 (Not declined), or other valid business decline responses.

The response message sent to acquirers will also include the PAN Mapping information, that is Token (DE 2) and funding PAN information (DE 48, subelement 33).

IPM MPE and 80-Byte Financial Institution Table (FIT) identification of MDES token BINs/account ranges

Dual Message System MDES service-eligible account ranges are identified on the IPM MPE Table IP0040T1 (Issuer Account Range) with value C (Mastercard Digital Enablement Service Account Range) in the Mapping Service Indicator Field.

Single Message System MDES service-eligible account ranges are identified on the 80-byte Financial Institution Table (FIT) File with position 52 (Mastercard Digital Enablement Service Flag), value Y (Mastercard Digital Enablement Service is performed).

Acquirers must not perform On-Us or intraprocessor processing for service-eligible account ranges since tokens within those account ranges will not be known to the issuing card systems.

Token file maintenance

Mastercard provides enhanced Issuer File Update Request/0302 and Issuer File Update Request Response/0312 message layouts to enable issuers to submit certain file maintenance updates for tokens.

MDES issuers can report PAN and PAN expiry date updates to AMS using the Issuer File Update Request/0302 - Maintenance (Token/PAN Update) messages, bulk file R311 (Account Management System File Updates), or customer service application programming interfaces (APIs).

When an issuer initiates PAN maintenance to the AMS MCC106 file using the Issuer File Update Request/0302 message or R311 Bulk File update, Mastercard will update the PAN to token mapping in AMS only if at least one token is in active or suspended status. Otherwise, only the PAN to PAR mapping update occurs if a PAR exists for that PAN.

If a PAN or PAN expiry date change is first submitted through the Automated Billing Updater (ABU), then ABU will automatically update AMS with this information.

Issuers should be aware that if they submit a PAN change first through ABU and then to AMS using Issuer File Update Request/0302 - Maintenance (Token/PAN Update) messages, AMS will send Issuer File Update Request Response/0312 reject messages to issuers.

Available maintenance request actions include deactivate, suspend, or resume a token, and PAN maintenance. Dual Message System and Single Message System issuers may use either the online or bulk method for token file maintenance.

Identifying token transactions

When MDES processes a token transaction successfully, Mastercard provides token information to the issuer in DE 48, subelement 33 (PAN Mapping File Information).

This table shows the main characteristics that can help identify the type of token transaction, where token type is:

- C = MDES Secure Element token
- H = MDES Cloud-based token
- F = MDES static token

Token transaction type	POS entry mode value	Token type	Other indicators
Contactless transactions	07 or 91	C or H	Presence of DE 35 (Track2 Data) and DE 55 (Integrated Circuit Card [ICC] System- Related Data) NOTE: Track2 data and Chip data are suppressed unless issuer opts in to receive this information through MDES Manager.
Consumer-Presented QR transactions	03	C or H	DE 48 (Additional Data - Private Use), subelement 21 (Acceptance Data), subfield 2 (Additional Terminal Capability Indicator) DE 55 (Integrated Circuit Card [ICC] System- Related Data)
E-Commerce transactions with full EMV DSRP data	09	C or H	DE 61, subfield 3 (POS Terminal Location): Whether the transaction occurred remotely (value 2) or in store (value 4). DE 55 (Integrated Circuit Card [ICC] SystemRelated Data)

Token transaction type	POS entry mode value	Token type	Other indicators
E-Commerce Transactions with UCAF DSRP data	81 or 10	C or H	The presence of DE 48, subelement 42, subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator). DE 61, subfield 3 (POS Terminal Location): Whether the transaction occurred remotely (value 2) or in store (value 4). NOTE: From 11 August 2020 acquirers are required to populate DSRP UCAF data in DE 104 (Digital Payment Data) subelement 001 (Digital Payment Cryptogram) instead of DE 48 (Additional Data) subelement 43 (Universal Cardholder Authentication Field - UCAF). NOTE: Issuers do not receive DSRP cryptogram in DE 104 (Digital Payment Data) subelement 001 (Digital Payment Cryptogram).
E-commerce transactions with Dynamic Token Verification Code (DTVC)	81 or 10 or 82	F	The presence of DTVC in DE 48, subelement 92 (CVC 2). DE 48, subelement 42, subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator): 210 NOTE: Acquirer submitted dynamic token verification code (DTVC) is not passed to issuer.
E-commerce transactions without dynamic data	10 or 81 or 82	F	DSRP Crypto and DTVC not present. DE 48, subelement 42, subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator): 210
Dynamic Magnetic Stripe Data Transactions	90	H	The presence of DE 48, subelement 49 (Time Validation Information) and its subfields. In addition, all the following are true for DE 48: • Subelement 33, subfield 1 is value C or H • Subelement 33, subfield 2 is not a space • Subelement 92 is present

Token transaction type	POS entry mode value	Token type	Other indicators
E-Commerce MoneySend Payments	81 or 10	C or H or F	DE 3, subelement 1 (transaction code): Value 28 - Payments DE 61, Subelement 10 (CAT Level): Value 6 (Electronic Commerce)

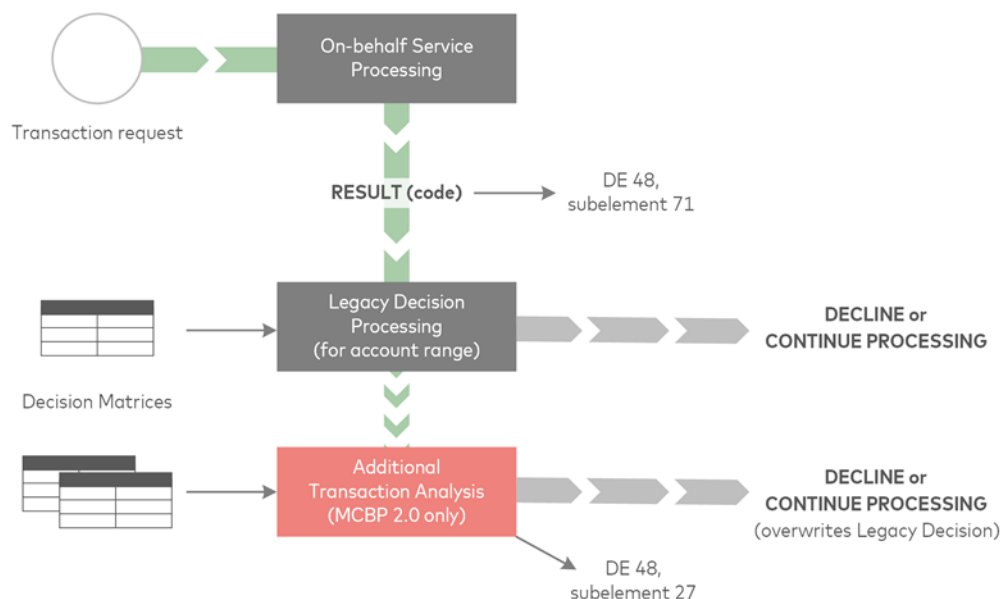
NOTE: The table does not show all of the transaction message characteristics. For details about the message data elements and fields, see the relevant sections in [Acquirer-specific information](#) and [Issuer-specific information](#).

MDES transaction validation and decision matrix

One of the benefits of MDES is that it can perform the applicable (cryptographic) validation on token transactions, minimizing the impact on the issuer's infrastructure. The validation result (code) from this on-behalf service processing is provided in DE 48, subelement 71 in the message sent to the issuer.

During issuer onboarding, the issuer can configure the actions (decline or continue processing) that Mastercard should take for some of the on-behalf service results, per token type; these choices form the legacy issuer decision matrix. The legacy decision processing outcome will be the applicable action for the on-behalf service result. For example, issuers can choose for Mastercard to decline MCBP transactions that have an invalid TVR/CVR (On-behalf Services value: 61T). In practice, the TVR and CVR values are checked against a specific bitmap that is used to determine if some values are set.

Figure 1: Decision processing flow



For MCBP 2.0 wallets, the issuer benefits from the additional MDES Transaction Analysis service, which provides specific decision matrices for MCBP 2.0 token transactions. The issuer can configure those matrices during the issuer enablement process (at the time of initial onboarding) and also as part of the issuer maintenance process. The Transaction Analysis result (outcome) will override the legacy decision processing outcome. For more information, see section [Transaction analysis service and user experiences for MCBP tokens](#).

Transaction analysis service and user experiences for MCBP tokens

MDES supports various user experiences for MCBP 2.0 token transactions that address market needs and deliver an improved and consistent user experience for all cards within the wallet.

In addition to the existing support for CDCVM based on Mobile PIN (validated online by MDES as part of authorization processing), other cardholder device cardholder verification methods (CDCVMs) are also supported, for instance device PIN, pattern, password or biometric methods such as fingerprint, iris, or face recognition features that are validated on the consumer's mobile device. For details about the user experiences and their applicability, refer to the *MDES—Issuer Implementation Guide*.

Wallet providers (and issuers acting in the role of a wallet provider) can select the user experience supported by their wallet for contactless transactions when onboarding a new wallet onto MDES. A CDCVM Always user experience is the rule for any Digital Secure Remote Payment (DSRP) transaction. Mastercard introduced DE 48 (Additional Data - Private Use), subelement 27 (Transaction Analysis), subfield 1 (Overview) and subfield 2 (Test Results) for the Single Message System and Dual Message Authorization System to deliver transaction analysis information to issuers for all transactions processed using MCBP 2.0. Mastercard has defined:

- A list of real-time validations that are performed on the transaction data
- A decision matrix used for assessing MCBP 2.0 token transactions

NOTE: DE 48, subelement 27 is not sent to issuers when processing a perfect transaction (that is, without any failed checks) or a transaction from an alternative network.

For additional information, refer to the *MDES Issuer Implementation Guide* and *MDES Wallet Provider Implementation Guide*.

Token transactions

Mastercard supports different types of token transaction, including contactless and electronic commerce (e-commerce). These transactions contain the cardholder's token (a digital substitute for their PAN) and associated data, which the acquirer sends to Mastercard as a transaction request. Mastercard validates the data and maps the token to the cardholder's PAN before forwarding the request to the issuer.

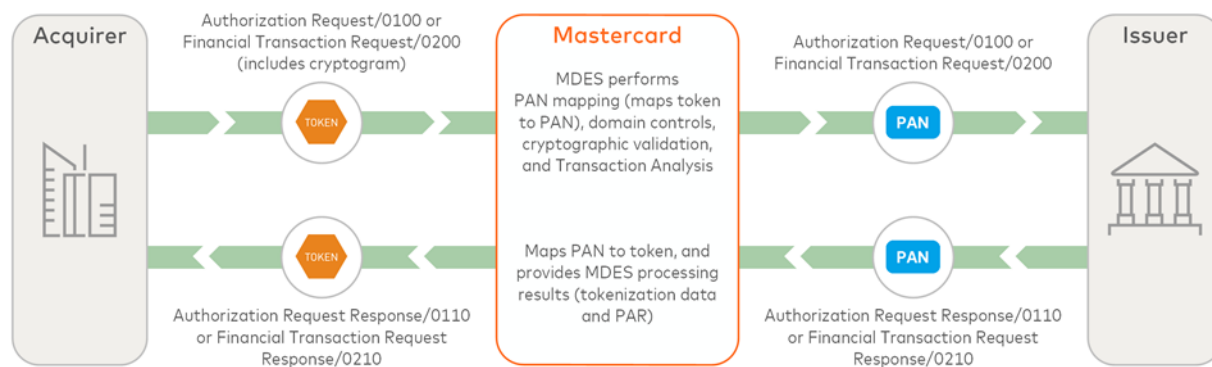
Process flows

This section summarizes the general message flows and processing for token transactions. For details about the message data elements and fields, see the relevant sections in [Acquirer-specific information](#) and [Issuer-specific information](#).

Standard authorization message process flow

This flow depicts successful processing for a token transaction.

Tokenized contactless and e-commerce transactions with DSRP Cryptogram



1. The acquirer sends an authorization request message, containing the token and a cryptogram, to the Mastercard Network.
2. MDES:
 - a. Attempts to map the token (sent by the acquirer) to an Account PAN in the PAN mapping database and, if successful, checks that the token is in good standing (not

- suspended, deactivated or expired). In the authorization request, MDES replaces the token and token expiration date with the PAN and PAN expiration date, and adds the Wallet ID (when applicable) and mapping information (such as the Token Type and Token Requestor ID).
- b. Performs token domain controls to ensure the integrity of the transaction and accuracy of key fields such as DE 22 (Point-of-Service [POS] Entry Mode) and dynamic data.
 - c. Performs the chip pre-validation or dynamic CVC 3 pre-validation or DSRP crypto validation or DTVC validation, as applicable.
 - d. Performs Transaction Analysis on Cloud device token transactions initiated from Mastercard Cloud-Based Payments (MCBP) 2.0 wallets and adds the result to the authorization request.
3. The Authorization Platform:
 - a. Retrieves the corresponding PAR value (if available) from the Mastercard PAR Vault, if Mastercard is the BIN Controller for the PAN, and includes the value in the transaction request. For more information, see Payment Account Reference (PAR).
 - b. Suppresses token cryptography-related data (track data, EMV and UCAF data, as applicable) from the authorization request message before sending it to issuer.
 4. The issuer responds with an authorization response message.
If the issuer or third party entity is the BIN Controller for the PAN, the issuer may include the corresponding PAR value in its response message. For more information, see Payment Account Reference (PAR).
 5. MDES maps the PAN back to the token.
 6. The Authorization Platform adds the PAN mapping information and the PAR to the authorization response message and sends it to the acquirer.
 7. The acquirer forwards the authorization information to the merchant.

Token to PAN mapping failure process flow

This flow depicts processing for a token transaction that was declined by MDES because PAN mapping was unsuccessful. This includes scenarios where a token has been temporarily placed in suspended status (for example, in the case of a misplaced device), deactivated (for example, in the event of a lost or stolen device) or has expired.

1. The acquirer sends an authorization request message, containing the token and a cryptogram or Dynamic Token Verification Code (DTVC), to the Mastercard Network.
2. MDES attempts to map the token (sent by the acquirer) to an Account PAN in the PAN mapping database and, if successful, checks that the token is in good standing.
3. When the token mapping or checking fails, the Authorization Platform sends:
 - An authorization response message to the acquirer
 - An advice message to the issuer

The Response Code (DE 39) in both messages is one of the following:

- 14 = Invalid card number (unable to map the token to the PAN)
- 62 = Restricted card (token suspended, deactivated or expired)
- 96 = System error (mapping failed because of technical reasons)

Chip pre-validation or dynamic CVC 3 pre-validation or DSRP validation failure process flow

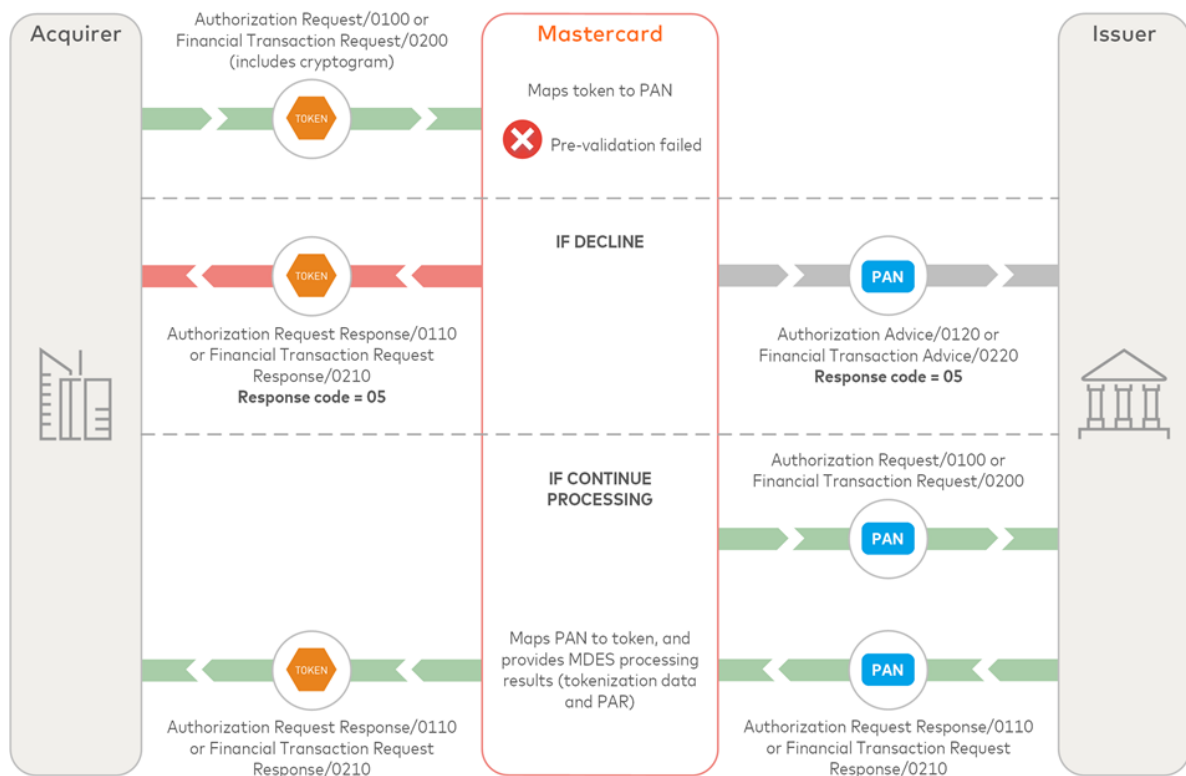
This flow depicts processing for a token transaction that fails MDES cryptographic validation. Issuers can designate the action Mastercard should take: decline or continue processing.

NOTE: This process flow does not apply to e-commerce transactions with static tokens, because there is no dynamic cryptographic data for MDES to validate.

Issuer supporting MCBP 2.0 wallets may benefit from the Transaction Analysis service and configure it during the enablement process; see [Transaction analysis service and user experiences for MCBP tokens](#).

The process flow is as follows:

Figure 2: Pre-validation failure



1. The acquirer sends an authorization request message, containing the token and a cryptogram, to the Mastercard Network.
2. MDES:
 - a. Attempts to map the token (sent by the acquirer) to an Account PAN in the PAN mapping database and, if successful, checks that the token is in good standing (not suspended, deactivated). In the authorization request, MDES replaces the token and token expiration date with the PAN and PAN expiration date, and adds the Wallet ID

(when applicable) and mapping information (such as the Token Type and Token Requestor ID).

- b. Performs token domain controls to ensure the integrity of the transaction and accuracy of key fields such as DE 22 (Point-of-Service [POS] Entry Mode) and dynamic data.
3. If the issuer has chosen for Mastercard to **decline** on their behalf, the Authorization Platform sends:
 - An authorization response message to the acquirer
 - A decline advice message to the issuer

The Response Code (DE 39) in both messages is 05 = Do not honor.

4. If the issuer has chosen for Mastercard to **continue processing**, the validation results are added to the authorization request message.

NOTE: The authorization request message contains similar data elements to those present for a transaction without cryptographic errors; see the Dual Message System (Authorization) and Single Message System section for the appropriate token transaction type in Issuer-Specific Information.

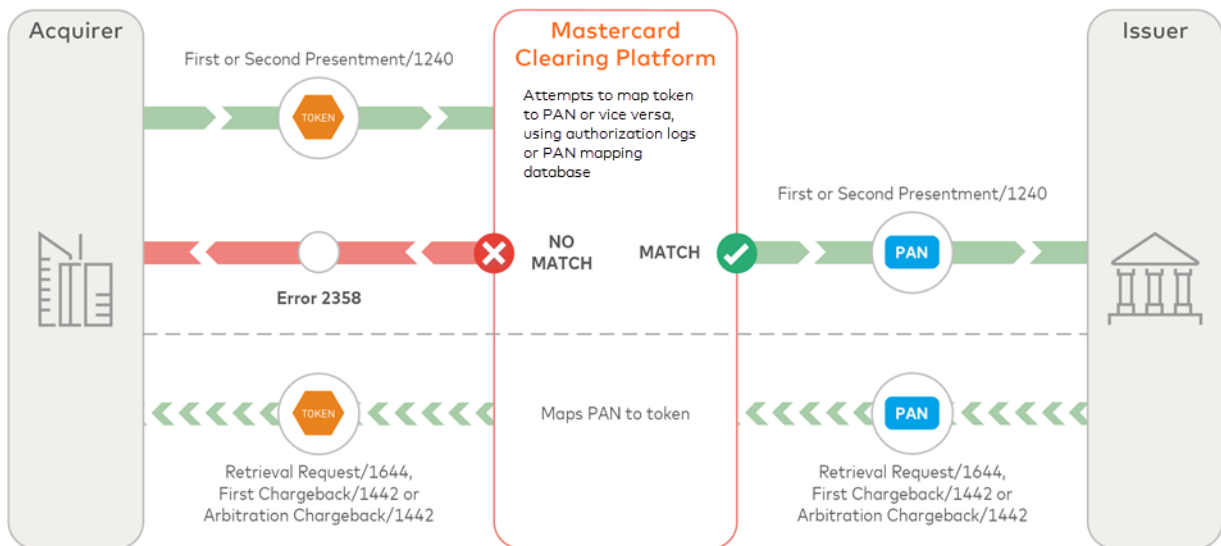
Processing continues as per the Standard Authorization Message Flow:

5. The Authorization Platform retrieves the corresponding PAR value (if available) from the Mastercard PAR Vault, if Mastercard is the BIN Controller for the PAN, and includes the value in the authorization request message it sends to the issuer.
6. The issuer responds with an authorization response message.
If the issuer or third party entity is the BIN Controller for the PAN, the issuer may include the corresponding PAR value in its response message. For more information, see Payment Account Reference (PAR).
7. MDES maps the PAN back to the token.
8. The Authorization Platform adds the PAN mapping information and the PAR to the authorization response message and sends it to the acquirer.
9. The acquirer forwards the authorization information to the merchant.

Clearing (Dual Message System) process flow

This flow depicts the clearing process for token transactions.

Figure 3: Clearing process flow



1. The acquirer sends a First Presentment/1240 message, containing the token, to the Clearing Platform.
2. The Clearing Platform attempts to map the token (sent by the acquirer) to an Account PAN, using either the authorization logs or a PAN mapping database.
If the mapping is successful (there is a match), the Clearing Platform replaces the token with the PAN, adds further token information (such as the Wallet ID and Token Requestor ID), and sends the First Presentment/1240 message to the issuer.

Otherwise, the Clearing Platform rejects the First Presentment/1240 message with error number 2358 (DE2 PAN INVALID. THE PAN MAPPING SERVICE CANNOT MAP DE2 TO ANOTHER ACCOUNT NUMBER).

3. Issuers can send subsequent retrieval request and chargeback messages to the Clearing Platform containing the cardholder's PAN. The Clearing Platform maps the cardholder's PAN to the token before sending the message to the acquirer.
For all Retrieval Request/1644, First Chargeback/1442 and Arbitration Chargeback/1442 messages, the Clearing Platform attempts to match the PAN (in DE 2) to the MDES First Presentment history file.

If DE 2 in the retrieval or chargeback message cannot be matched to a First Presentment/1240 message in the MDES First Presentment History File, the Clearing Platform assumes that it did not perform MDES at the time of the First Presentment/1240 message. The Clearing Platform continues processing the retrieval or chargeback message and delivers it to the acquirer with the PAN in DE 2. If, upon receipt, the acquirer does not recognize the transaction because the First Presentment/1240 had actually been submitted with an

MDES token, the acquirer must follow existing dispute resolution rules and procedures that apply when the PAN is not recognized.

4. With Second Presentment/1240 messages, the acquirer provides the token.
5. The Clearing Platform attempts to match each Second Presentment/1240 message to the existing Chargeback history file.

If there is a match, and the Chargeback history file indicates that DE 2 contains an MDES token, the Clearing Platform maps the token to the PAN and uses the PAN in its processing. The Clearing Platform sends the Second message Presentment/1240 to the issuer.

Otherwise, the Clearing Platform rejects the Second Presentment/1240 message.

Suspended or deactivated tokens (Clearing)

When a token has been suspended or deactivated from use, Mastercard will permit these transactions to pass through to the issuer for clearing.

If the cardholder disputes the transaction, the issuer can charge back under message reason code 4808 (Requested/Required Authorization Not Obtained). This processing would be consistent with that applied for non-token transactions. Mastercard will not block the transactions in Clearing but rather allow the dispute resolution process to occur. For transactions including a token, Mastercard will not enforce error number 2358 (DE2 PAN INVALID. THE PAN MAPPING SERVICE CANNOT MAP DE2 TO ANOTHER ACCOUNT NUMBER) and will allow the deactivated or suspended tokens to pass through to the issuer for clearing.

Contactless transactions

MDES can provision secure element tokens and cloud-based tokens for use in mobile devices.

A card present transaction occurs when a cardholder uses the token to initiate a contactless transaction by tapping their NFC-enabled mobile device at a contactless terminal. The cardholder's token and associated chip or dynamic card validation code 3 (CVC 3) data are transmitted to the terminal. The token and data are forwarded to the acquirer.

Online PIN support

MDES enables cardholders to use the same online PIN as their digitized card for contactless transactions initiated by a mobile device using a token.

When a token is used at the point of interaction, the online PIN block sent by the acquirer is based on the token. As part of MDES, Mastercard will convert this PIN block to an issuer PIN block based on the PAN before sending the message to the issuer.

NOTE: Since 1 May 2019, Mastercard has been making the cardholder experience at contactless ATMs more seamless by requiring the support of Online PIN for unattended cash for newly-provisioned digitized Mastercard and Maestro cards on mobile payment devices that are digitized through MDES.

Mastercard Consumer-Presented QR Transactions

A Mastercard Consumer-Presented QR Transaction is an EMV chip Transaction effected through the presentment of a QR Code by the Cardholder, using a Mobile Payment Device, and the capture of the QR Code by the Merchant containing the Transaction Data required to initiate a Transaction

For each Mastercard Consumer-Presented QR Transaction:

- There is no maximum Transaction amount.
- The Transaction must be authorized online by the Issuer.
- The Transaction must be completed with CDCVM. CDCVM is the only valid CVM for Mastercard Consumer-Presented QR Transactions.
- The Merchant must send a properly identified Authorization Request with EMV data.

NOTE: For information about Mastercard Consumer-Presented QR Transactions, refer to the *Mastercard Cloud-Based Payments (MCBP)* documentation and the *Mastercard Consumer-Presented QR Requirements* manual.

E-Commerce Transactions with Digital Secure Remote Payment (DSRP) Cryptograms

Tokenized e-commerce transactions with Digital Secure Remote Payment (DSRP) cryptogram allows card-not-present merchants to benefit from the dynamic secure data generated by M/Chip applications.

Mastercard supports the following DSRP transaction types:

- DSRP containing full EMV data. This transaction contains dynamic data (a cryptogram) generated using EMV-based cryptography by a payment application to secure the transaction. The cardholder's token and associated chip data and cryptogram are transmitted to the merchant or the merchant payment gateway, which forwards the data to the acquirer.
- DSRP containing Universal Cardholder Authentication Field™ data. This transaction through an e-commerce merchant contains UCAF data.

All tokenized e-commerce transactions with DSRP cryptograms require cardholder verification, and the cryptogram provides evidence that cardholder verification has been performed (Mastercard validates the cryptogram when it receives the transaction). Such transactions can be initiated from any device that can perform cardholder verification, including mobile devices and properly secured web-based implementations.

For instance, typical mobile e-commerce scenarios where the cardholder is using a mobile device browser or a specific merchant application to purchase goods and services.

The use of a mobile application also offers the potential for shopping experiences where DSRP is used in circumstances where traditionally a face-to-face transaction using a point-of-sale device would have been conducted. For example, a merchant may provide an application that allows the cardholder to use the camera on a mobile device to scan the barcodes of goods in a brick

and mortar store. When the cardholder has completed shopping, rather than going to a cashier to check out and pay, the cardholder may pay using DSRP through the mobile application, and leave the store with the goods without going to a cashier to check out.

Refer to the [Guidance for acquirers on typical e-Commerce purchase submissions](#) for comprehensive guidance.

Refer to [Guidance for acquirers on typical e-Commerce MITs and CITs transactions](#) for guidance on CIT and MIT message values for MDES Tokenized Remote Commerce transactions.

E-commerce Transactions with Dynamic Token Verification Code (DTVC)

Mastercard has enhanced MDES processing to allow merchants and acquirers with legacy transaction processing systems that are not yet upgraded to support tokenized e-commerce transactions with DSRP UCAF or DSRP full EMV data, to submit static token transactions without the presence of a DSRP cryptogram.

Refer to the [Guidance for acquirers on typical e-Commerce purchase submissions](#) for comprehensive guidance.

Refer to [Guidance for acquirers on typical e-Commerce MITs and CITs transactions](#) for guidance on CIT and MIT message values for MDES Tokenized Remote Commerce transactions.

Static tokens

MDES can generate static tokens for use by acquirers supporting e-commerce payments.

MDES maps the static token in the e-commerce transaction, to the cardholder's PAN before forwarding the request to the issuer. No dynamic payment credentials are introduced in the transaction.

Static tokens can be identified by token type F in DE 48, subelement 33, subfield 1.

Static tokens can be requested and maintained only through an API available to participating Token Requestors as part of their token services implementation. Token Requestors and acquirers that process for merchants implementing a PAN tokenization strategy should contact their Mastercard representative for further information. For additional information on the MDES API, refer to the Technical Resource Center (TRC) on [Mastercard Connect](#).

To further strengthen the security of the static token transaction, Mastercard introduced support for DTVC, an alternative to 3-digit static CVC 2 that allows merchants to process tokenized transactions. Mastercard during the authorization processing will recognize and validate the DTVC before sending the request to the issuer for approval.

Dynamic magnetic stripe transactions

For transactions initiated through magnetic stripe readers, all acquirers must be prepared to receive additional data that may be present on Dual Message System (Authorization and

Clearing) and Single Message System response messages, including chargebacks and retrieval requests, if an MDES token is used to initiate a transaction.

MDES supports the dynamic magnetic stripe POS Entry Modes to accommodate devices that use the magnetic stripe reader at a terminal to pass dynamic transaction data for payment processing. Mastercard performs the appropriate MDES cryptography validation based on whether the payment data was stored in the secure element of the device or a secure software application accessible through Mastercard Cloud-Based Payments (MCBP).

Merchant-initiated Transactions (MITs)

In an e-commerce partial shipment scenario, a merchant may split the shipment of the ordered merchandise into multiple shipments by submitting a separate authorization for each shipment. The merchant must identify the subsequent transactions as a partial shipment.

When a merchant has the need to process recurring payments with MDES tokens, the authorization for the recurring payment must be identified as a recurring payment transaction and include the partial shipment indicators. DSRP cryptogram is optional in Partial shipment transactions, however Mastercard attempts to validate the crypto when present and decline if the validation fails. For more details, see the [Acquirer-specific information](#) and [Issuer-specific information](#) sections.

A new indicator is introduced to specify the type of cardholder-initiated transaction (CIT) or merchant-initiated transaction (MIT) within Dual Message System transaction requests (Authorization), Single Message System transaction requests, and applicable Clearing messages. It conveys to the issuer that the merchant and the cardholder have an established relationship and an agreement to use stored payment credentials as part of the transaction process. It also helps provide transparency into the transaction type, making it easier for issuers to identify legitimate transactions, and make better informed authorization decisions. Merchants, acquirers, and payment service providers that populate the CIT and MIT indicators should see higher approval rates and reduced customer servicing costs associated with transaction declines. Refer definition of DE 48, subelement 22, subfield 5 in *Mastercard Network Processing Dual Message Authorization System Guide* or *Mastercard Network Processing Single Message System Guide* for valid values.

Refer to the [Guidance for acquirers on typical e-Commerce purchase submissions](#) for comprehensive guidance.

Refer to [Guidance for acquirers on typical e-Commerce MITs and CITs transactions](#) for guidance on CIT and MIT message values for MDES Tokenized Remote Commerce transactions.

Account status inquiries, refunds and incremental authorizations

Mastercard has relaxed domain controls for account status inquiries (DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status) = 8), refunds (DE 3 [Processing Code], subfield 1 [Cardholder Transaction Type Code]: 20 = Purchase Return/Refund) and incremental authorizations (DE 48 [Additional Data - Private Use], subelement 63 [Trace ID] is present).

Any POS Entry Mode can be submitted for these transactions despite the original transaction being contactless, DSRP, or Dynamic Magnetic Stripe data. This applies to the following payment message types:

- Authorization Request/0100
- Financial Transaction Request/0200

Acquirers and issuers participating in MDES must be aware that ASIs, refunds and incremental authorizations:

- Do not require cryptography. DSRP cryptograms are optional for refunds and incremental authorizations, but Mastercard will validate the DSRP cryptogram if provided, ensuring that it is unique for each transaction request.
- Do not require a specific POS Entry Mode.
- Can be initiated or received with POS Entry Modes other than contactless, DSRP, and Dynamic Magnetic Stripe data.

NOTE: Issuers should continue to review transactions that do not have cryptography validation and confirm that refunds or incremental authorizations correspond to an original transaction detail.

Mastercard MoneySend transactions

Effective 12 June 2018, transactions that are uniquely identified by data element (DE) 3, subfield 1, value 28 (Payment Transaction) are processed without cryptogram data, such as, DE 48, subelement 43, or DE 104, subelement 001.

This applies to the following message types, specifically for Mastercard MoneySend transactions (DE 22, subfield 1, value 81) or Credential on File transactions (DE 22, subfield 1, value 10):

- Authorization Request/0100
- Financial Transaction Request/0200

Payment transaction authorization requests involving tokens without cryptograms and not identified as payment transactions (DE 3, subfield 1, value 28) will be rejected with a format error (DE 39, value 30 and DE 44, value 048).

Payment Account Reference (PAR)

EMVCo introduced PAR to provide an industry-aligned approach designed to help link all transactions associated with a specific account, whether PAN based or token based, without using the PAN as the linkage mechanism.

Mastercard is the BIN Controller (as defined by EMVCo standards) for PANs issued from the Mastercard product ranges of 222100–272099 and 510000–559999 and the Maestro product ranges of 639000–639099 and 670000–679999. Mastercard currently assigns a PAR to a given MDES tokenized PAN issued from these ranges.

Mastercard implemented PAR by supporting DE 56 in a limited set of Dual Message System (Authorization) and Single Message System messages, including authorization and financial request and response messages (both transactional and pre-digitization Tokenization Complete Notifications), and administrative advice pre-digitization Tokenization Complete Notifications.

This allowed Mastercard to communicate Mastercard-generated PAR values to MDES issuers and all acquirers in transactions containing a PAN associated with a BIN for which Mastercard is the BIN Controller and a PAR is available. When the issuer or third party entity is the BIN Controller and the issuer includes a PAR in the authorization request response or financial transaction request response message, Mastercard will send that PAR to the acquirer.

Mastercard supports the inclusion of DE 56 in the following messages when PAR is available for the applicable PAN and Mastercard is the BIN Controller.

Authorization	Single Message System	Mastercard is the BIN Controller
Authorization Request/0100	Financial Transaction Request/0200	Yes
Authorization Request Response/0110	Financial Transaction Request Response/0210	Yes
Authorization Request/0100—Tokenization Complete Notification	Financial Transaction Request/0200—Tokenization Complete Notification	Yes
Authorization Advice/0120—Acquirer-Generated	Financial Transaction Advice/0220—Debit Mastercard Stand-In	Yes
Authorization Advice/0120—System-Generated	Financial Transaction Advice/0220—Maestro Preauthorization Completion	Yes
Authorization Advice Response/0130—Issuer-Generated	Financial Transaction Advice Response/0230	No
Reversal Request/0400	Non-Financial Transaction Advice/0220	Yes
Reversal Request Response/0410	Issuer Reversal Advice/0422—Exception, System Initiated	No
Reversal Advice/0420	Acquirer Reversal Advice/0420—Acquirer Initiated	Yes
	Acquirer Reversal Advice/0420—Time-out-Induced, Acquirer Initiated	Yes

Authorization	Single Message System	Mastercard is the BIN Controller
	Acquirer Reversal Advice/0420— Exception, System Initiated	Yes
	Acquirer Reversal Advice Response/0430—System Initiated	No
Administrative Advice/0620— Issuer Token Notification Advice for Tokenization Complete Notification	Administrative Advice/0620— Issuer Token Notification Advice for Tokenization Complete Notification	Yes

Issuer PAR Requests

Issuers may request PAR manually, outside of the MDES tokenization process, for any account for which Mastercard is the BIN Controller using the `getPaymentAccountReference` service within the Payment Account Management API.

- PAR values are created for PANs when specifically requested by the issuer if Mastercard is the BIN Controller.
- Any issuer that requests the assignment of PAR values for their PANs must be capable of supporting the presence of DE 56 in the applicable Dual Message System (Authorization) and Single Message System messages.
- Mastercard includes PAR in messages sent to the issuer for any transactions involving a PAN for which a PAR value has been generated.
- Issuers should not request a PAR from Mastercard for PANs for which Mastercard is not the BIN Controller.

NOTE:

PAR use is not limited to MDES. Issuers are able to use PAR with other applications to link all transactions associated with a specific account.

Issuer Life Cycle Management of PAN-PAR Mapping

Issuers are required to properly manage PAN-PAR relationships stored by Mastercard when an account life cycle event occurs involving a PAN replacement or account closure. By properly managing the PAN-PAR relationship when a life cycle event occurs, jointly Mastercard and issuers can ensure the integrity of PAR within the payments ecosystem.

For example, if a consumer's plastic card is stolen and the PAN (for which Mastercard is the BIN Controller) is replaced, the issuer must inform Mastercard of the new PAN to ensure that the original PAR is associated with the new PAN. Or, when the issuer permanently closes a consumer's account, the issuer must inform Mastercard of the account closure to ensure the PAN-PAR relationship is eliminated.

NOTE:

Issuers that recycle PANs for reissuance to subsequent consumers must ensure that obsolete PAN-PAR mapping relationships are eliminated within their systems and from Mastercard's PAN-PAR mapping file. This will ensure that the PAN is associated with a unique PAR value with each reuse of that PAN.

When Mastercard is the BIN Controller

When Mastercard is the BIN Controller for the PAN, issuers may choose from the following methods to manage PAR for PAN replacements and account closures.

PAN Replacement and Account Closure Methods:

- Automatic Billing Updater (ABU)
- Online or batch (bulk file type R311) Issuer File Update Request/0302 Maintenance (PAN-PAR Mapping via File Name MCC111)
- Payment Account Management (PAM) API

PAN Replacement Only Methods for MDES Issuers:

- Online or batch (bulk file type R311) Issuer File Update Request/0302 Maintenance (Token-PAN Mapping via File Name MCC 106)
- MDES Customer Service Application or MDES Customer Service API

AMS AM700010-II PAR Daily Account File Activity Report

Issuers that are connected to the Dual Message System (Authorization) have the option to request the AMS report (AM700010-II PAR Daily Account File Activity Report). This report contains PAN-PAR file maintenance activity requested by an issuer specifically via online or batch Issuer File Update Request/0302 Maintenance messages, using DE 101 File Name value MCC111. The report will be sent daily (on any day for which maintenance activity occurs), either by bulk file delivery or eService delivery, as requested by the issuer. The report is billed to issuers weekly via the following MCBS billing event codes (based on report delivery method):

- 2RP6328B AM700010-II PAR Daily Account File Activity Report Bulk
- 2RP6328E AM700010-II PAR Daily Account File Activity Report eService

Issuers can request the report by contacting their Global Customer Support representative.

Chapter 3 Acquirer-specific information

The following is information specific to acquirers for MDES. Acquirers must be prepared to receive the Authorization, Clearing, and Single Message System data element values described.

IPM MPE and 80-byte Financial Institution Table (FIT) File.....	35
Contactless transactions.....	35
Dual Message System (Authorization) and Single Message System.....	35
250-byte batch data file.....	39
Single Message Transaction Manager.....	40
Online PIN.....	40
Dual Message System (Clearing).....	41
Mastercard Consumer-Presented QR Transactions.....	42
Dual Message System (Authorization) and Single Message System.....	43
250-byte batch data file.....	47
Single Message Transaction Manager.....	47
Dual Message System (Clearing).....	47
E-commerce transactions with Digital Secure Remote Payment (DSRP) Crypto.....	49
Dual Message System (Authorization) and Single Message System - EMV Data.....	49
Dual Message System (Clearing) - EMV Data.....	55
Dual Message System (Authorization) and Single Message System - UCAF data.....	56
Dual Message System (Clearing) - UCAF data.....	65
E-commerce transactions with Dynamic Token Verification Code (DTVC).....	69
Dual Message System (Authorization) and Single Message System.....	69
Dual Message System (Clearing).....	75
Dynamic magnetic stripe transactions.....	76
Merchant-initiated Transactions (MITs).....	80
MoneySend Payment Transactions.....	82
Dual Message System (Authorization) and Single Message System - e-Commerce MoneySend Funding Transactions.....	83
Dual Message System (Clearing) - E-Commerce MoneySend Funding Transactions.....	89
Dual Message System (Authorization) and Single Message System - e-Commerce MoneySend Payment Transactions.....	90
Dual Message System (Clearing) - E-Commerce MoneySend Payment Transactions.....	96
Dual Message System (Authorization) and Single Message System - Contactless MoneySend Funding Transactions.....	97
Dual Message System (Clearing) - Contactless MoneySend Funding Transactions.....	102

Dual Message System (Authorization) and Single Message System - Contactless MoneySend
Payment Transactions.....103

Dual Message System (Clearing) - Contactless MoneySend Payment Transactions..... 108

Purchase mail order/telephone order transactions.....109

IPM MPE and 80-byte Financial Institution Table (FIT) File

Transactions initiated with MDES tokens are routed to the Mastercard Network based on Dual Message System service-eligible account ranges that are identified on the IPM MPE Table IP0040T1 (Issuer Account Range) with value C (Mastercard Digital Enablement Service Account Range) in the Mapping Service Indicator field.

Single Message System service-eligible account ranges are identified on the 80-byte Financial Institution Table File with position 52 (Mastercard Digital Enablement Service Flag), value Y (Mastercard Digital Enablement Service is performed).

Acquirers must not perform On-Us or intra-processor processing for service-eligible account ranges because the token account numbers within those account ranges will not be known to the issuing card systems.

Contactless transactions

MDES can provision secure element tokens and cloud-based tokens for use in mobile devices.

A card present transaction occurs when a cardholder uses the token to initiate a contactless transaction by tapping their NFC-enabled mobile device at a contactless terminal. The cardholder's token and associated chip or dynamic card validation code 3 (CVC 3) data are transmitted to the terminal. The token and data are forwarded to the acquirer.

NOTE: Mastercard is enabling cardholders to use Near Field Communications (NFC)-enabled devices and wallets to initiate tokenized transactions at contactless ATMs from Release 18Q4.

NOTE: Since 1 May 2019, Mastercard has been making the cardholder experience at contactless ATMs more seamless by requiring the support of Online PIN for unattended cash for newly-provisioned digitized Mastercard and Maestro cards on mobile payment devices that are digitized through MDES.

Dual Message System (Authorization) and Single Message System

This section describes the data that acquirers must be prepared to send to, and receive from Mastercard when supporting contactless transactions generated from a mobile device using a token.

The *Data elements and values for contactless transactions* table provides the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210

Table 4: Data elements and values for contactless transactions

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's token	√	√
14	Date, Expiration	Token expiration date	√	•
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: Value either: • 07 = PAN auto-entry through Contactless M/Chip • 91 = PAN auto-entry through contactless magnetic stripe	√	•
23	Card Sequence Number	Applicable for DE 22, subfield 1, value 07	√	•
35	Track 2 Data	Either this data element or DE 45 is present. Contains the full track data (without alteration or truncation) that has been read from the data on the mobile device.	√	•
39	Response Code	If the token to PAN mapping and pre-validation services are successful, the transaction is submitted to the issuer for authorization. Otherwise, one of the following values might be received: • 05 = Do not honor; returned when the issuer has chosen for Mastercard to decline on their behalf and chip pre-validation or dynamic card validation code 3 (CVC 3) pre-validation failed • 14 = Invalid card number; returned when MDES is unable to map the token to the PAN • 30 = Format error • 62 = Restricted card; returned when a token has been suspended or deactivated from use • 96 = System error; returned when the mapping failed because of technical reasons For the full list of codes, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or <i>Mastercard Network Processing Single Message System Guide</i> .	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
45	Track 1 Data	Either this data element or DE 35 is present. Contains the full track data (without alteration or truncation) that has been read from the data on the mobile device.	√	•
48		Additional Data - Private Use		
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: If applicable	•	•
	33 - PAN Mapping File Information	The following subfields might be present in transaction response messages when the token has been successfully mapped to the PAN.		
		Subfield 1 - Account Number Indicator: Value M = PAN	•	√
		Subfield 2 - Account Number: PAN	•	√
		Subfield 3 - Expiration Date: PAN expiration date	•	√
		Subfield 5 - Token Assurance Method: Assigned token assurance method value (if applicable)	•	√
		Subfield 6 - Token Requestor ID: Token Requestor ID (as applicable)	•	√
		Subfield 8 - Storage Technology: Storage technology (as applicable)	•	√
	74 - Additional Processing Information	When the chip pre-validation result is not value V (Valid), contains the following subfields.		
		Subfield 1 - Processing Indicator: Value 02 = Mastercard On-behalf Service - M/Chip Cryptogram Pre-validation	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 2 - Processing Information: One of the following values: • A = Valid Application Cryptogram (AC); Application Transaction Counter (ATC) outside allowed range • E = Valid Application Cryptogram; ATC Replay • F = Format error in DE 55 • G = Application cryptogram is valid but is not an ARQC • I = Application cryptogram invalid • K = No matching key file for this PAN, PAN expiry date and KDI combination • T = Application cryptogram is valid but TVR/CVR was invalid • U = Application cryptogram could not be validated due to technical error • X = Security platform timeout • Z = Security platform processing error	•	√
	87 - Card Validation Code Result	When the MDES dynamic CVC 3 pre-validation is not successful, contains one of the following values: • E = Length of unpredictable number was not a valid length • P = Unable to process • Y = Invalid	•	√
55	Integrated Circuit Card (ICC) System-Related Data	Cryptogram (applicable for DE 22, subfield 1, value 07)	√	•
	91 - Issuer Authentication Data	Present when MDES chip pre-validation has been performed and an Authorization Response Cryptogram (ARPC) has been generated. An ARPC will be generated when the MDES chip pre-validation result is one of the following: • Valid • Application Cryptogram is valid but is not an ARQC (Authorization Request Cryptogram) • Application Cryptogram is valid, but TVR/CVR was invalid	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR), provided when: - Mastercard is the BIN Controller and a PAR value is associated with the PAN in the AMS - The issuer is the BIN Controller and has associated a PAR value with the PAN	•	√
105	Multi-Use Transaction Identification Data	If applicable	•	√
119	MDES Service Indicators, subelement 004: MDES Service Indicators for acquirers	If applicable. Refer to the MDES service indicators for more details.	•	√

DE 48, subelement 33 subfields are also included in:

- Authorization Advice Response/0130
- Reversal Request Response/0410
- Issuer Reversal Advice/0422

The acquirer forwards the token and other authorization or financial transaction response information including the PAR (if available) to the merchant. If the acquirer provides the PAN to the merchant, the acquirer must include only the last four digits of the PAN received in DE 48, subelement 33, subfield 2. Exceptions to this are permitted for transit merchants associated with the following card acceptor business codes (MCCs):

- 4111 = Transportation - Suburban and Local Commuter Passenger
- 4131 = Bus Lines
- 4784 = Bridge and Road Fees, Tolls
- 7523 = Automobile Parking Lots and Garages

250-byte batch data file

Single Message System acquirers will receive the following in the 250-byte batch data file when MDES has processed a transaction successfully.

- Financial (FREC)/Non-financial (NREC) and Financial/Non-financial Exception (EREC) records will contain the token in the PAN field (positions 33-51).
- Financial/Non-financial Additional Data Addendum (ADDR) will contain the cardholder's PAN in the Mapped PAN Account field (positions 120-138).
- Financial/Non-financial Additional Data Addendum (ADDR) will contain value M (PAN) in the Mapped PAN Indicator field (position 139).

If a PAR value is found, the Single Message System will provide the 29-byte PAR value in a field on the Batch Data File.

Single Message System acquirers will receive the following responses in the 250-byte Batch Data File when MDES PAN mapping is not successful. PAN mapping is considered not successful when any of the following occur:

- A PAN is not on file for a token.
- PAN mapping cannot be performed for technical reasons.
- A token has been suspended or deactivated.

The Non-financial (NREC) record will contain the token in the PAN field (positions 33-51) and the applicable advice reason code in the Advice Reason Code field (positions 108-114). The Non-financial Additional Data Addendum (ADDR) will contain blanks in the Mapped PAN Account field (positions 120-138) and a blank in the Mapped PAN Indicator field (position 139).

Single Message Transaction Manager

Exception item processing can be accessed through Single Message Transaction Manager using the cardholder's PAN or token.

Online PIN

Acquirers will continue to send the online PIN block in the Authorization Request/0100 or Financial Transaction Request/0200 messages as they do currently. When a token is used at the point of interaction, the online PIN block in DE 52 (Personal ID Number Data) or in DE 110 (Encryption Data) will be based on the token.

MDES will convert the PIN block to an issuer PIN block based on the PAN before sending the request message to the issuer.

If MDES cannot convert the PIN data, the acquirer is sent the following elements and values.

- Authorization Request Response/0110:
 - DE 39 (Response Code): Value 88 = Cryptographic failure
 - DE 48, subelement 80 (PIN Service Code): TI = The Authorization Platform was unable to translate the PIN
- Financial Transaction Request Response/0210:
 - DE 39 (Response Code): Value 63 = Error in decryption of PIN block
 - DE 48, subelement 80 (PIN Service Code): Not present

Dual Message System (Clearing)

Acquirers of contactless transactions must send First and Second Presentment/1240 messages with the following data elements and values.

Table 5: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's token
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value either: - A = PAN auto-entry through contactless magnetic stripe - M = PAN auto-entry through Contactless M/Chip
105	Multi-Use Transaction Identification Data	If applicable
PDS 0058	Token Assurance Method	Optional
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	Optional in the first presentment. Not expected in the second presentment.
PDS 0207	Wallet Identifier	If applicable

Acquirers receive PDS 0001 in these messages when MDES has processed the transaction:

- First Chargeback/1442
- Arbitration Chargeback/1442
- Retrieval Request/1644

Table 6: PDS 0001 messages after transaction processing

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: Value TC = Subfield 2 contains the PAN. Mapping was based on a source other than the authorization logs

DE/PDS	Name/Subelement	Comments/Subfield
Subfield 2 - Account Number: PAN		

On a First Presentment/1240 message, if there is no match on the authorization logs or PAN-mapping database for the token, the Clearing Platform rejects the First Presentment/1240 message with error number 2358 (DE2 PAN INVALID. THE PAN MAPPING SERVICE CANNOT MAP DE2 TO ANOTHER ACCOUNT NUMBER).

Process flow - Suspended or deactivated token (Clearing)

When a token has been suspended or deactivated from use, Mastercard will permit these transactions to pass through to the issuer for clearing; if the cardholder disputes the transaction, the issuer can chargeback under Message Reason Code 4808 (Requested/Required Authorization Not Obtained). This processing would be consistent with that applied for non-token transactions. Mastercard will not block the transactions in Clearing but rather allow the dispute resolution process to occur. For transactions including a token, Mastercard will not enforce error number 2358 (DE2 PAN INVALID. THE PAN MAPPING SERVICE CANNOT MAP DE2 TO ANOTHER ACCOUNT NUMBER) and will allow the deactivated or suspended tokens to pass through to the issuer for clearing.

Mastercard Consumer-Presented QR Transactions

A Mastercard Consumer-Presented QR Transaction is an EMV chip Transaction effected through the presentment of a QR Code by the Cardholder, using a Mobile Payment Device, and the capture of the QR Code by the Merchant containing the Transaction Data required to initiate a Transaction

For each Mastercard Consumer-Presented QR Transaction:

- There is no maximum Transaction amount.
- The Transaction must be authorized online by the Issuer.
- The Transaction must be completed with CDCVM. CDCVM is the only valid CVM for Mastercard Consumer-Presented QR Transactions.
- The Merchant must send a properly identified Authorization Request with EMV data.

NOTE: For information about Mastercard Consumer-Presented QR Transactions, refer to the Mastercard Cloud-Based Payments (MCBP) documentation and the Mastercard Consumer-Presented QR Requirements manual.

Dual Message System (Authorization) and Single Message System

This section describes the data that acquirers must be prepared to send to, and receive from, Mastercard when supporting Consumer-Presented QR Transactions generated from a mobile device using a token.

The *Data elements and values for contactless transactions* table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210

Table 7: Data elements and values for contactless transactions

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's token	√	√
14	Date, Expiration	Token expiration date	√	•
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: Value 03 = PAN Auto-entry through bar code reader	√	•
35	Track 2 Data	This data element contains the full track data (without alteration or truncation) that has been read from the data on the mobile device.	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
39	Response Code	If the token to PAN mapping and pre-validation services are successful, the transaction is submitted to the issuer for authorization. Otherwise, one of the following values might be received: • 05 = Do not honor; returned when the issuer has chosen for Mastercard to decline on their behalf and chip pre-validation or dynamic card validation code 3 (CVC 3) pre-validation failed • 14 = Invalid card number; returned when MDES is unable to map the token to the PAN • 30 = Format error • 62 = Restricted card; returned when a token has been suspended or deactivated from use • 96 = System error; returned when the mapping failed because of technical reasons For the full list of codes, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or <i>Mastercard Network Processing Single Message System Guide</i>.	•	√
48	Additional Data - Private Use			
	21 - Acceptance Data	Subfield 2 - Additional Terminal Capability Indicator, identifies Mastercard Consumer-Presented QR and barcode terminal capabilities	√	•
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: If applicable	•	•
	33 - PAN Mapping File Information	The following subfields might be present in transaction response messages when the token has been successfully mapped to the PAN.		
		Subfield 1 - Account Number Indicator: Value M = PAN	•	√
		Subfield 2 - Account Number: PAN	•	√
		Subfield 3 - Expiration Date: PAN expiration date	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 5 - Token Assurance Method: Assigned token assurance method (if applicable)	•	√
		Subfield 6 - Token Requestor ID: Token Requestor ID (as applicable)	√	√
		Subfield 8 - Storage Technology: Storage technology (as applicable)	•	√
	74 - Additional Processing Information	When the chip pre-validation result is not value V (Valid), contains the following subfields.		
		Subfield 1 - Processing Indicator: Value 02 = Mastercard On-behalf Service - M/Chip Cryptogram Pre-validation	•	√
		Subfield 2 - Processing Information: One of the following values: • A = Valid Application Cryptogram (AC); Application Transaction Counter (ATC) outside allowed range • E = Valid Application Cryptogram; ATC Replay • F = Format error in DE 55 • G = Application cryptogram is valid but is not an ARQC • I = Application cryptogram invalid • K = No matching key file for this PAN, PAN expiry date and KDI combination • T = Application cryptogram is valid but TVR/CVR was invalid • U = Application cryptogram could not be validated due to technical error • X = Security platform timeout • Z = Security platform processing error	•	√
	87 - Card Validation Code Result	When the MDES dynamic CVC 3 pre-validation is not successful, contains one of the following values: • E = Length of unpredictable number was not a valid length • P = Unable to process • Y = Invalid	•	√
55	Integrated Circuit Card (ICC) System-Related Data	Cryptogram	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
	91 - Issuer Authentication Data	Present when MDES chip pre-validation has been performed and an Authorization Response Cryptogram (ARPC) has been generated. An ARPC will be generated when the MDES chip pre-validation result is one of the following: - Valid - Application Cryptogram is valid but is not an ARQC (Authorization Request Cryptogram) - Application Cryptogram is valid, but TVR/CVR was invalid	•	√
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR), provided when: - Mastercard is the BIN Controller and a PAR value is associated with the PAN in the AMS - The issuer is the BIN Controller and has associated a PAR value with the PAN	•	√
105	Multi-Use Transaction Identification Data	If applicable	•	√
119	MDES Service Indicators, subelement 004: MDES Service Indicators for acquirers	If applicable. Refer to the MDES service indicators for more details.	•	√

DE 48, subelement 33 subfields are also included in:

- Authorization Advice Response/0130
- Reversal Request Response/0410
- Issuer Reversal Advice/0422

The acquirer forwards the token and other authorization or financial transaction response information including the PAR (if available) to the merchant. If the acquirer provides the PAN to the merchant, the acquirer must include only the last four digits of the PAN received in DE 48, subelement 33, subfield 2. Exceptions to this are permitted for transit merchants associated with the following card acceptor business codes (MCCs):

- 4111 = Transportation - Suburban and Local Commuter Passenger
- 4131 = Bus Lines
- 4784 = Bridge and Road Fees, Tolls
- 7523 = Automobile Parking Lots and Garages

250-byte batch data file

Single Message System acquirers will receive the following in the 250-byte Batch Data File when MDES has processed a transaction successfully:

- Financial (FREC)/Non-financial (NREC) and Financial/Non-financial Exception (EREC) records will contain the token in the PAN field (positions 33-51).
- Financial/Non-financial Additional Data Addendum (ADDR) will contain the cardholder's PAN in the Mapped PAN Account field (positions 120-138).
- Financial/Non-financial Additional Data Addendum (ADDR) will contain value M (PAN) in the Mapped PAN Indicator field (position 139).

If a PAR value is found, the Single Message System will provide the 29-byte PAR value in a field on the Batch Data File. Single Message System acquirers will receive the following responses in the 250-byte Batch Data File when MDES PAN mapping is not successful. PAN mapping is considered not successful when any of the following occur:

- A PAN is not on file for a token.
- PAN mapping cannot be performed for technical reasons.
- A token has been suspended or deactivated.

The Non-financial (NREC) record will contain the token in the PAN field (positions 33-51) and the applicable advice reason code in the Advice Reason Code field (positions 108-114). The Non-financial Additional Data Addendum (ADDR) will contain blanks in the Mapped PAN Account field (positions 120-138) and a blank in the Mapped PAN Indicator field (position 139). Concept definition.

Single Message Transaction Manager

Exception item processing can be accessed through Single Message Transaction Manager using the cardholder's PAN or token.

Dual Message System (Clearing)

Acquirers of contactless transactions must send First and Second Presentment/1240 messages with the following data elements and values.

Table 8: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's token

DE/PDS	Name/Subelement	Comments/Subfield
22	Point of Service Data Code	Subfield 7—Card Data—Input Mode: Value: M = PAN auto-entry through contactless M/Chip
105	Multi-Use Transaction Identification Data	If applicable
PDS 0058	Token Assurance Method	Optional
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	Optional in first presentment; not expected in second presentment
PDS 0207	Wallet Identifier	If applicable

Acquirers will receive PDS 0001 in these messages when MDES has processed the transaction:

- First Chargeback/1442
- Arbitration Chargeback/1442
- Retrieval Request/1644

Table 9: PDS 0001 messages after transaction processing

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: Value TC = Subfield 2 contains the PAN. Mapping was based on a source other than the authorization logs
Subfield 2 - Account Number: PAN		

On a First Presentment/1240 message, if there is no match on the authorization logs or PAN-mapping database for the token, the Clearing Platform will reject the First Presentment/1240 message with error number 2358 (DE2 PAN INVALID. THE PAN MAPPING SERVICE CANNOT MAP DE2 TO ANOTHER ACCOUNT NUMBER).

Process flow - Suspended or deactivated token (Clearing)

When a token has been suspended or deactivated from use, Mastercard will permit these transactions to pass through to the issuer for clearing; if the cardholder disputes the

transaction, the issuer can chargeback under Message Reason Code 4808 (Requested/Required Authorization Not Obtained). This processing would be consistent with that applied for non-token transactions. Mastercard will not block the transactions in Clearing but rather allow the dispute resolution process to occur. For transactions including a token, Mastercard will not enforce error number 2358 (DE2 PAN INVALID. THE PAN MAPPING SERVICE CANNOT MAP DE2 TO ANOTHER ACCOUNT NUMBER) and will allow the deactivated or suspended tokens to pass through to the issuer for clearing.

E-commerce transactions with Digital Secure Remote Payment (DSRP) Crypto

Tokenized e-commerce transactions enable secure transactions for remote payments through the submission of Digital Secure Remote Payment (DSRP) cryptograms generated by a payment application.

Payment applications use EMV based cryptography to generate cryptograms. These cryptograms in the transactions provide evidence that the cardholder verification is performed.

NOTE: For information relating to partial shipments and recurring payments, see [Merchant-initiated Transactions \(MITs\)](#).

Dual Message System (Authorization) and Single Message System - EMV Data

This section describes the data that acquirers must be prepared to send to, and receive from, Mastercard when supporting merchants in DSRP (full EMV).

The *Data elements and values for contactless transactions* table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210

Table 10: Data elements and values for contactless transactions

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's token	√	√
14	Date, Expiration	Token expiration date	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: Value 09 = PAN/Token entry via electronic commerce containing DSRP cryptogram in DE 55 (Integrated Circuit Card [ICC] System-Related Data)	√	•
23	Card Sequence Number		√	•
39	Response Code	If the token to PAN mapping and pre-validation services are successful, the transaction is submitted to the issuer for authorization. Otherwise, one of the following values might be received: • 05 = Do not honor; returned when the issuer has chosen for Mastercard to decline on their behalf and pre-validation failed • 14 = Invalid card number; returned when MDES is unable to map the token to the PAN • 30 = Format error • 62 = Restricted card; returned when a token has been suspended or deactivated from use • 96 = System error; returned when the mapping failed because of technical reasons For the full list of codes, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or <i>Mastercard Network Processing Single Message System Guide</i>.	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
48	Additional Data - Private Use	Transaction Category Code (TCC) value as applicable. Refer <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or <i>Mastercard Network Processing Single Message System Guide</i> for valid values.	√	√
	22 - Multi-Purpose Merchant Indicator	Subfield 1 Low-Risk Merchant Indicator - for Acquirers in the EEA, the United Kingdom, San Marino and Gibraltar when applicable in compliance with PSD 2 regulation. Subfield 5 Cardholder/Merchant Initiated Transaction Indicator - As applicable		
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: If applicable	√	•
	33 - PAN Mapping File Information	The following subfields might be present in transaction response messages when the token has been successfully mapped to the PAN. Subfield 1 - Account Number Indicator: Value M = PAN Subfield 2 - Account Number: PAN Subfield 3 - Expiration Date: PAN expiration date Subfield 5 - Token Assurance Method: Assigned token assurance method value (if applicable) Subfield 6 - Token Requestor ID: Token Requestor ID (if applicable) Subfield 8 - Storage Technology: Storage technology (if applicable)	• • • • • •	√ √ √ √ √ √
	43 - 3-D Secure Electronic Commerce Verification Service [Visa, JCB, Diners Club, and American Express]	UCAF with the encoded AAV value if provided by the merchant		
	74 - Additional Processing Information	When the chip pre-validation result is not value V (Valid), contains the following subfields.		

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 1 - Processing Indicator: Value 02 = Mastercard On-behalf Service - M/Chip Cryptogram Pre-validation	•	√
		Subfield 2 - Processing Information: One of the following values: • A = Valid Application Cryptogram (AC); Application Transaction Counter (ATC) outside allowed range • E = Valid Application Cryptogram; ATC Replay • F = Format error in DE 55 • G = Application Cryptogram is valid but is not an ARQC • I = Application Cryptogram invalid • K = No matching key file for this PAN, PAN expiry date and KDI combination • T = Application Cryptogram is valid but TVR/CVR was invalid • U = Application Cryptogram could not be validated due to technical error • X = Security platform timeout • Z = Security platform processing error	•	√
	87 - Card Validation Code Result	When the MDES dynamic CVC 3 pre-validation is not successful, contains one of the following values: • E = Length of unpredictable number was not a valid length • P = Unable to process • Y = Invalid	•	√
55	Integrated Circuit Card (ICC) System-Related Data	Cryptogram for the DSRP transaction	√	•
	91 - Issuer Authentication Data	Present when an Authorization Response Cryptogram (ARPC) has been generated. An ARPC will be generated when the MDES chip pre-validation result is one of the following: • Valid • Application Cryptogram is valid, but is not an ARQC (Authorization Request Cryptogram) • Application Cryptogram is valid, but TVR/CVR was invalid	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
56	01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR), provided when: - Mastercard is the BIN Controller and a PAR value is associated with the PAN in the AMS - The issuer is the BIN Controller and has associated a PAR value with the PAN	•	√
61	POS Data	Subfield 3 - POS Terminal Location: Value either: 2 = Off premises of card acceptor facility (cardholder terminal including home PC, mobile phone, PDA) 4 = On premises of card acceptor facility (cardholder terminal including home PC, mobile phone, PDA)	√	•
		Subfield 4 - POS Cardholder Presence: Value 5 = Electronic order (home PC, Internet, mobile phone, PDA)	√	•
		Subfield 7 - POS Transaction Status with value of either: 0 (Normal request) 1 (Deferred Authorization) 4 (Preauthorization request) 8 (Account Status Inquiry)		
		Subfield 10 - Cardholder-Activated Terminal Level: Value 6 = Authorized Level 6 CAT: Electronic commerce	√	•
105	Multi-Use Transaction Identification Data	If applicable	•	√
119	MDES Service Indicators, subelement 004: MDES Service Indicators for acquirers	If applicable. Refer to the MDES service indicators for more details.	•	√

DE 48, subelement 33, subfields are also included in the following messages:

- Authorization Advice Response/0130
- Financial Transaction Active Response/0230
- Reversal Request Response/0410
- Issuer Reversal Advice/0422
- Acquirer Reversal Advice Response/0430

The acquirer forwards the token and other authorization or financial transaction response information, including PAR (if available), to the merchant. If the acquirer provides the PAN to the merchant, the acquirer must include only the last four digits of the PAN received in DE 48, subelement 33, subfield 2. Exceptions to this are permitted for transit merchants associated with the following card acceptor business codes (MCCs):

- 4111 = Transportation - Suburban and Local Commuter Passenger
- 4131 = Bus Lines
- 4784 = Bridge and Road Fees, Tolls
- 7523 = Automobile Parking Lots and Garages

250-byte Batch Data File

Single Message System acquirers will receive the following in the 250-byte Batch Data File after MDES has been successfully applied to a transaction:

- Financial (FREC)/Non-financial (NREC) and Financial/Non-financial Exception (EREC) records will contain the token in the PAN field (positions 33–51).
- Financial/Non-financial Additional Data Addendum (ADDR) records will contain the cardholder's PAN in the Mapped PAN Account field (positions 120–138).
- Financial/Non-financial Additional Data Addendum (ADDR) records will contain value M (PAN) in the Mapped PAN Indicator field (position 139).

Single Message System acquirers will receive the following responses in the 250-byte Batch Data File when MDES PAN mapping is not successful. PAN mapping is considered not successful when any of the following occur:

- A PAN is not on file for a token
- PAN mapping cannot be performed for technical reasons
- A token has been suspended or deactivated

The Non-financial (NREC) record will contain the token in the PAN field (positions 33-51) and the applicable advice reason code in the Advice Reason Code field (positions 108-114).

The Non-financial Additional Data Addendum (ADDR) will contain blanks in the Mapped PAN Account field (positions 120-138) and a blank in the Mapped PAN Indicator field (position 139).

Single Message Transaction Manager

Exception item processing can be accessed via Single Message Transaction Manager using the cardholder's PAN. Acquirers can locate this number in DE 48, subelement 33, subfield 2 of the Financial Transaction Request Response/0210 message and the Mapped PAN Account field (positions 120-138) of the Financial/Non-financial Additional Data Addendum Record (ADDR) - Optional of the 250-byte Batch Data File.

Transactions are currently not searchable in Single Message Transaction Manager using the token account number.

Dual Message System (Clearing) - EMV Data

Acquirers that support merchants using DSRP transactions (full EMV) must send First and Second Presentment/1240 messages with the following data elements and values.

Table 11: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's token
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value R = PAN entry via electronic commerce, including remote chip
55	Integrated Circuit Card System-Related Data	If applicable
105	Multi-Use Transaction Identification Data	If applicable
PDS 0058	Token Assurance Method	Optional
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	Optional in first presentment; not expected in second presentment
PDS 0207	Wallet Identifier	If applicable

Acquirers will receive PDS 0001 in these messages when MDES has processed the transaction:

- First Chargeback/1442
- Arbitration Chargeback/1442
- Retrieval Request/1644

Table 12: PDS 0001 messages after transaction processing

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: Value TC = Subfield 2 contains the PAN. Mapping was based on a source other than the authorization logs

DE/PDS	Name/Subelement	Comments/Subfield
Subfield 2 - Account Number: PAN		

On a First Presentment/1240 message, if there is no match on the authorization logs or PAN mapping database for the token, the Clearing Platform will reject the First Presentment/1240 message with error number 2358 (DE2 PAN INVALID. THE PAN MAPPING SERVICE CANNOT MAP DE2 TO ANOTHER ACCOUNT NUMBER).

Process Flow - Suspended or Deactivated Token (Clearing)

When a token has been suspended or deactivated from use, Mastercard will permit these transactions to pass through to the issuer for clearing; if the cardholder disputes the transaction, the issuer can chargeback under message reason code 4808 (Requested/Required Authorization Not Obtained). This processing would be consistent with that applied for non-token transactions. Mastercard will not block the transactions in Clearing but rather allow the dispute resolution process to occur. For transactions including a token, Mastercard will not enforce error number 2358 (DE2 PAN INVALID. THE PAN MAPPING SERVICE CANNOT MAP DE2 TO ANOTHER ACCOUNT NUMBER) and will allow the deactivated or suspended tokens to pass through to the issuer for clearing.

Dual Message System (Authorization) and Single Message System - UCAF data

This section describes the data that acquirers must be prepared to send to, and receive from, Mastercard when supporting e-commerce merchants not yet EMV-compliant and offering cardholders the ability to initiate DSRP. This applies when the transaction is being initiated with a Secure Element (SE) or Cloud token type.

The "UCAF data" table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210

Table 13: UCAF data

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's token	√	√
14	Date, Expiration	Token expiration date	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: Value 81 = PAN/Token entry via electronic commerce with optional <i>SecureCode</i> -AAV or DSRP cryptogram in UCAF; or value 10 = Credential on File	√	•
39	Response Code	If the token to PAN mapping and pre-validation services are successful, the transaction is submitted to the issuer for authorization. Otherwise, one of these values might be received: • 05 = Do not honor; returned when the issuer has chosen for Mastercard to decline on their behalf and pre-validation failed • 14 = Invalid card number; returned when MDES is unable to map the token to the PAN • 30 = Format error • 62 = Restricted card; returned when a token has been suspended or deactivated from use • 96 = System error; returned when the mapping failed because of technical reasons For the full list of codes, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or <i>Mastercard Network Processing Single Message System Guide</i>.	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
48	Additional Data - Private Use	Transaction Category Code (TCC) value as applicable. Refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or <i>Mastercard Network Processing Single Message System Guide</i> for valid values.		
	22 - Multi-Purpose Merchant Indicator	Subfield 1 - Low-Risk Merchant Indicator: As applicable, for Acquirers in the EEA, the United Kingdom, San Marino and Gibraltar in compliance with PSD2 regulation		
		Subfield 5 - Cardholder/Merchant Initiated Transaction Indicator - As applicable	√	•
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: Wallet ID (if applicable)	•	•
	32 - Mastercard Assigned ID	When available	√	•
NOTE: Subfield 1 - in Financial Transaction Messages				
	33 - PAN Mapping File Information	These subfields might be present in transaction response messages when the token has been successfully mapped to the PAN. These subelements support the mapping between the virtual account data and actual account data.		
		Subfield 1 - Account Number Indicator: Value M = PAN	•	√
		Subfield 2 - Account Number: PAN	•	√
		Subfield 3 - Expiration Date: PAN expiration date	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 5 - Token Assurance Method: Assigned token assurance method value (if applicable) Values: 10 = Card Issuer Account Verification (Successful TAR, AS, ASI response with no authentication) 11 = Card Issuer Interactive Cardholder Authentication - 1 Factor (Token authenticated using MDES Authentication Service (OTP)) 14 = Card Issuer Asserted Authentication (Push Provisioning into without additional authentication beyond Issuer's own application) 20 = Push Provisioned token with MDES Authentication service after token creation 21 = Push Provisioned token with Identity Check™ authentication after token creation 22 = Token authenticated via Identity Check™ Risk-based authentication (no cardholder challenge) 23 = Token authenticated via Identity Check™ Interactive step-up (cardholder challenge) 24 = Token with device-asserted authentication	•	√
		Subfield 6 - Token Requestor ID: Token Requestor ID (if applicable)	•	√
		Subfield 8 - Storage Technology: Storage technology (if applicable)	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
	42 - Electronic Commerce Indicators	Subfield 1 - Electronic Commerce Security Level Indicator and UCAF Collection Indicator. NOTE: For more information about the SLI values, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide Data Element 48 (Additional Data)</i>, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator). NOTE: Mastercard will validate and, when necessary, modify the Security Level Indicator (SLI) in DE 48, subelement 42, subfield 1. When Mastercard modifies the SLI, acquirers will receive the modified SLI, original SLI, and reason for the modification in DE 48, subelement 42, subfields 1, 2, and 3, respectively. Otherwise, acquirers will only receive subfield 1.	√	√
		Subfield 2 - Original Electronic Commerce Security Level Indicator and UCAF Collection Indicator: Refer to note above	•	√
		Subfield 3 - Reason for UCAF Collection Indicator Downgrade: Refer to note above	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
	43 - Universal Cardholder Authentication Field (UCAF)	The encoded <i>SecureCode</i> AAV Cryptogram or the DSRP UCAF cryptogram generated on the cardholder device and collected by the merchant. Payment transactions do not require a cryptogram in DE 48, subelement 43 when they are uniquely identified by including DE 3, subfield 1, value 28 (Payment Transaction). NOTE: From 11 August 2020 acquirers are required to populate DRSP data in DE 104 (Digital Payment Data) subelement 001 (Digital Payment Cryptogram) instead of DE 48 (Additional Data) subelement 43 (Universal Cardholder Authentication Field - UCAF). Acquirers can identify the dynamic data type using the leading character of the cryptogram. Refer to <i>AN 6984 Identifying Between Mastercard Identity Check AAV and Digital Secure Remote Payment Cryptogram</i> for details.	√	•
	48 - Digital Commerce Solution Indicator	These subfields might be present in transactions associated with specific programs such as Click to Pay, SQR, Pay By Link. Subfield 1=Digital Commerce Domain Identifier • 0 = Multi-Domain • 1 = Mastercard • 2 = Acquirer • 3 = Issuer • 4-9 = Reserved for future use		

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 2=Digital Commerce Program Type 0 = Multi-Program 1 = SQR 2 = C2P 3 = In-store Biometric 4 = Next Gen POI 5 = Mastercard QR: Pay by Link 6 = Mastercard QR: X QR 7 = Tap To Own (to be assigned) 8 = Agentic Payment (to be assigned) 9 =EV Charging (to be assigned)		
		Subfield 3=Digital Commerce Program Details 10 = SQR 20 = C2P 30 = Biometrics 40= Next Gen POI 51= Merchant Presented PBL 52= Consumer Presented PBL 61 = Merchant Presented QR 62= Consumer Presented QR 70 = Tap To Own 80 = Agentic Payment 91 = EV Contactless 92 = EV COF		
		Subfield 4=Digital Commerce Check-In Details. Value: Open Text		
74 - Additional Processing Information		When the chip pre-validation result is 'Application Cryptogram is valid but is not an ARQC', contains these subfields.	•	√
		Subfield 1 - Processing Indicator: Value 02 = Mastercard On-behalf Service - M/Chip Cryptogram Pre-validation		
		Subfield 2 - Processing Information: Value G = Application Cryptogram is valid but is not an ARQC		

DE	Name/Subelement	Comments/Subfield	Req	Res
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR), provided when: • Mastercard is the BIN Controller and a PAR value is associated with the PAN in the AMS • The issuer is the BIN Controller and has associated a PAR value with the PAN	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
61	POS Data	Subfield 3 - POS Terminal Location: Value either: 2 = Off premises of card acceptor facility (cardholder terminal including home PC, mobile phone, PDA) 4 = On premises of card acceptor facility (cardholder terminal including home PC, mobile phone, PDA)	√	•
		Subfield 4 - POS Cardholder Presence: Whether the cardholder is present at the point of service and the condition (if the cardholder is not present). Value either: 2 = Card not present (mail/facsimile order) 3 = Cardholder not present (phone or Automated Response Unit [ARU]) 4 = Recurring transactions 5 = Cardholder not present (Electronic order [home PC, Internet, mobile phone, PDA])	√	•
		Subfield 7 - POS Transaction Status: Value either: 0 = Normal request (original presentment) 4 = Preauthorization request 8 = Account Status Inquiry Service (ASI)	√	•
		Subfield 10 - Cardholder-Activated Terminal Level: Whether the cardholder activated the terminal with the use of the card and the CAT security level; value 6 = Authorized Level 6 CAT: Electronic commerce.theres	√	•
		Subfield 11 - POS Card Data Terminal Input Capability Indicator: The terminal capabilities for transferring the data on the card into the terminal. One of the these values: 0 = Unknown or unspecified 1 = No terminal used (voice/ARU authorization) 6 = Key entry only	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
104	Digital Payment Data	Subfield 001 - Digital Payment Cryptogram NOTE: CITs with Secure Element (SE) or Cloud tokens that have no DSRP cryptogram in the DPD field (DE 104, subfield 001) are subject to decline.	√	•
		Subfield 003 - Remote Commerce Acceptor Identifier It can be optionally provided by Acquirers. Contains a merchant identifier such as the merchant business website URL or reverse domain name as presented to the consumer during checkout. The merchant identifier is provided encoded as Base64.	√	•
105	Multi-Use Transaction Identification Data	If applicable	•	√
119	MDES Service Indicators, subelement 004: MDES Service Indicators for acquirers	If applicable. Refer to the MDES service indicators for more details.	•	√

NOTE: EMV 3-D Secure (EMV 3-D Secure) Identity Check Insights requests previously known as "Data only", allow merchants to submit EMV 3-D Secure data to Mastercard without requesting issuer authentication. These Identity Check Insights requests can also be submitted with Mastercard Digital Enablement Service (MDES) tokens. Mastercard generates an authentication risk assessment in the existing Digital Transaction Insights security fields in authorization data element (DE) 48 (Additional Data - Private Use), subelement 56 (Security Services Additional Data for Issuers), subfield 1 (Security Services Indicator) and subfield 2 (Security Services Data) that issuers will receive and may consider during their authorization decision process. Identity Check Insights initiated with MDES tokens will have the SLI value of 214.

Dual Message System (Clearing) - UCAF data

Acquirers that submit DSRP containing UCAF as e-commerce transactions initiated on a Secure Element or Cloud token type must send First and Second Presentment/1240 messages with the following data elements and values.

Table 14: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's token

DE/PDS	Name/Subelement	Comments/Subfield
22	Point of Service Data Code	Subfield 4 - Terminal Operating Environment: Value either: • 2 = On card acceptor premises; unattended terminal • 4 = Off card acceptor premises; unattended
		Subfield 7 - Card Data - Input Mode: Value S = Electronic Commerce
105	Multi-Use Transaction Identification Data	If applicable

DE/PDS	Name/Subelement	Comments/Subfield
PDS0020	Digital Commerce Solution Indicator	These subfields might be present in transactions associated to specific programs such as Click to Pay, SQR, Pay By Link. Subfield 1=Digital Commerce Domain Identifier • 0 = Multi-Domain • 1 = Mastercard • 2 = Acquirer • 3 = Issuer • 4-9 = Reserved for future use Subfield 2=Digital Commerce Program Type • 0 = Multi-Program • 1 = SQR • 2 = C2P • 3 = In-store Biometric • 4 = Next Gen POI • 5 = Mastercard QR: Pay by Link • 6 = Mastercard QR: X QR • 7 = Tap To Own (to be assigned) • 8 = Agentic Payment (to be assigned) • 9 =EV Charging (to be assigned) Subfield 3=Digital Commerce Program Details • 10 = SQR • 20 = C2P • 30 = Biometrics • 40= Next Gen POI • 51= Merchant Presented PBL • 52= Consumer Presented PBL • 61 = Merchant Presented QR • 62= Consumer Presented QR • 70 = Tap To Own • 80 = Agentic Payment • 91 = EV Contactless • 92 = EV COF Subfield 4=Digital Commerce Check-In Details. Value: Open Text
PDS 0023	Terminal Type	Value CT6 = Electronic commerce transaction

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0052	Electronic Commerce Security Level Indicator	Refer to Authorization Request Response/0110 or Financial Transaction Request Response/0210. Acquirers must use the modified SLI from the Authorization Request Response/0110 or Financial Transaction Request Response/0210 response.
PDS 0058	Token Assurance Method	Optional
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	Optional in first presentment; not expected in second presentment
PDS 0207	Wallet Identifier	If applicable

DE 55 (Integrated Circuit Card System-Related Data) will not be present.

Acquirers will receive the following elements in these messages when MDES has processed the transaction:

- First Chargeback/1442
- Arbitration Chargeback/1442
- Retrieval Request/1644

Table 15: PDS 0001 messages after transaction processing

DE/PDS	Name/Subelement	Comments/Subfield
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value S = Electronic Commerce NOTE: Acquirers should note Error Number 2651: DE22 S7 INVALID. FOR MASTERCARD DIGITAL ENABLEMENT SERVICE-ELIGIBLE ACCOUNT RANGES, MUST BE PRESENT AND EQUAL TO A, M, R, OR S.
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: Value TC = Subfield 2 contains the PAN. Mapping was based on a source other than the authorization logs
		Subfield 2 - Account Number: PAN

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0052	Electronic Commerce Security Level Indicator	Refer to Authorization Request Response/0110 or Financial Transaction Request Response/0210

NOTE: The transaction scenario described above does not qualify for Full UCAF interchange in the U.S. region. Acquirers in the U.S. region should not submit these transactions for the Full UCAF interchange rate programs.

E-commerce transactions with Dynamic Token Verification Code (DTVC)

Mastercard has enhanced MDES processing to allow merchants and acquirers with legacy transaction processing systems that are not yet upgraded to support tokenized ecommerce transactions with DSRP UCAF data or DSRP full EMV data, to submit static token transactions without the presence of DSRP cryptogram in either UCAF or DPD field.

MDES generate static tokens, previously known as Card on File tokens, for use by acquirers supporting e-commerce payments. MDES maps the static token, in the e-commerce transaction, to the cardholder's PAN before forwarding the request to the issuer.

To further strengthen the security of the static token transaction, Mastercard introduced support for dynamic token verification code (DTVC), an alternative to 3-digit static CVC 2 that allows merchants to process tokenized transactions. Mastercard during the authorization processing will recognize and validate the DTVC before sending the request to the issuer for approval.

NOTE: For information relating to partial shipments and recurring payments, see [Merchant-initiated Transactions \(MITs\)](#).

Dual Message System (Authorization) and Single Message System

This section describes the data that acquirers must be prepared to send to, and receive from, Mastercard when supporting authorization or financial transaction requests for e-commerce purchases (POS Entry Mode 10, 81 or 82) with no UCAF data.

As tokenized e-commerce transactions may not have a cryptogram, acquirers must be prepared to construct the transaction request message without a cryptogram in DE 48 subelement 43, DE 104 subelement 001 or DE 55.

The "UCAF data" table shows the data elements and values that can be present for these message types:

- Authorization Request/0100

- Authorization Request Response/0110
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210

Table 16: UCAF data

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	MDES token (secure element or cloud-based)	√	√
14	Date, Expiration	Dynamic Expiration Date for the e-commerce purchase (when received from the merchant). It is used by Mastercard internal processing only; it is not passed to the issuer.	√	•
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: Value 81 = PAN/Token entry through electronic commerce with optional <i>SecureCode</i> -AAV or DSRP cryptogram in UCAF; or value 10 = Credential on File; or value 82 = PAN Auto Entry via Server (issuer, acquirer, or third-party vendor system).	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
39	Response Code	If the token to PAN mapping and pre-validation services are successful, the transaction is submitted to the issuer for authorization. Otherwise, one of the following values might be received: Subfield 1 - POS Terminal PAN Entry Mode: Value 81 = PAN/Token entry through e-commerce with optional • 05 = Do not honor; returned when the issuer has chosen for Mastercard to decline on their behalf and pre-validation failed • 14 = Invalid card number; returned when MDES is unable to map the token to the PAN • 30 = Format error • 62 = Restricted card; returned when a token has been suspended or deactivated from use • 96 = System error; returned when the mapping failed because of technical reasons For full list of applicable values, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or <i>Mastercard Network Processing Single Message System Guide</i>.	•	√
48	Additional Data - Private Use	Transaction Category Code (TCC) value as applicable. Refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or <i>Mastercard Network Processing Single Message System Guide</i> for valid values.		
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: Wallet ID (if applicable)	•	•
	33 - PAN Mapping File Information	The following subfields might be present when the token has been successfully mapped to the PAN. Subfield 1 - POS Terminal PAN Entry Mode: Value Subfield 1 - Account Number Indicator: Value M = PAN	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 2 - Account Number: PAN		
		Subfield 3 - Expiration Date: PAN expiration date		
		Subfield 5 - Token Assurance Method: Assigned token assurance method value (if applicable) Values: • 10 = Card Issuer Account Verification (Successful TAR, AS, ASI response with no authentication) • 11 = Card Issuer Interactive Cardholder Authentication - 1 Factor (Token authenticated using MDES Authentication Service (OTP)) • 14 = Card Issuer Asserted Authentication (Push Provisioning into without additional authentication beyond Issuer's own application) • 20 = Push Provisioned token with MDES Authentication service following token creation • 21 = Push Provisioned token with Identity Check™ authentication following token creation • 22 = Token authenticated via Identity Check™ Risk-based authentication (no cardholder challenge) • 23 = Token authenticated through Identity Check™ Interactive step-up (cardholder challenge) • 24 = Token with device-asserted authentication		
		Subfield 6 - Token Requestor ID: Token Requestor ID (as applicable)		
		Subfield 8 - Storage Technology: Storage technology (as applicable)		

DE	Name/Subelement	Comments/Subfield	Req	Res
	42 - Electronic Commerce Indicators	Subfield 1 - Electronic Commerce Security Level Indicator and UCAF Collection Indicator. NOTE: For more information about the SLI values, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide Data Element 48 (Additional Data)</i>, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator). NOTE: Mastercard will validate and, when necessary, modify the Security Level Indicator (SLI) in DE 48, subelement 42, subfield 1. When Mastercard modifies the SLI, acquirers will receive the modified SLI, original SLI, and reason for the modification in DE 48, subelement 42, subfields 1, 2, and 3, respectively. Otherwise, acquirers will only receive subfield 1.	√	√
		Subfield 2 - Original Electronic Commerce Security Level Indicator and UCAF Collection Indicator: See note above	•	√
		Subfield 3 - Reason for UCAF Collection Indicator Downgrade: See note above	•	√
	43 - Universal Cardholder Authentication Field (UCAF)	If applicable. MDES will not decline a transaction submitted without a DSRP cryptogram in this subelement as it may carry the SecureCode AAV Cryptogram instead.	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
	48 - Digital Commerce Solution Indicator	These subfields might be present in transactions associated to specific programs such as Click to Pay, SQR, Pay By Link. - Subfield 1=Digital Commerce Domain Identifier - Subfield 2=Digital Commerce Program Type - Subfield 3=Digital Commerce Program Details - Subfield 4=Digital Commerce Check-In Details For full list of applicable values, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or <i>Mastercard Network Processing Single Message System Guide</i>.		
	87 - Card Validation Code Result	Provided when the request message had a CVC 2 in DE 48, subelement 92	•	√
	92 - CVC 2	Optional. Dynamic Token Verification Code/CVC 2 for the e-commerce purchase (when received from the merchant). It is used by Mastercard internal processing only; it is not passed to the issuer. Value can be either Funding PAN CVC 2 or Dynamic Token Verification Code (DTVC).	√	•
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR), provided when: - Mastercard is the BIN Controller and a PAR value is associated with the PAN in the AMS - The issuer is the BIN Controller and has associated a PAR value with the PAN	•	√
105	Multi-Use Transaction Identification Data	Is applicable	•	√
119	MDES Service Indicators, subelement 004: MDES Service Indicators for acquirers	If applicable. Refer to the MDES service indicators for more details.	•	√

Acquirers can reference GCMS Table IP0040T1 (Issuer Account Range) or the 80-byte Financial Institution Table (FIT) to identify MDES token account ranges/BINs.

Dual Message System (Clearing)

Acquirers that support e-commerce transactions (POS Entry Mode 81) containing no UCAF data (DE 48, subelement 43) must send First and Second Presentment/1240 messages with the following data elements and values.

Table 17: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's token
PDS0020	Digital Commerce Solution Indicator	These subfields might be present in transactions associated to specific programs such as Click to Pay, SQR, Pay By Link. • Subfield 1=Digital Commerce Domain Identifier • Subfield 2=Digital Commerce Program Type • Subfield 3=Digital Commerce Program Details • Subfield 4=Digital Commerce Check-In Details For full list of applicable values, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i>.
22	Point of Service Data Code	Subfield 4 - Terminal Operating Environment: Value either: • 2 = On card acceptor premises; unattended terminal • 4 = Off card acceptor premises; unattended Subfield 7 - Card Data - Input Mode: Value S = Electronic Commerce
105	Multi-Use Transaction Identification Data	If applicable
PDS 0023	Terminal Type	Value CT6 = Electronic commerce transaction
PDS 0052	Electronic Commerce Security Level Indicator	Acquirers must use the modified SLI from the Authorization Request Response/0110 or Financial Transaction Request Response/0210 response.
PDS 0058	Token Assurance Method	Optional
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	Optional in first presentment; not expected in second presentment
PDS 0207	Wallet Identifier	If applicable

Acquirers will receive these elements in these messages when MDES has processed the transaction:

- First Chargeback/1442
- Arbitration Chargeback/1442
- Retrieval Request/1644

Table 18: PDS 0001 messages after transaction processing

DE/PDS	Name/Subelement	Comments/Subfield
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value S = Electronic Commerce NOTE: Acquirers should note Error Number 2651: DE22 S7 INVALID. FOR MASTERCARD DIGITAL ENABLEMENT SERVICE-ELIGIBLE ACCOUNT RANGES, MUST BE PRESENT AND EQUAL TO A, M, R, OR S.
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: Value TC = Subfield 2 contains the PAN. Mapping was based on a source other than the authorization logs Subfield 2 - Account Number: PAN
PDS 0052	Electronic Commerce Security Level Indicator	Acquirers must use the modified SLI from the Authorization Request Response/0110 or Financial Transaction Request Response/0210 response.

Dynamic magnetic stripe transactions

All acquirers supporting magnetic stripe terminals must be aware that an MDES token may be used to initiate transactions at magnetic stripe readers at their terminals.

Dual Message System (Authorization) and Single Message System

The *Data elements and values for Authorization messages* table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210

Table 19: Data elements and values for Authorization messages

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's token	√	√
14	Date, Expiration	Token expiration date	√	•
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: Value 90 = PAN auto-entry via magnetic stripe	√	•
35	Track 2 Data	Either this data element or DE 45 is present	√	•
39	Response Code	If the token to PAN mapping and pre-validation services are successful, the transaction is submitted to the issuer for authorization. Otherwise, one of the following values might be received: • 05 = Do not honor; returned when the issuer has chosen for Mastercard to decline on their behalf and dynamic CVC 3 pre-validation failed • 14 = Invalid card number; returned when MDES is unable to map the token to the PAN • 30 = Format error • 62 = Restricted card; returned when a token has been suspended or deactivated from use • 96 = System error; returned when the mapping failed because of technical reasons For the full list of codes, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or <i>Mastercard Network Processing Single Message System Guide</i>.	•	√
45	Track 1 Data	Either this data element or DE 35 is present	√	•
48	Additional Data - Private Use			
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: If applicable	•	•

DE	Name/Subelement	Comments/Subfield	Req	Res
	33 - PAN Mapping File Information	The following subfields might be present in transaction response messages when the token has been successfully mapped to the PAN.	•	√
		Subfield 1 - Account Number Indicator: Value M = PAN		
		Subfield 2 - Account Number: PAN		
		Subfield 3 - Expiration Date: PAN expiration date		
		Subfield 5 - Token Assurance Method: Assigned token assurance method value (if applicable)		
		Subfield 6 - Token Requestor ID: Token Requestor ID (as applicable)		
		Subfield 8 - Storage Technology: Storage technology (as applicable)		
	87 - Card Validation Code Result	When the MDES dynamic CVC 3 pre-validation is not successful, contains one of the following values: • E = Length of unpredictable number was not a valid length • P = Unable to process • Y = Invalid	•	√
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR), provided when: • Mastercard is the BIN Controller and a PAR value is associated with the PAN in the AMS • The issuer is the BIN Controller and has associated a PAR value with the PAN	•	√
105	Multi-Use Transaction Identification Data	If applicable	•	√
119	MDES Service Indicators, subelement 004: MDES Service Indicators for acquirers	If applicable. Refer to the MDES service indicators for more details.	•	√

Acquirers also receive DE 48 (Additional Data - Private Use), subelement 33 (PAN Mapping File Information) and its associated subfields in the following messages:

- Authorization Advice Response/0130
- Financial Transaction Advice Response/0230
- Reversal Request Response/0410
- Issuer Reversal Advice/0422

The acquirer forwards the token and other authorization response information, including PAR (if available), to the merchant. If the acquirer provides the PAN to the merchant, the acquirer must include only the last four digits of the PAN received in DE 48, subelement 33, subfield 2. Exceptions to this are permitted for transit merchants associated with the card acceptor business codes (MCCs) listed below:

- 4111 = Transportation - Suburban and Local Commuter Passenger
- 4131 = Bus Lines
- 4784 = Bridge and Road Fees, Tolls
- 7523 = Automobile Parking Lots and Garages

Dual Message System (Clearing)

Acquirers of token transactions using the magnetic stripe readers at terminals must send First and Second Presentment/1240 messages with the following data elements and values.

Table 20: Data elements and values for First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's token
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value B = Magnetic stripe reader input; track data captured and passed unaltered
105	Multi-Use Transaction Identification Data	If applicable
PDS 0058	Token Assurance Method	Optional
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	Optional in first presentment; not expected in second presentment
PDS 0207	Wallet Identifier	If applicable

Acquirers will receive PDS 0001 in these messages when MDES has processed the transaction:

- First Chargeback/1442
- Arbitration Chargeback/1442
- Retrieval Request/1644

Table 21: PDS 0001 messages after transaction procession

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: Value TC = Subfield 2 contains the PAN. Mapping was based on a source other than the authorization logs
		Subfield 2 - Account Number: PAN

Merchant-initiated Transactions (MITs)

This section describes the data that acquirers must be prepared to send to, and receive from, Mastercard when supporting partial shipment and recurring payments for MDES processing. These transactions apply to DSRP and Tokenized E-Commerce with Dynamic Payment Credentials.

Acquirers may need to verify that they can receive and process the data in DE 48 (Additional Data - Private Use), subelement 42 (Electronic Commerce Indicators) from merchants that may not be currently providing these subelements.

The *Data elements and values for Authorization messages* table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110

All mandatory data elements apply, even if not specifically shown here.

Table 22: Data elements and values for Authorization messages

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's token	√	√
4	Amount, Transaction	The amount of funds the cardholder requested in the local currency of the acquirer or source location of the transaction	√	√
14	Date, Expiration	Token expiration date	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: 81 = PAN/Token entry via electronic commerce with optional SecureCode-AAV or DSRP cryptogram in UCAF 10 = Credential on File 82 = PAN Auto Entry via Server (issuer, acquirer, or third party vendor system).	√	•
48	Additional Data - Private Use			
	22 - Multi-Purpose Merchant Indicator	Subfield 5 - Cardholder/Merchant Initiated Transaction Indicator : M1XX (Recurring payment MITs) or M2XX (Industry practice MITs) - As applicable	√	•
	33 - PAN Mapping File Information	Provided for MDES transactions	•	√
	42 - Electronic Commerce Indicators	Subfield 1 - Electronic Commerce Security Level Indicator and UCAF Collection Indicator.	√	√
		NOTE: For more information about the SLI values, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide Data Element 48 (Additional Data)</i>, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator).		
61	Point of Service Data	Subfield 3 - POS Terminal Location: Value either: 2 = Off premises of card acceptor facility (cardholder terminal including home PC, mobile phone, PDA) 4 = On premises of card acceptor facility (cardholder terminal including home PC, mobile phone, PDA) Refer to Guidance for acquirers on typical e-Commerce MITs and CITs transactions .	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 4 - POS Cardholder Presence: Whether the cardholder is present at the point of service and the condition (if the cardholder is not present); value either: 4 = Standing order/recurring transactions 5 = Electronic order (home PC, Internet, mobile phone, PDA)	√	•
		Subfield 10 - Cardholder-Activated Terminal Level: Whether the cardholder activated the terminal with the use of the card and the CAT security level; value 6 = Authorized Level 6 CAT: Electronic commerce	√	•
105	Multi-Use Transaction Identification Data	If applicable	•	√
119	MDES Service Indicators, subelement 004: MDES Service Indicators for acquirers	If applicable. Refer to the MDES service indicators for more details.	•	√

MoneySend Payment Transactions

The Mastercard® MoneySend™ and Funding Transactions service enables person-to- person transfers, account-to-account transfers, agent cash out, credit card bill payment, business to consumer disbursement, government/non-government organization to consumer disbursement, and business money transfers by allowing consumers to use their Mastercard®, Debit Mastercard®, Cirrus®, or Maestro® card on MDES enabled wallets to send and access funds.

The MoneySend and Funding Transactions service also allows use of multiple channels to initiate transactions such as an ATM, a bank branch, a stand-alone kiosk, mobile, or over the Internet.

NOTE: Originating institution identified by the ICA in DE 32 (Acquiring Institution ID code) must be registered for MoneySend. For information about the Mastercard MoneySend service, refer to the *Mastercard Network Processing Dual Message Authorization System Guide* .

Dual Message System (Authorization) and Single Message System - e-Commerce MoneySend Funding Transactions

This section describes the data that acquirers must be prepared to send to, and receive from, Mastercard when supporting MoneySend funding transactions generated from a mobile device using a token.

The following table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210

Table 23: Data elements and values for MoneySend funding transactions

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Token	√	√
3	Processing Code	Subfield 1 - Cardholder Transaction Type code: Value 00 = Funding Transaction	√	√
14	Date, Expiration	Token expiration date	√	•
18	Merchant Type	MCC 6538 (MoneySend Funding) MCC 6540 (Funding Transaction) MCC 4829 (Money Transfer) Refer to the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> for details.	√	√
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode • 81 = PAN/Token entry through electronic commerce with optional SecureCode-AAV or DSRP cryptogram in UCAF • 10 = Credential on file	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
39	Response Code	If the token to PAN mapping and pre-validation services are successful, the transaction is submitted to the issuer for authorization. Otherwise, one of the following values might be received: • 05 = Do not honor; returned when the issuer has chosen for Mastercard to decline on their behalf and chip pre-validation or dynamic card validation code 3 (CVC 3) pre-validation failed • 14 = Invalid card number; returned when MDES is unable to map the token to the PAN • 30 = Format error • 62 = Restricted card; returned when a token has been suspended or deactivated from use • 96 = System error; returned when the mapping failed because of technical reasons For the full list of codes, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or <i>Mastercard Network Processing Single Message System Guide</i>.	•	√
48	Additional Data - Private Use			
	TCC	Transaction Category Code • R = Face-to-face retail (applicable to MCCs 6538 and 6540) • T = Non-face-to-face retail (applicable to MCCs 6538 and 6540) • U = Unique Transaction – Quasi Cash Disbursement (applicable to MCC 4829)	√	√
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: If applicable	•	√
	33 - PAN Mapping File Information	The following subfields might be present in transaction response messages when the token has been successfully mapped to the PAN.		
		Subfield 1 - Account Number Indicator: Value M = PAN	•	√
		Subfield 2 - Account Number: PAN	•	√
		Subfield 3 - Expiration Date: PAN expiration date	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 5 - Token Assurance Method: Assigned token assurance method (if applicable)	√	√
		Subfield 6 -Token Requestor ID: Token Requestor ID (as applicable)	√	√
		Subfield 8 - Storage Technology: Storage technology (as applicable)	•	√
42 - Electronic Commerce Indicators		Subfield 1 - Electronic Commerce Security Level Indicator and UCAF Collection Indicator.	√	√
		NOTE: For more information about the SLI values, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide Data Element 48 (Additional Data), subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator).</i>		
		NOTE: Mastercard will validate and, when necessary, modify the Security Level Indicator (SLI) in DE 48, subelement 42, subfield 1. When Mastercard modifies the SLI, acquirers will receive the modified SLI, original SLI, and reason for the modification in DE 48, subelement 42, subfields 1, 2, and 3, respectively. Otherwise, acquirers will only receive subfield 1.		
		Subfield 2 - Original Electronic Commerce Security Level Indicator and UCAF Collection Indicator	•	√
		Subfield 3 - Reason for UCAF Collection Indicator Downgrade	•	√
43 - Universal Cardholder Authentication Field (UCAF)		The encoded SecureCode AAV Cryptogram collected by the merchant.	√	•
		NOTE: From August 11, 2020 acquirers are required to populate DRSP UCAF data in DE 104 (Digital Payment Data) subelement 001 (Digital Payment Cryptogram) instead of DE 48 (Additional Data) subelement 43 (Universal Cardholder Authentication Field - UCAF).		

DE	Name/Subelement	Comments/Subfield	Req	Res
	74 - Additional Processing Information	When the chip pre-validation result is not value V (Valid), contains the following subfields. Subfield 1 - Processing Indicator: Value 02 = Mastercard On-behalf Service - M/Chip Cryptogram Pre-validation Subfield 2 - Processing Information: One of the following values: • A = Valid Application Cryptogram (AC); Application Transaction Counter (ATC) outside allowed range • E = Valid Application Cryptogram; ATC Replay • F = Format error in DE 55 • G = Application cryptogram is valid but is not an ARQC • I = Application cryptogram invalid • K = No matching key file for this PAN, PAN expiry date and KDI combination • T = Application cryptogram is valid but TVR/CVR was invalid • U = Application cryptogram could not be validated due to technical error • X = Security platform timeout • Z = Security platform processing error		
	77 - Transaction Type Identifier	• C07 = General Person-to-Person • C52 = General Transfer to Own Account • C53 = Agent Cash Out • C54 = Payment of Own Credit Card Bill • C55 = Business Disbursement • C56 = Government/Non-profit Disbursement • C57 = Rapid Merchant Settlement • C65 = General Business to Business Transfer • F07 = General Person-to-Person Transfer • F08 • F52 = General Transfer to Own Account • F53 = Agent Cash Out • F54 = Payment of Own Credit Card Bill • F55 = Business Disbursement • F61 = Transfer to Own Staged Digital Wallet Account • F64 = Transfer to Own Debit or Prepaid Account • F65 = General Business-to-Business Transfer	√	√

DE	Name/Subelement	Comments/Subfield	Req	Res
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR), provided when: - Mastercard is the BIN Controller and a PAR value is associated with the PAN in the AMS - The issuer is the BIN Controller and has associated a PAR value with the PAN	•	√
61	POS Data	Subfield 04 - POS Cardholder Presence: Value 5 = Cardholder not present (Electronic order [home PC, Internet, mobile phone, PDA])	√	√
		Subfield 05 - POS Card Presence: Value 1 = Card not present	√	√
		Subfield 10 - Cardholder-Activated Terminal Level: Value 6 = Electronic commerce	√	√
104	Digital Payment Data	If applicable. Payment transactions do not require a cryptogram Subfield 001 - Digital Payment Cryptogram: DSRP crypto The encoded DSRP UCAF cryptogram generated on the cardholder device and collected by the merchant.	•	•
		Subfield 002 - Estimated Amount: contains the estimated purchase amount of the transaction when the payment request is presented to the consumer during checkout. This amount may not always be the final amount. The estimated amount is encoded and included within the cryptogram. NOTE: The estimated amount is in the transaction currency defined in DE 49 (Currency Code, Transaction).	•	•
		Subfield 003 - Remote Commerce Acceptor Identifier It can be optionally provided by Acquirers. Contains a merchant identifier such as the merchant business website URL or reverse domain name as presented to the consumer during checkout. The merchant identifier is provided encoded as Base64.	•	•
105	Multi-Use Transaction Identification Data	If applicable	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
108	Additional Transaction Reference	Required for all MoneySend and Funding transactions. As applicable for MoneySend use cases. Refer to the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> for details.	√	√
119	MDES Service Indicators, subelement 004: MDES Service Indicators for acquirers	If applicable. Refer to the MDES service indicators for more details.	•	√

250-byte batch data file

Single Message System acquirers will receive the following in the 250-byte Batch Data File when MDES has processed a transaction successfully:

- Financial (FREC)/Non-financial (NREC) and Financial/Non-financial Exception (EREC) records will contain the token in the PAN field (positions 33-51).
- Financial/Non-financial Additional Data Addendum (ADDR) will contain the cardholder's PAN in the Mapped PAN Account field (positions 120-138).
- Financial/Non-financial Additional Data Addendum (ADDR) will contain value M (PAN) in the Mapped PAN Indicator field (position 139).

If a PAR value is found, the Single Message System will provide the 29-byte PAR value in a field on the Batch Data File. Single Message System acquirers will receive the following responses in the 250-byte Batch Data File when MDES PAN mapping is not successful. PAN mapping is considered not successful when any of the following occur:

- A PAN is not on file for a token
- PAN mapping cannot be performed for technical reasons
- A token has been suspended or deactivated

The Non-financial (NREC) record will contain the token in the PAN field (positions 33-51) and the applicable advice reason code in the Advice Reason Code field (positions 108-114). The Non-financial Additional Data Addendum (ADDR) will contain blanks in the Mapped PAN Account field (positions 120-138) and a blank in the Mapped PAN Indicator field (position 139). Concept definition.

Single Message Transaction Manager

Exception item processing can be accessed through Single Message Transaction Manager using the cardholder's PAN or token.

Dual Message System (Clearing) - E-Commerce MoneySend Funding Transactions

Acquirers of payment transactions must send First and Second Presentment/1240 messages with the following data elements and values.

Table 24: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's token
3	Processing Code	Value 00 = Funding
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value S = Electronic Commerce or Value 7 = Credential on File
26	Card Acceptor Business Code (MCC)	MCC 6538 (MoneySend Funding) or MCC 6540 (Funding Transaction) or MCC 4829 (Money Transfer)
105	Multi-Use Transaction Identification Data	If applicable
PDS 0023	Terminal Type	Value CT6 = Electronic commerce transaction
PDS 0043	Transaction Type Identifier	As applicable for use cases. Refer to the definitions in the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> and <i>Mastercard Network Processing Dual Message Authorization System Guide</i> .
PDS 0058	Token Assurance Method	Optional
PDS 0198	Device Type	Optional in first presentment; not expected in second presentment
PDS 0765	MoneySend Receiver Data	If applicable

Acquirers will receive PDS 0001 in these messages when MDES has processed the transaction:

- First Chargeback/1442
- Arbitration Chargeback/1442
- Retrieval Request/1644

Table 25: PDS 0001 messages after transaction processing

DE/PDS	Name/Subelement	Comments/Subfield
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value S = Electronic Commerce

DE/PDS	Name/Subelement	Comments/Subfield
105	Multi-Use Transaction Identification Data	If applicable
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: Value TC = Subfield 2 contains the PAN. Mapping was based on a source other than the authorization logs Subfield 2 - Account Number: PA
PDS 0052	Electronic Commerce Security Level Indicator	Refer to Authorization Request Response/0110 or Financial Transaction Request Response/0210.

Dual Message System (Authorization) and Single Message System - e-Commerce MoneySend Payment Transactions

This section describes the data that acquirers must be prepared to send to, and receive from, Mastercard when supporting MoneySend payment transactions generated from a mobile device using a token.

The following table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210

Table 26:

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Token	√	√
3	Processing Code	Subfield 1 - Cardholder Transaction Type code: Value 28 = Payment Transaction	√	√
14	Date, Expiration	Token expiration date	√	•
18	Merchant Type	MCC 6538 (MoneySend Funding) MCC 6537 (MoneySend Intercountry) Refer to the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> for details.	√	√

DE	Name/Subelement	Comments/Subfield	Req	Res
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode 81 = PAN/Token entry through electronic commerce with optional SecureCode-AAV or DSRP cryptogram in UCAF 10 = Credential on file	√	•
39	Response Code	If the token to PAN mapping and pre-validation services are successful, the transaction is submitted to the issuer for authorization. Otherwise, one of the following values might be received: 05 = Do not honor; returned when the issuer has chosen for Mastercard to decline on their behalf and chip pre-validation or dynamic card validation code 3 (CVC 3) pre-validation failed 14 = Invalid card number; returned when MDES is unable to map the token to the PAN 30 = Format error 62 = Restricted card; returned when a token has been suspended or deactivated from use 96 = System error; returned when the mapping failed because of technical reasons For the full list of codes, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or <i>Mastercard Network Processing Single Message System Guide</i> .	•	√
48	Additional Data - Private Use			
	TCC	P = Payment Transaction	√	√
		NOTE: If TCC is not supplied by the acquirer, Mastercard will insert a "P".		
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: If applicable	•	√
	33 - PAN Mapping File Information	The following subfields might be present in transaction response messages when the token has been successfully mapped to the PAN.		
		Subfield 1 - Account Number Indicator: Value M = PAN	•	√
		Subfield 2 - Account Number: PAN	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 3 - Expiration Date: PAN expiration date	•	√
		Subfield 5 - Token Assurance Method: Assigned token assurance method (if applicable)	√	√
		Subfield 6 -Token Requestor ID: Token Requestor ID (as applicable)	√	√
		Subfield 8 - Storage Technology: Storage technology (as applicable)	•	√
42 - Electronic Commerce Indicators		Subfield 1 - Electronic Commerce Security Level Indicator and UCAF Collection Indicator.	√	√
		NOTE: For more information about the SLI values, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide Data Element 48 (Additional Data), subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator).</i>		
		Subfield 2 - Original Electronic Commerce Security Level Indicator and UCAF Collection Indicator	•	√
		Subfield 3 - Reason for UCAF Collection Indicator Downgrade	•	√
43 - Universal Cardholder Authentication Field (UCAF)		The encoded SecureCode AAV Cryptogram collected by the merchant.	√	•
		NOTE: Payment transactions do not require a cryptogram. If provided, Mastercard will modify the Security Level Indicator (SLI) in DE 48, subelement 42, subfield 1. Acquirers will receive the modified SLI, original SLI, and reason for the modification in DE 48, subelement 42, subfields 1, 2, and 3, respectively.		

DE	Name/Subelement	Comments/Subfield	Req	Res
	74 - Additional Processing Information	Applicable when a UCAF or DSRP cryptogram is captured on a payment transaction. When the chip pre-validation result is not value V (Valid), contains the following subfields. Subfield 1 - Processing Indicator: Value 02 = Mastercard On-behalf Service - M/Chip Cryptogram Pre-validation Subfield 2 - Processing Information: One of the following values: • A = Valid Application Cryptogram (AC); Application Transaction Counter (ATC) outside allowed range • E = Valid Application Cryptogram; ATC Replay • F = Format error in DE 55 • G = Application cryptogram is valid but is not an ARQC • I = Application cryptogram invalid • K = No matching key file for this PAN, PAN expiry date and KDI combination • T = Application cryptogram is valid but TVR/CVR was invalid • U = Application cryptogram could not be validated due to technical error • X = Security platform timeout • Z = Security platform processing error		
	77 - Transaction Type Identifier	• C07 = General Person-to-Person • C52 = General Transfer to Own Account • C53 = Agent Cash Out • C54 = Payment of Own Credit Card Bill • C55 = Business Disbursement • C56 = Government/Non-profit Disbursement • C57 = Rapid Merchant Settlement • C65 = General Business to Business Transfer	√	√
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR), provided when: • Mastercard is the BIN Controller and a PAR value is associated with the PAN in the AMS • The issuer is the BIN Controller and has associated a PAR value with the PAN	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
61	POS Data	Subfield 04 - POS Cardholder Presence: Value 5 = Cardholder not present (Electronic order [home PC, Internet, mobile phone, PDA])	√	√
		Subfield 05 - POS Card Presence: Value 1 = Card not present	√	√
		Subfield 10 - Cardholder-Activated Terminal Level: Value 6 = Electronic commerce	√	√
104	Digital Payment Data	Subfield 001 -Digital Payment Cryptogram The encoded DSRP UCAF cryptogram generated on the cardholder device and collected by the merchant. NOTE: Payment transactions do not require a cryptogram. If provided, Mastercard will modify the Security Level Indicator (SLI) in DE 48, subelement 42, subfield 1. Acquirers will receive the modified SLI, original SLI, and reason for the modification in DE 48, subelement 42, subfields 1, 2, and 3, respectively.	•	•
		Subfield 002 - Estimated Amount: contains the estimated purchase amount of the transaction when the payment request is presented to the consumer during checkout. This amount may not always be the final amount. The estimated amount is encoded and included within the cryptogram. NOTE: The estimated amount is in the transaction currency defined in DE 49 (Currency Code, Transaction).	•	•
		Subfield 003 - Remote Commerce Acceptor Identifier It can be optionally provided by Acquirers. Contains a merchant identifier such as the merchant business website URL or reverse domain name as presented to the consumer during checkout. The merchant identifier is provided encoded as Base64.	•	•
105	Multi-Use Transaction Identification Data	If applicable	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
108	Additional Transaction Reference	Required for all MoneySend and Funding transactions. As applicable for MoneySend use cases. Refer to the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> for details.	√	√
119	MDES Service Indicators, subelement 004: MDES Service Indicators for acquirers	If applicable. Refer to the MDES service indicators for more details.	•	√

250-byte batch data file

Single Message System acquirers will receive the following in the 250-byte Batch Data File when MDES has processed a transaction successfully:

- Financial (FREC)/Non-financial (NREC) and Financial/Non-financial Exception (EREC) records will contain the token in the PAN field (positions 33-51).
- Financial/Non-financial Additional Data Addendum (ADDR) will contain the cardholder's PAN in the Mapped PAN Account field (positions 120-138).
- Financial/Non-financial Additional Data Addendum (ADDR) will contain value M (PAN) in the Mapped PAN Indicator field (position 139).

If a PAR value is found, the Single Message System will provide the 29-byte PAR value in a field on the Batch Data File. Single Message System acquirers will receive the following responses in the 250-byte Batch Data File when MDES PAN mapping is not successful. PAN mapping is considered not successful when any of the following occur:

- A PAN is not on file for a token
- PAN mapping cannot be performed for technical reasons
- A token has been suspended or deactivated

The Non-financial (NREC) record will contain the token in the PAN field (positions 33-51) and the applicable advice reason code in the Advice Reason Code field (positions 108-114). The Non-financial Additional Data Addendum (ADDR) will contain blanks in the Mapped PAN Account field (positions 120-138) and a blank in the Mapped PAN Indicator field (position 139). Concept definition.

Single Message Transaction Manager

Exception item processing can be accessed through Single Message Transaction Manager using the cardholder's PAN or token.

Dual Message System (Clearing) - E-Commerce MoneySend Payment Transactions

Acquirers of payment transactions must send First and Second Presentment/1240 messages with the following data elements and values.

Table 27: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's token
3	Processing Code	Value 00 = Funding
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value S = Electronic Commerce or Value 7 = Credential on File
26	Card Acceptor Business Code (MCC)	MCC 6536 (MoneySend Intracountry) or MCC 6537 (MoneySend Intercountry)
105	Multi-Use Transaction Identification Data	If applicable
PDS 0023	Terminal Type	Value CT6 = Electronic commerce transaction
PDS 0043	Transaction Type Identifier	As applicable for use cases. Refer to the definitions in the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> and <i>Mastercard Network Processing Dual Message Authorization System Guide</i> .
PDS 0058	Token Assurance Method	Optional
PDS 0198	Device Type	Optional in first presentment; not expected in second presentment
PDS 0765	MoneySend Receiver Data	If applicable, provide the payee full name and address

Acquirers will receive PDS 0001 in these messages when MDES has processed the transaction:

- First Chargeback/1442
- Arbitration Chargeback/1442
- Retrieval Request/1644

Table 28: PDS 0001 messages after transaction processing

DE/PDS	Name/Subelement	Comments/Subfield
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value S = Electronic Commerce or Value 7 = Credential on File

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: Value TC = Subfield 2 contains the PAN. Mapping was based on a source other than the authorization logs Subfield 2 – Account Number: PAN
PDS 0052	Electronic Commerce Security Level Indicator	Refer to Authorization Request Response/0110 or Financial Transaction Request Response/0210

Dual Message System (Authorization) and Single Message System - Contactless MoneySend Funding Transactions

This section describes the data that acquirers must be prepared to send to, and receive from, Mastercard when supporting MoneySend funding transactions generated from a mobile device using a token.

The following table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210

Table 29: Data elements and values for MoneySend transactions

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Token	√	√
3	Processing Code	Subfield 1 - Cardholder Transaction Type code: Value 00 = Funding Transaction	√	√
14	Date, Expiration	Token expiration date	√	•
18	Merchant Type	MCC 6538 (MoneySend Funding) MCC 6540 (Funding Transaction) MCC 4829 (Money Transfer) Refer to the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> for details.	√	√

DE	Name/Subelement	Comments/Subfield	Req	Res
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode 07 = PAN auto-entry through Contactless M/Chip 90 = PAN auto-entry through magnetic stripe	√	•
23	Card Sequence Number	Applicable for DE 22, subfield 1, value 07	√	•
35	Track2 Data	Either this data element or DE 45 is present Contains the full track data (without alteration or truncation) that has been read from the data on the mobile device.	√	•
39	Response Code	If the token to PAN mapping and pre-validation services are successful, the transaction is submitted to the issuer for authorization. Otherwise, one of the following values might be received: 05 = Do not honor; returned when the issuer has chosen for Mastercard to decline on their behalf and chip pre-validation or dynamic card validation code 3 (CVC 3) pre-validation failed 14 = Invalid card number; returned when MDES is unable to map the token to the PAN 30 = Format error 62 = Restricted card; returned when a token has been suspended or deactivated from use 96 = System error; returned when the mapping failed because of technical reasons For the full list of codes, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or <i>Mastercard Network Processing Single Message System Guide</i> .	•	√
45	Track 1 data	Either this data element or DE 35 is present. Contains the full track data (without alteration or truncation) that has been read from the data on the mobile device.	•	•
48	Additional Data - Private Use			

Acquirer-specific information

Dual Message System (Authorization) and Single Message System - Contactless MoneySend
Funding Transactions

DE	Name/Subelement	Comments/Subfield	Req	Res
	TCC	Transaction Category Code • R = Face-to-face retail (applicable to MCCs 6538 and 6540) • T = Non-face-to-face retail (applicable to MCCs 6538 and 6540) • U = Unique Transaction - Quasi Cash Disbursement (applicable to MCC 4829)	√	√
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: If applicable	•	√
	33 - PAN Mapping File Information	The following subfields might be present in transaction response messages when the token has been successfully mapped to the PAN.		
		Subfield 1 - Account Number Indicator: Value M = PAN	•	√
		Subfield 2 - Account Number: PAN	•	√
		Subfield 3 - Expiration Date: PAN expiration date	•	√
		Subfield 5 - Token Assurance Method: Assigned token assurance method (if applicable)	√	√
		Subfield 6 -Token Requestor ID: Token Requestor ID (as applicable)	√	√
		Subfield 8 - Storage Technology: Storage technology (as applicable)	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
	74 - Additional Processing Information	When the chip pre-validation result is not value V (Valid), contains the following subfields. Subfield 1 - Processing Indicator: Value 02 = Mastercard On-behalf Service - M/Chip Cryptogram Pre-validation Subfield 2 - Processing Information: One of the following values: • A = Valid Application Cryptogram (AC); Application Transaction Counter (ATC) outside allowed range • E = Valid Application Cryptogram; ATC Replay • F = Format error in DE 55 • G = Application cryptogram is valid but is not an ARQC • I = Application cryptogram invalid • K = No matching key file for this PAN, PAN expiry date and KDI combination • T = Application cryptogram is valid but TVR/CVR was invalid • U = Application cryptogram could not be validated due to technical error • X = Security platform timeout • Z = Security platform processing error		
	77 - Transaction Type Identifier	• C07 = General Person-to-Person • C52 = General Transfer to Own Account • C53 = Agent Cash Out • C54 = Payment of Own Credit Card Bill • C55 = Business Disbursement • C56 = Government/Non-profit Disbursement • C57 = Rapid Merchant Settlement • C65 = General Business to Business Transfer • F07 = General Person-to-Person Transfer • F08 • F52 = General Transfer to Own Account • F53 = Agent Cash Out • F54 = Payment of Own Credit Card Bill • F55 = Business Disbursement • F61 = Transfer to Own Staged Digital Wallet Account • F64 = Transfer to Own Debit or Prepaid Account • F65 = General Business-to-Business Transfer	√	√

DE	Name/Subelement	Comments/Subfield	Req	Res
55	Integrated Circuit Card (ICC) System-Related Data	Cryptogram (applicable for DE 22, subfield 1, value 07)	√	•
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR), provided when: • Mastercard is the BIN Controller and a PAR value is associated with the PAN in the AMS • The issuer is the BIN Controller and has associated a PAR value with the PAN	•	√
61	POS Data	Subfield 04 - POS Cardholder Presence: Value 0 = Cardholder present	√	√
		Subfield 05 - POS Card Presence: Value 0 = Card present	√	√
		Subfield 10 - Cardholder-Activated Terminal Level: As applicable	√	√
105	Multi-Use Transaction Identification Data	If applicable	•	√
108	Additional Transaction Reference	Required for all MoneySend and Funding transactions. As applicable for MoneySend use cases. Refer to the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> for details.	√	√
119	MDES Service Indicators, subelement 004: MDES Service Indicators for acquirers	If applicable. Refer to the MDES service indicators for more details.	•	√

250-byte batch data file

Single Message System acquirers will receive the following in the 250-byte Batch Data File when MDES has processed a transaction successfully:

- Financial (FREC)/Non-financial (NREC) and Financial/Non-financial Exception (EREC) records will contain the token in the PAN field (positions 33-51).
- Financial/Non-financial Additional Data Addendum (ADDR) will contain the cardholder's PAN in the Mapped PAN Account field (positions 120-138).
- Financial/Non-financial Additional Data Addendum (ADDR) will contain value M (PAN) in the Mapped PAN Indicator field (position 139).

If a PAR value is found, the Single Message System will provide the 29-byte PAR value in a field on the Batch Data File. Single Message System acquirers will receive the following responses in

the 250-byte Batch Data File when MDES PAN mapping is not successful. PAN mapping is considered not successful when any of the following occur:

- A PAN is not on file for a token
- PAN mapping cannot be performed for technical reasons
- A token has been suspended or deactivated

The Non-financial (NREC) record will contain the token in the PAN field (positions 33-51) and the applicable advice reason code in the Advice Reason Code field (positions 108-114). The Non-financial Additional Data Addendum (ADDR) will contain blanks in the Mapped PAN Account field (positions 120-138) and a blank in the Mapped PAN Indicator field (position 139). Concept definition.

Single Message Transaction Manager

Exception item processing can be accessed through Single Message Transaction Manager using the cardholder's PAN or token.

Dual Message System (Clearing) - Contactless MoneySend Funding Transactions

Acquirers of payment transactions must send First and Second Presentment/ 1240 messages with the following data elements and values.

Table 30: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's token
3	Processing Code	Value 00 = Funding
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: As applicable
26	Card Acceptor Business Code (MCC)	MCC 6538 (MoneySend Funding) or MCC 6540 (Funding Transaction) or MCC 4829 (Money Transfer)
105	Multi-Use Transaction Identification Data	If applicable
PDS 0023	Terminal Type	As applicable
PDS 0043	Transaction Type Identifier	As applicable for use cases. Refer to the definitions in the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> and <i>Mastercard Network Processing Dual Message Authorization System Guide</i> .
PDS 0058	Token Assurance Method	Optional
PDS 0198	Device Type	Optional in first presentment; not expected in second presentment

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0765	MoneySend Receiver Data	If applicable

Acquirers will receive PDS 0001 in these messages when MDES has processed the transaction:

- First Chargeback/1442
- Arbitration Chargeback/1442
- Retrieval Request/1644

Table 31: PDS 0001 messages after transaction processing

DE/PDS	Name/Subelement	Comments/Subfield
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: As applicable
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: Value TC = Subfield 2 contains the PAN. Mapping was based on a source other than the authorization logs Subfield 2 - Account Number: PAN

Dual Message System (Authorization) and Single Message System - Contactless MoneySend Payment Transactions

This section describes the data that acquirers must be prepared to send to, and receive from, Mastercard when supporting MoneySend payment transactions generated from a mobile device using a token.

The following table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210

Table 32: Data elements and values for MoneySend transactions

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Token	√	√
3	Processing Code	Subfield 1 - Cardholder Transaction Type code: Value 28 = Payment Transaction	√	√

DE	Name/Subelement	Comments/Subfield	Req	Res
14	Date, Expiration	Token expiration date	√	•
18	Merchant Type	MCC 6536 (MoneySend Intracountry) MCC 6537 (MoneySend Intercountry) Refer to the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> for details.	√	√
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode • 07 = PAN auto-entry through Contactless M/Chip • 90 = PAN auto-entry through magnetic stripe	√	•
23	Card Sequence Number	Applicable for DE 22, subfield 1, value 07	√	•
35	Track2 Data	Either this data element or DE 45 is present Contains the full track data (without alteration or truncation) that has been read from the data on the mobile device.	√	•
39	Response Code	If the token to PAN mapping and pre-validation services are successful, the transaction is submitted to the issuer for authorization. Otherwise, one of the following values might be received: • 05 = Do not honor; returned when the issuer has chosen for Mastercard to decline on their behalf and chip pre-validation or dynamic card validation code 3 (CVC 3) pre-validation failed • 14 = Invalid card number; returned when MDES is unable to map the token to the PAN • 30 = Format error • 62 = Restricted card; returned when a token has been suspended or deactivated from use • 96 = System error; returned when the mapping failed because of technical reasons For the full list of codes, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or <i>Mastercard Network Processing Single Message System Guide</i> .	•	√
45	Track 1 data	Either this data element or DE 35 is present. Contains the full track data (without alteration or truncation) that has been read from the data on the mobile device.	•	•

DE	Name/Subelement	Comments/Subfield	Req	Res
48	Additional Data - Private Use			
	TCC	P = Payment Transaction	√	√
		NOTE: If TCC is not supplied by the acquirer, Mastercard will insert a "P".		
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: (if applicable)	•	√
	33 - PAN Mapping File Information	The following subfields might be present in transaction response messages when the token has been successfully mapped to the PAN.		
		Subfield 1 - Account Number Indicator: Value M = PAN	•	√
		Subfield 2 - Account Number: PAN	•	√
		Subfield 3 - Expiration Date: PAN expiration date	•	√
		Subfield 5 - Token Assurance Method: Assigned token assurance method (if applicable)	√	√
		Subfield 6 -Token Requestor ID: Token Requestor ID (as applicable)	√	√
		Subfield 8 - Storage Technology: Storage technology (as applicable)	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
	74 - Additional Processing Information	When the chip pre-validation result is not value V (Valid), contains the following subfields. Subfield 1 - Processing Indicator: Value 02 = Mastercard On-behalf Service - M/Chip Cryptogram Pre-validation Subfield 2 - Processing Information: One of the following values: • A = Valid Application Cryptogram (AC); Application Transaction Counter (ATC) outside allowed range • E = Valid Application Cryptogram; ATC Replay • F = Format error in DE 55 • G = Application cryptogram is valid but is not an ARQC • I = Application cryptogram invalid • K = No matching key file for this PAN, PAN expiry date and KDI combination • T = Application cryptogram is valid but TVR/CVR was invalid • U = Application cryptogram could not be validated due to technical error • X = Security platform timeout • Z = Security platform processing error		
	77 - Transaction Type Identifier	• C07 = MoneySend Person-to-Person • C52 = MoneySend Account-to-Account Transfers • C53 = MoneySend Agent Cash Out • C54 = MoneySend Credit Card Bill Payment • C55 = MoneySend Business Disbursement • C56 = MoneySend Government/ Nonprofit Disbursement • C57 = MoneySend Acquirer Merchant Settlement • C58 = MoneySend Cash2ATM • C59 = MoneySend Cash2Card • C65 = MoneySend Business to Business Transfer	√	√
55	Integrated Circuit Card (ICC) System-Related Data	Cryptogram (applicable for DE 22, subfield 1, value 07)	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR), provided when: - Mastercard is the BIN Controller and a PAR value is associated with the PAN in the AMS - The issuer is the BIN Controller and has associated a PAR value with the PAN	•	√
61	POS Data	Subfield 04 - POS Cardholder Presence: Value 0 = Cardholder present	√	√
		Subfield 05 - POS Card Presence: Value 0 = Card present	√	√
		Subfield 10 - Cardholder-Activated Terminal Level: As applicable	√	√
105	Multi-Use Transaction Identification Data	If applicable	•	√
108	Additional Transaction Reference	Required for all MoneySend and Funding transactions. As applicable for MoneySend use cases. Refer to the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> for details.	√	√
119	MDES Service Indicators, subelement 004: MDES Service Indicators for acquirers	If applicable. Refer to the MDES service indicators for more details.	•	√

250-byte batch data file

Single Message System acquirers will receive the following in the 250-byte Batch Data File when MDES has processed a transaction successfully:

- Financial (FREC)/Non-financial (NREC) and Financial/Non-financial Exception (EREC) records will contain the token in the PAN field (positions 33-51).
- Financial/Non-financial Additional Data Addendum (ADDR) will contain the cardholder's PAN in the Mapped PAN Account field (positions 120-138).
- Financial/Non-financial Additional Data Addendum (ADDR) will contain value M (PAN) in the Mapped PAN Indicator field (position 139).

If a PAR value is found, the Single Message System will provide the 29-byte PAR value in a field on the Batch Data File. Single Message System acquirers will receive the following responses in the 250-byte Batch Data File when MDES PAN mapping is not successful. PAN mapping is considered not successful when any of the following occur:

- A PAN is not on file for a token

- PAN mapping cannot be performed for technical reasons
- A token has been suspended or deactivated

The Non-financial (NREC) record will contain the token in the PAN field (positions 33-51) and the applicable advice reason code in the Advice Reason Code field (positions 108-114). The Non-financial Additional Data Addendum (ADDR) will contain blanks in the Mapped PAN Account field (positions 120-138) and a blank in the Mapped PAN Indicator field (position 139). Concept definition.

Single Message Transaction Manager

Exception item processing can be accessed through Single Message Transaction Manager using the cardholder's PAN or token.

Dual Message System (Clearing) - Contactless MoneySend Payment Transactions

Acquirers of payment transactions must send First and Second Presentment/ 1240 messages with the following data elements and values.

Table 33: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's token
3	Processing Code	Value 28 = Payment Transaction
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: As applicable
26	Card Acceptor Business Code (MCC)	MCC 6536 (MoneySend Intracountry) or MCC 6537 (MoneySend Intercountry)
105	Multi-Use Transaction Identification Data	If applicable
PDS 0023	Terminal Type	As applicable
PDS 0043	Transaction Type Identifier	As applicable for use cases. Refer to the definitions in the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> and <i>Mastercard Network Processing Dual Message Authorization System Guide</i> .
PDS 0058	Token Assurance Method	Optional
PDS 0198	Device Type	Optional in first presentment; not expected in second presentment
PDS 0765	MoneySend Receiver Data	If applicable, provide the payee full name and address

Acquirers will receive PDS 0001 in these messages when MDES has processed the transaction:

- First Chargeback/1442
- Arbitration Chargeback/1442
- Retrieval Request/1644

Table 34: PDS 0001 messages after transaction processing

DE/PDS	Name/Subelement	Comments/Subfield
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: As applicable
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1- Account Number Type: Value TC = Subfield 2 contains the PAN. Mapping was based on a source other than the authorization logs

Purchase mail order/telephone order transactions

This section describes the data that acquirers must be prepared to send to, and receive from, Mastercard when supporting tokenized purchase mail order/telephone order (MO/TO) transactions.

The data elements (DE) for Tokenized Purchase MO/TO transactions table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210

All mandatory data elements apply, even if not specifically shown in the table.

Table 35: DE for tokenized purchase MO/TO transactions

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's token	✓	✓
3	Processing Code	Subfield 1 - Cardholder Transaction Type Code: 00 = Purchase	✓	✓
4	Amount, Transaction	The amount of funds the cardholder requested in the local currency of the acquirer or source location of the transaction.	✓	✓
14	Date, Expiration	Token expiration date	✓	

DE	Name/Subelement	Comments/Subfield	Req	Res
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: 10 = Credential on File. 81 = PAN/Token entry through electronic commerce with optional SecureCode-AAV or DSRP cryptogram in UCAF.	✓	
48	Additional Data - Private Use			
	33 - PAN Mapping File Information	Provided for MDES transactions		✓
	42 - Electronic Commerce Indicators	Subfield 1 - Electronic Commerce Security Level Indicator and UCAF Collection Indicator. CITs with Static token: 210 = Present in unauthenticated transactions, or in transactions where an Identity Check merchant has chosen not to undertake Identity Check, or where Identity Check failed authentication and the merchant desires to proceed with the transaction. Also present in transactions with a DTVC. CITs with Secure Element or Cloud token: 246 = Tokenized transaction with DSRP cryptogram. MITs: 247 = Tokenized Merchant Initiated Transaction. NOTE: For more information about the SLI values, see the <i>Mastercard Network Processing Dual Message Authorization System Guide Data Element 48 (Additional Data)</i>, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator).	✓	✓

DE	Name/Subelement	Comments/Subfield	Req	Res
61	Point of Service Data	Subfield 4 - POS Cardholder Presence: CITs: • 2 = Cardholder not present (mail/facsimile order) • 3 = Cardholder not present (phone or Automated Response Unit [ARU]) MITs: • 4 = Standing order/recurring transactions • 5 = Cardholder not present (Electronic order [home PC, Internet, mobile phone, PDA])	✓	✓
		Subfield 7 - POS Transaction Status: 0 = Normal request (original presentment)	✓	
104	Digital Payment Data	Subfield 001 - Digital Payment Cryptogram • CIT with Secure Element and Cloud Tokens require a cryptogram. • MIT Recurring Payment or Installment with Secure Element and Cloud Tokens can have a cryptogram.	✓	
105	Multi-Use Transaction Identification Data	If applicable	•	✓
119	MDES Service Indicators, subelement 004: MDES Service Indicators for acquirers	If applicable. Refer to the MDES service indicators for more details.	•	✓

Chapter 4 Issuer-specific information

The following is information specific to issuers for MDES. Issuers must be prepared to receive the Authorization, Clearing, and Single Message System data element values described in this section, with the exception of the pre- and post-digitization messages which are optional. Mastercard encourages issuers to perform functionality testing to support MDES. For further information, refer to the Mastercard Network Processing Dual Message Authorization System Guide and Mastercard Network Processing Single Message System Guide.

Pre-digitization and post-digitization network messages.....	114
Single Message System.....	118
IPM MPE and 80-byte Financial Institution Table (FIT) File.....	118
Funding PAN and funding PAN expiry through ABU.....	118
Bulk files.....	118
Account Management System File updates (bulk file R311).....	118
Contactless transactions.....	119
Dual Message System (Authorization) and Single Message System.....	119
Token Mapping Failures, Suspended/Deactivated Tokens, and Pre-validation Failures.....	124
Reports.....	129
Online PIN.....	130
Dual Message System (Clearing).....	131
Mastercard Consumer-Presented QR Transactions.....	132
Dual Message System (Authorization) and Single Message System.....	133
Token Mapping Failures, Suspended/Deactivated Tokens, and Pre-Validation Failures.....	138
Reports.....	141
Dual Message System (Clearing).....	141
E-commerce transactions with Digital Secure Remote Payment (DSRP) Crypto.....	143
Dual Message System (Authorization) and Single Message System EMV Data.....	143
250-byte batch data file.....	147
Single Message Transaction Manager.....	148
Dual Message System (Clearing) - EMV Data.....	148
Dual Message System (Authorization) and Single Message System - UCAF data.....	150
Dual Message System (Clearing) - UCAF data.....	160
Token Mapping Failures, Suspended/Deactivated Tokens, and Pre-validation Failures.....	163
E-Commerce Transactions with Dynamic Token Verification Code (DTVLC).....	169
Dual Message System (Authorization) and Single Message System.....	170
Token Mapping Failures, Suspended/Deactivated Tokens, and Pre-Validation Failures.....	175
Dual Message System (Clearing).....	176

Dynamic magnetic stripe transactions.....	178
Merchant-initiated Transactions (MITs).....	183
Stand-In authorization.....	189
MoneySend Transactions.....	190
Dual Message System (Authorization) and Single Message System - e-Commerce MoneySend Funding Transactions.....	190
Dual Message System (Clearing) - e-Commerce MoneySend Funding Transactions.....	196
Dual Message System (Authorization) and Single Message System - e-Commerce MoneySend Payment Transactions.....	198
Dual Message System (Clearing) - e-Commerce MoneySend Payment Transactions.....	202
Dual Message System (Authorization) and Single Message System - Contactless MoneySend Funding Transactions.....	204
Dual Message System (Clearing) - Contactless MoneySend Funding Transactions.....	209
Dual Message System (Authorization) and Single Message System - Contactless MoneySend Payment Transactions.....	211
Dual Message System (Clearing) - Contactless MoneySend Payment Transactions.....	216
Purchase mail order/telephone order transactions.....	218

Pre-digitization and post-digitization network messages

The message types described in this section are available to issuers to enable the real-time decisioning of whether a card can be digitized and for the subsequent notification of information related to digitization or token lifecycle events.

Issuers participating in MDES can opt in to receive tokenization requests or notifications through Authorization Request/0100 (for every purpose), Financial Transaction Request/0200 (for every purpose), or Administrative Advice/0620 messages (for notifications only) depending on the issuer's current configuration that:

- Enable the issuer to provide a real-time decision whether Mastercard can designate a token on their behalf (TER and TAR).
- Provide the issuer with an activation code that the issuer can deliver to the cardholder (ACN).
- Inform the issuer of the token designated to a PAN (TCN).
- Inform the issuer of events during and after digitization such as activation code entry failures or when a token is suspended, resumed, or deactivated (TVN).

NOTE: ASI is mandatory for the eligibility check (token authorization) unless the issuer opts to receive Authorize Service API messages (from MDES Pre-Digitization API).

Issuers that have opted to receive these digitization messages must be prepared to receive authorization or financial transaction requests with DE 61 (POS Data), subfield 7 (POS Transaction Status), value 9 (Tokenization Request/Notification) and DE 124 (Member-Defined Data). Issuers must be prepared to respond to these messages with the specified message layout.

Aside from receiving the ASI during token authorization, issuer receipt of any of the real-time messages for tokenization is optional.

For further information, refer to the *MDES - Issuer Implementation Guide* available on TRC.

Consumer Identifier

Issuers participating in MDES can receive a consumer identifier (when supplied by the Wallet Provider) as DE 124, position 212–299 in the Authorization Request/0100 TAR message, for the Dual Message System (Authorization).

NOTE:

The Single Message System pre-digitization message does not currently support inclusion of the consumer identifier.

This feature will be enabled on a market-by-market and wallet-by-wallet basis. Issuers and processors will not receive the new consumer identifier in DE 124 unless they explicitly opt to receive it.

Local Use Only Account Ranges

As of 18.Q1, the services for sending Authorization Request/0100 MDES digitization network messages and ASI messages have been modified to allow MDES to conditionally provide the local values within the following data elements for Local Use Only account ranges (see the note):

- DE 43 (Card Acceptor Name/Location for All Transactions), subfield 5 (Card Acceptor State or Country Code)
- DE 49 (Currency Code, Transaction)
- DE 61 (POS Data), subfield 13 (POS Country Code)

Because these data elements are common to all ASI messages and Authorization Request/0100 digitization messages for Local Use Only account ranges, they apply to all message types mentioned in DE 124, namely TER, TAR, ACN, TCN, and TVN. This means that when an issuer has opted to receive such messages as Authorization Request/0100 for a Local Use Only account range (see the note), those messages are initiated from an ICA associated with the country of the issuer (instead of being associated with the United States, which has the code 840) so that the messages are accepted inside the country.

Tokenization Eligibility Request (TER)

This message may be sent to the issuer when a card is within an account range eligible for digitization. Only limited data is provided in DE 124. However, this message gives the issuer advanced notice that a cardholder has requested digitization so that issuer systems are prepared to receive the subsequent TAR messages with further details.

For information relating to receiving this message as Authorization Request/0100 for Local Use Only account ranges, see Local Use Only Account Ranges.

The issuer will provide a DE 39 (Response Code) value that indicates how MDES should proceed, as follows:

- Continue. DE 39 = 00; the digitization process will continue.
- Decline. DE 39 = 05; card digitization is declined.
- Continue, but require additional authentication. DE 39 = 85; the digitization process will continue but additional authentication must be performed prior to activation.

The issuer must be prepared to not receive the TER even when having opted-in to receive the message. For certain wallet programs, the TER is not sent. Refer to the Token Implementation Plans for obtaining the specifics of each wallet program.

Tokenization Authorization Request (TAR)

This message is sent to the issuer when digitization of a PAN is requested. All data received in support of the digitization request is provided in the message in DE 124.

For information relating to receiving this message as Authorization Request/0100 for Local Use Only account ranges, see Local Use Only Account Ranges.

An issuer will provide a DE 39 (Response Code) value that indicates how MDES should proceed, as follows:

- Approve. DE 39 = 00; the digitization process will proceed to activation.
- Decline. DE 39 = 05; card digitization is declined.
- Approve, but require additional authentication. DE 39 = 85; the digitization process will proceed, depending on the wallet program. For certain wallet programs, additional authentication may be performed prior to activation. Alternatively, depending on the wallet program, tokens may be provisioned in an active state, prior to cardholder authentication being performed. Refer to the *Token Implementation Plans* for obtaining the specifics of each wallet program.

If TAR is not selected, ASI will be used unless the issuer opted-in to receive Authorize Service API messages (from MDES Pre-Digitization API).

If the issuer's response to the TAR is not received within standard authorization response time, MDES will use the pre-defined eligibility rules and default eligibility settings to determine a digitization decision, or use the response code from the previous TER, if applicable. For more information, refer to the Card Eligibility Message Timeout and Retry Logic for Wallet Programs section of the *MDES - Issuer Implementation Guide*.

When the issuer has indicated in the TER 'Continue, but require additional authentication,' or in the TAR 'Approve, but require additional authentication,' the issuer may provide one or more activation code distribution methods within the response.

Within the response, the issuer may also optionally provide a card product configuration identifier that is associated with the graphical and text card art assets to be used in the display of the token within the payment application. Otherwise, the issuer's pre-defined default artwork established during issuer enablement at Account Range level will be used, if applicable.

When Mastercard is not the BIN Controller, issuers that have opted-in to receive Authorization Request/0100 or Financial Transaction Request/0200 messages as TAR must be prepared to send DE 56 (Payment Account Data) in the Authorization Request Response/0110 - Tokenization Authorization or Financial Transaction Request Response/0210 - Tokenization Authorization messages.

Alternatively, when Mastercard is not the BIN Controller, issuers that have not opted to receive Authorization Request/0100 or Financial Transaction Request/0200 messages as TAR must be prepared to send DE 56 (Payment Account Data) in the Authorization Request Response/0110 - Account Status Inquiry or Financial Transaction Request Response/0210—Account Status Inquiry messages.

Activation Code Notification (ACN)

This message is sent to the issuer when an activation code needs to be delivered to the cardholder through a channel managed by the issuer.

For information relating to receiving this message as Authorization Request/0100 for Local Use Only account ranges, see Local Use Only Account Ranges.

An activation code may be provided by MDES when one of the following occurs:

- The issuer has responded to a TER with a 'Continue, but require additional authentication' response.
- The issuer has responded to a TAR with an 'Approve, but require additional authentication' response.
- The issuer's pre-defined eligibility rules and default decision values, established during issuer enablement, indicate that additional authentication is required.

Tokenization Complete Notification (TCN)

This message is sent to advise the issuer that a card has been successfully digitized and is able to perform contactless, DSRP transactions, or e-commerce transactions with a static token. This message contains all known information regarding the cardholder, the digitization process outcome, and the allocated token.

For information relating to receiving this message as Authorization Request/0100 for Local Use Only account ranges, see Local Use Only Account Ranges.

Issuers that have opted-in to receive TCN as Authorization Request/0100 or Financial Transaction Request/0200 messages will receive DE 56 (Payment Account Data) when Mastercard is the BIN Controller.

Alternatively, issuers that have chosen to receive Administration Advice/0620 messages for Issuer Token Notification Advice for TCN will receive DE 56 when Mastercard is the BIN Controller.

Tokenization Event Notification (TVN)

This message is sent to the issuer for the following reasons:

- Advise the issuer when a digitization activation code has not matched.
- Communicate to the issuer changes to the status of a token (for example, when the token has been suspended or deactivated).

For information relating to receiving this message as Authorization Request/0100 for Local Use Only account ranges, see Local Use Only Account Ranges.

Data element definitions

- DE 56 (Payment Account Data), subelement 01 (Payment Account Data), subfield 01 (Payment Account Reference)
- DE 60 (Advice Reason Code), subfield 2 (Advice Detail Code)
- DE 61 (Point-of-Service Data), subfield 7 (POS Transaction Status), value 9 (Tokenization Request/Notification)
- DE 120 (Record Data)

Single Message System

Issuers must support message layouts for MDES. For additional details, refer to the *Single Message System Specifications* manual.

IPM MPE and 80-byte Financial Institution Table (FIT) File

Transactions are routed to the Mastercard Network based on Dual Message System service-eligible account ranges that are identified on the IPM MPE Table IP0040T1 (Issuer Account Range) with value C (Mastercard Digital Enablement Service Account Range) in the Mapping Service Indicator field.

Single Message System service-eligible account ranges are identified on the 80-byte Financial Institution Table File with position 52 (Mastercard Digital Enablement Service Flag), value Y (Mastercard Digital Enablement Service is performed).

Funding PAN and funding PAN expiry through ABU

ABU and MDES issuers can submit PAN and PAN expiry date updates to ABU that are automatically applied to the AMS MDES token PAN mapping database.

MDES issuers can report PAN and PAN expiry date updates to AMS using the Issuer File Update Request/0302 - Maintenance (Token/PAN Update) messages, bulk file R311, or customer service APIs. If a PAN or PAN expiry date change is first submitted through ABU, ABU then will automatically update AMS with this information.

If the PAN change is updated first through ABU and subsequently to AMS using Issuer File Update Request/0302 - Maintenance (Token/PAN Update) messages, AMS sends Issuer File Update Request Response/0312 reject messages to issuers.

Bulk files

The following bulk files are applicable to MDES issuers.

Account Management System File updates (bulk file R311)

This file is used for MCC106 (Mastercard Digital Enablement Service Mapping File PAN Update).

For additional information on bulk files applicable to MDES issuers, refer to the *Account Management System User Manual*.

Contactless transactions

When the cardholder uses a token to initiate a transaction at a contactless terminal, Mastercard will validate the accompanying chip or dynamic CVC 3 data and map the token to the cardholder's PAN before forwarding the request to the issuer.

NOTE: Mastercard is enabling cardholders to use Near Field Communications (NFC)-enabled devices and wallets to initiate tokenized transactions at contactless ATMs.

NOTE: Since 1 May 2019, Mastercard has been making the cardholder experience at contactless ATMs more seamless by requiring the support of Online PIN for unattended cash for newly-provisioned digitized Mastercard and Maestro cards on mobile payment devices that are digitized through MDES.

Dual Message System (Authorization) and Single Message System

This section describes the messages that issuers receive and send when MDES has processed a contactless transaction successfully.

The following table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Authorization Advice/0120
- Authorization Advice Response/0130
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210
- Financial Transaction Advice/0220
- Financial Transaction Advice Response/0230
- Reversal Request/0400
- Reversal Advice/0420

NOTE: If a token transaction fails MDES cryptographic validation but the issuer has chosen for Mastercard to continue processing, the issuer will receive a request message with similar data elements and values to those in the following table. Mastercard will add the validation failure results (to DE 48, subelements 27 and/or 71) and the ATC Information (DE 48, subelement 34). For information about cryptographic validation failures, see Chip or Dynamic Card Validation Code 3 (CVC 3) Pre-Validation Failures.

Table 36: Data elements and values for contactless transactions

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's PAN	√	√

DE	Name/Subelement	Comments/Subfield	Req	Res
14	Date, Expiration	PAN expiration date	√	•
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: Value either: • 07 = PAN auto-entry through Contactless M/Chip • 91 = PAN auto-entry through contactless magnetic stripe	√	•
23	Card Sequence Number	As applicable	√	•
35	Track 2 Data	Not present when MDES validates the chip or dynamic CVC 3 data on behalf of the issuer NOTE: Issuers may opt in through MDES Manager to receive track data in DE 35 (Track 2 Data)	•	•
45	Track 1 Data	Not present when MDES validates the chip or dynamic CVC 3 data on behalf of the issuer NOTE: Issuers may opt in through MDES Manager to receive track data in DE 45 (Track 1 Data)	•	•
48	Additional Data - Private Use			
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: If applicable NOTE: New Wallet Identifier values may be provided in this field without prior written notice.	√	•
	27 - Transaction Analysis	As applicable to MCBP 2.0. Not present when the transaction comes from an alternative network.	√	•
		Subfield 1 - Overview		
		Subfield 2 - Test Results		
	30 - Token Transaction Identifier	If applicable	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
	33 - PAN Mapping File Information	Conditional in issuer responses. Issuers should provide the entire subelement 33 on transit transactions when using third-party tokenization services.		√
		Subfield 1 - Account Number Indicator: Value either: • C = MDES Secure Element Token • H = MDES Cloud-based Payments Token	√	*
		Subfield 2 - Account Number: Token	√	*
		Subfield 3 - Expiration Date: Token expiration date	√	*
		Subfield 5 - Token Assurance Method: Token assurance method value (as applicable)	√	
		Subfield 6 - Token Requestor ID: Token Requestor ID (as applicable)	√	
		Subfield 8 - Storage Technology: Storage technology (as applicable)	√	
	34 - ATC Information	Conditional. Not present for decline advice messages.	√	•
	71 - On-behalf Services	MDES validation results, which are the relevant combinations of the following subfields. NOTE: This data element may include different results when there are validation failures but the issuer has chosen for Mastercard to continue processing; see Chip or Dynamic Card Validation Code 3 (CVC 3) Pre-Validation Failures.	√	•
		Subfield 1: Value 50 = MDES PAN Mapping		
		Subfield 2: Value C = Conversion of Token to PAN completed successfully		
		Subfield 3: Mastercard use only with a space or value		

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 1: Value 51 = MDES Chip Pre-Validation Subfield 2: Value either: • V = Valid ARQC/TC and ATC and TVR/CVR • G = Application Cryptogram is valid but not an ARQC nor a TC, status of TVR/CVR unknown Subfield 3: Mastercard use only with a space or value		
		Subfield 1: Value 52 = MDES CVC 3 Pre-Validation Subfield 2: Value V = Valid Subfield 3: Mastercard use only with a space or value		
		Subfield 1: Value 61 = MDES Cloud-based Payments Chip Pre-Validation Service Subfield 2: Value either: • V = Valid • P = Valid MD AC; invalid UMD AC (Invalid Mobile PIN) Subfield 3: Mastercard use only with a space or value		
		Subfield 1: Value 62 = MDES Cloud-based Payments Magnetic Stripe Pre-Validation Service Subfield 2: Value V = Valid Subfield 3: Mastercard use only with a space or value		
	79 - Chip CVR/TVR Bit Error Results	Not present when MDES performs chip pre-validation	•	•
55	Integrated Circuit Card (ICC) System-Related Data	Not present when MDES validates the chip or dynamic CVC 3 data on behalf of the issuer	•	•
NOTE: Issuers may opt in through MDES Manager to receive DE 55 data.				

DE	Name/Subelement	Comments/Subfield	Req	Res
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR): - Authorization Request/0100 and Financial Transaction Request/0200 messages: Mastercard provides the PAR when Mastercard is the BIN Controller, and the AMS stores a PAR for the PAN. - Responses: If Mastercard is not the BIN Controller, the issuer may include DE 56 (Payment Account Data) in the response message. If Mastercard is the BIN Controller, the Dual Message System (Authorization) or Single Message System will ignore DE 56 received from the issuer and will set the PAR value retrieved from AMS in DE 56.	√	√
104	Digital Payment Data, subelement 005 - Digital Authentication Data	As applicable NOTE: Issuers can opt-in to receive DE 104 (Digital Payment Data) subelements through MDES Manager.	√	•
105	Multi-Use Transaction Identification Data	If applicable	√	•
119	MDES Service Indicators, subelement 003: MDES Service Indicators for issuers	If applicable. Refer to the MDES service indicators for more details. NOTE: Issuers can opt-in to receive this element through MDES Manager.	√	•

* Issuers who issue contactless cards or devices that provide a PAN within a transaction different from what is embossed on the card, must ensure that the following is provided in DE 48, subelement 33 of Authorization Request Response/0110 messages when responding to MCC 4111, 4131, 4784, 7523 transactions:

- Subfield 1 - Account Number Indicator: Value E
- Subfield 2 - Account Number: The embossed number
- Subfield 3 - Expiration Date: The card expiration date of the embossed number

250-byte Batch Data File

When transactions are processed by MDES, Single Message System issuers receive the following in the 250-byte Batch Data File:

- The Financial (FREC)/Non-financial (NREC) and Financial/Non-financial Exception (EREC) records will contain the cardholder's PAN in the PAN field (positions 33-51).
- The Financial/Non-financial Additional Data Addendum (ADDR) will contain the token in the Mapped PAN Account field (positions 120-138).
- The Financial/Non-financial Additional Data Addendum (ADDR) will contain value C (the PAN in the Mapped PAN Account field contains the MDES token) in the Mapped PAN Indicator field (position 139).

Token Mapping Failures, Suspended/Deactivated Tokens, and Pre-validation Failures

Mastercard might send decline advice messages to issuers when MDES cannot successfully complete token to PAN mapping or when Mastercard declines the transaction on behalf of the issuer (due to a cryptographic validation failure).

Decline advice messages are sent as either:

- Authorization Advice/0120 - All Dual Message System (Authorization) issuers will receive the declined advice messages.
- Financial Transaction Advice/0220 - Only those Single Message System issuers that have opted to receive declined advice messages will be sent online financial transaction advice messages. However, all Single Message System issuers will receive declined advices in the Non-Financial Record (NREC) of the 250-byte Batch Data File.

Token to PAN mapping failures

When MDES cannot successfully complete token to PAN mapping (for example, if the token to PAN mapping is not found, or the token is suspended or deactivated), Mastercard will send a decline response to the acquirer and an advice message to the issuer. Since the PAN mapping failed, DE 2 may contain the token PAN in the advice notification to the Issuer.

The *Data element values without errors* table shows data element values that are different from a transaction message that does not have mapping errors or cryptographic errors. The values can be present for these message types:

- Authorization Advice/0120
- Authorization Advice Response/0130
- Financial Transaction Advice/0220
- Financial Transaction Advice Response/0230

Table 37: Data element values without errors

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's token	√	√
14	Date, Expiration	Token expiration date (if applicable)	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
39	Response Code	One of the applicable values that the Authorization Platform/Single Message System sent to the acquirer: • 14 = Invalid card number (unable to map the token to the PAN) • 30 = Format error • 62 = Restricted card (for suspended or deactivated token) • 96 = System error (mapping failed because of technical reasons)	√	√
48	Additional Data - Private Use			
	33 - PAN Mapping File Information	Not present, unless the mapping failed because the token is suspended or deactivated	•	•
	34 - ATC Information	Not present	•	•
	71 - On-behalf Services	Subfield 1 - On-behalf Service: Value 50 = MDES PAN Mapping	√	•
		Subfield 2 - On-behalf Result 1: Value either: • I = Invalid Token • F = Format Error • U = Unable to process	√	•
		Subfield 3 - On-behalf Result 2: Mastercard use only with a space or value	√	•
	79 - Chip CVR/TVR Bit Error Results	Not present	•	•
60	Advice Reason Code	Subfield 1 - Advice Reason Code: Value 141 = MDES Advice to Issuer	√	•
		Subfield 2 - Advice Detail Code: One of the following values: • 0201 = Reject: Invalid Token–PAN mapping relationship • 0202 = Reject: Token in suspended status • 0203 = Reject: Token deactivated	√	•
		Subfield 3 - Advice Detail Text: Textual information relating to the value in DE 60, subfield 2.	√	•

Cryptographic pre-validation failures

When a token transaction fails MDES cryptographic validation, the issuer decision matrix or MDES Transaction Analysis service (for MCBP 2.0 tokens) can determine whether Mastercard declines the transaction or continues processing. For more information, see MDES Transaction Validation and Decision Matrix.

If Mastercard declines on behalf of the issuer, a decline advice message is sent to the issuer, and a decline response message is sent to the acquirer. Otherwise, if processing continues, the validation failure results are provided in the authorization request message sent to the issuer.

NOTE: The authorization request message contains similar data elements to those present for a transaction without cryptographic errors; see Dual Message System (Authorization) and Single Message System.

The following table shows data element values that are different from a transaction message that does not have mapping errors or cryptographic errors. The values can be present for these decline advice messages:

- Authorization Advice/0120
- Authorization Advice Response/0130
- Financial Transaction Advice/0220
- Financial Transaction Advice Response/0230

Table 38: Data element values for decline advice messages

DE	Name/Subelement	Comments/Subfield	Req	Res
39	Response Code	Value 05 (Do not honor) that the Authorization Platform/Single Message System sent to the acquirer	√	√
48	Additional Data—Private Use			
	27 - Transaction Analysis	As applicable to MCBP 2.0. Not present when the transaction comes from an alternative network.	√	•
		Subfield 1 - Overview		
		Subfield 2 - Test Results		
	34 - ATC Information	Not present	•	•
	Failure values in subelement 71 - On-behalf Services	Subfield 1 - On-behalf Service: One of the following values: • 51 = MDES Chip Pre-Validation • 61 = MDES Cloud-based Payments Chip Pre-Validation Service	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 2 - On-behalf Result 1: One of the following values: EMV: • MDES chip pre-validation failure (subfield 1 = 51): – A = ATC outside allowed range (applicable when ATC value is dynamic [varying] value) – E = ATC Replay – F = Format Error – I = Invalid Cryptogram – K = No matching key file for this PAN, PAN expiry date and KDI combination – T = Valid ARQC/TC and ATC; TVR/CVR invalid – U = Unable to process • MDES cloud-based payments chip pre-validation failure (subfield 1 = 61): – F = Format Error – K = No matching key file for this PAN, PAN expiry date and KDI combination – P = Valid MD AC; invalid UMD AC (Invalid Mobile PIN)	√	•
		Subfield 3 - On-behalf Result 2: Mastercard use only with a space or value	√	•
	79 - Chip CVR/TVR Bit Error Results	As applicable	√	•
60	Advice Reason Code	Subfield 1 - Advice Reason Code: Value 141 = MDES Advice to Issuer	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 2 - Advice Detail Code: One of the following values:	√	•
		MDES chip pre-validation failure:		
		• 0032 = Reject: Chip Data Processing Error • 0034 = Reject: Chip validation failed • 0035 = Reject: TVR/CVR validation failed • 0037 = Reject: No matching key File for this PAN, PAN expiry date and KDI combination - Validation of ARQC and CVR/TVR not performed, status unknown • 0043 = Reject: ATC outside allowed range • 0046 = Reject: ATC Replay • 0059 = DE 55 format error (for MDES chip validation)		
		MCBP chip pre-validation failure:		
		• 0009 = Reject: Invalid Time Validation (Dual Message System [Authorization] only) • 0059 = DE 55 format error (for MDES chip validation) • 0204 = Reject: ATC Invalid - Not in List of Currently Active Single Use Keys • 0205 = Reject: ATC Replay • 0206 = Reject: Invalid MD AC and UMD AC (Invalid Mobile PIN) • 0207 = Reject: Valid MD AC; Invalid UMD AC (Mobile PIN Try Counter Max Limit not Reached, Token not Suspended) • 0208 = Reject: Invalid MD AC; Valid UMD AC • 0209 = Reject: Valid MD AC; Invalid UMD AC (Mobile PIN Try Counter Max Limit Reached, Token Suspended) • 0211 = Reject: TVR/CVR validation failed • 0212 = Reject: Unable to Process		

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 3 - Advice Detail Text: Textual information relating to the value in DE 60, subfield 2.	√	•
105	Multi-Use Transaction Identification Data	If applicable	√	•
119	MDES Service Indicators, subelement 003: MDES Service Indicators for issuers	If applicable. Refer to the MDES service indicators for more details. NOTE: Issuers can opt-in to receive this element through MDES Manager.	√	•

250-byte batch data file

When MDES cannot successfully complete token to PAN mapping (for example, if the token to PAN mapping is not found, or the token is suspended or deactivated):

- The Non-financial (NREC) record will contain the token in the PAN field (positions 33-51) and the applicable advice reason code in the Advice Reason Code field (positions 108-114).
- The Non-financial Additional Data Addendum (ADDR) will contain blanks in the Mapped PAN Account field (positions 120-138) and a blank in the Mapped PAN Indicator field (position 139).

When a token transaction fails cryptography validation within MDES and Mastercard sends a decline response to the acquirer:

- The Non-financial (NREC) record will contain the cardholder's PAN in the PAN field (positions 33-51) and the applicable advice reason code in the Advice Reason Code field (positions 108-114).
- The Non-financial Additional Data Addendum (ADDR) will contain the token in the Mapped PAN Account field (positions 120-138).
- The Non-financial Additional Data Addendum (ADDR) will contain value C (the PAN in the Mapped PAN Account field contains the MDES token) in the Mapped PAN Indicator field (position 139).

Reports

The Global Parameters section of the Authorization Parameter Summary Report (SI737010-AA) displays the service. A sample of this report can be found in the Authorization Manual.

The Daily Account File Activity Report for MDES (AM700010-CC) shows file maintenance activity on tokens (add, update, deactivate, delete, suspend, resume).

The Detail Account Report (AM730010-AA) includes a section for MDES to show tokens on file with Mastercard and their associated status (active, suspended, deactivated).

Samples of the Daily Account File Activity Report and Detail Account Report can be found in the *Account Management System User Manual*.

The following Single Message System reports supporting MDES reflect the tokenization request/notification messages if there is associated activity:

- Daily Control Report (SWCHD363)
- Monthly Control Report (SWCHM363)
- Transaction Interchange Month-to-Date Report (SWCHM03)

Online PIN

MDES enables cardholders to use the same online PIN as their digitized card for contactless transactions initiated by a mobile device using a token.

When a token is used at the point of interaction, Mastercard converts the online PIN block provided by the acquirer, in DE 52 (Personal ID Number Data) or DE 110 (Encryption Data), to an issuer PIN block based on the PAN before sending the Authorization Request/0100 or Financial Transaction Request/0200 message to the issuer or performing PIN validation on behalf of the issuer.

A transaction qualifies to have PIN block translation performed if all these conditions are true:

- The transaction is an Authorization Request/0100 message or Financial Transaction Request/0200 message
- The transaction is processed as MDES
- DE 52, or DE 110 is present

Mastercard provides PIN validation on-behalf services as follows:

- Dual Message System issuers. PIN pre-validation, or PIN validation in Stand-In Processing
- Single Message System issuers. PIN validation in Stand-In Processing

Issuers that participate in MDES and perform their own PIN validation must be aware that Mastercard will not send track data in DE 35 (Track 2 Data) or DE 45 (Track 1 Data) on Dual Message System (Authorization) or Single Message System transactions to issuers when MDES has been performed. Therefore, these issuers must maintain the PIN offset/PVV in their own systems instead of expecting PIN offset/PVV in the track data.

NOTE:

Mastercard will send DE 23 (Card Sequence Number) to issuers on Dual Message System (Authorization) and Single Message System transactions when card sequence numbers have been explicitly provided to MDES, and the issuer has opted-in to receive DE 23, and the POS Entry Mode is either value 07 (PAN auto-entry through Contactless M/Chip) or 09 (PAN/Token entry through electronic commerce containing DSRP cryptogram in DE 55 [Integrated Circuit Card [ICC] System-Related Data]). Issuers may use DE 23 as part of their own PIN validation processing. Otherwise, Mastercard will not send DE 23, and issuers must assess the impact in relation to their PIN validation processing.

Mastercard will not send an authorization or financial transaction advice message to the issuer when the online PIN block translation has failed.

Dual Message System (Clearing)

Issuers will receive the following data elements and values in First and Second Presentment/1240 messages when MDES has processed a contactless transaction successfully.

Table 39: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's PAN
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value either: - A = PAN auto-entry through contactless magnetic stripe - M = PAN auto-entry through Contactless M/Chip
23	Card Sequence Number	As applicable
55	Integrated Circuit Card System-Related Data	If provided by the acquirer NOTE: Even though DE 55 may not have been passed in Authorization, DE 55 may be passed in Clearing to the issuer if it is present and has been provided.
105	Multi-Use Transaction Identification Data	If applicable

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: One of the following values: • C (value C followed by a space) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • CC (value CC) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. • H (value H followed by a space) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • HC (value HC) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log.
		Subfield 2 - Account Number: Token
PDS 0058	Token Assurance Method	If applicable
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	If applicable
PDS 0207	Wallet Identifier	If applicable

Issuers must format subsequent retrieval request and chargeback messages to the Clearing Platform with the following:

- DE 2 with the cardholder's PAN
- DE 22, subfield 7, value A or value M

Transaction Research Tool

Values for PDS 0001 (Mastercard Mapping Service Account Number) will be displayed on the Clearing Transaction Research Tool.

Mastercard Consumer-Presented QR Transactions

A Mastercard Consumer-Presented QR Transaction is an EMV chip Transaction effected through the presentment of a QR Code by the Cardholder, using a Mobile Payment Device, and the

capture of the QR Code by the Merchant containing the Transaction Data required to initiate a Transaction

For each Mastercard Consumer-Presented QR Transaction:

- There is no maximum Transaction amount
- The Transaction must be authorized online by the Issuer
- The Transaction must be completed with CDCVM. CDCVM is the only valid CVM for Mastercard Consumer-Presented QR Transactions
- The Merchant must send a properly identified Authorization Request with EMV data

NOTE:

For information about Mastercard Consumer-Presented QR Transactions, refer to the Mastercard Cloud-Based Payments (MCBP) documentation and the Mastercard Consumer-Presented QR Requirements manual.

Dual Message System (Authorization) and Single Message System

This section describes the messages that issuers receive and send when MDES has processed a contactless transaction successfully.

The *Data elements and values for Authorization messages* table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Authorization Advice/0120
- Authorization Advice Response/0130
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210
- Financial Transaction Advice/0220
- Financial Transaction Advice Response/0230
- Reversal Request/0400
- Reversal Advice/0420

NOTE: If a token transaction fails MDES cryptographic validation but the issuer has chosen for Mastercard to continue processing, the issuer will receive a request message with similar data elements and values to those in the following table. Mastercard will add the validation failure results (to DE 48, subelements 27 and 71) and the ATC Information (DE 48, subelement 34). For information about cryptographic validation failures, see Cryptographic Pre-Validation Failures.

Table 40: Data elements and values for Authorization messages

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's PAN	√	√
14	Date, Expiration	PAN expiration date	√	•
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: Value 03 = PAN Auto-entry through bar code reader	√	•
23	Card Sequence Number	As applicable	√	•
35	Track 2 Data	Not present when MDES validates the chip data on behalf of the issuer NOTE: Issuers may opt in through MDES Manager to receive track data in DE 35 (Track 2 Data)	•	•
48	Additional Data - Private Use			
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: If applicable NOTE: New Wallet Identifier values may be provided in this field without prior written notice.	√	•
	27 - Transaction Analysis	As applicable to MCBP 2.0. Not present when the transaction comes from an alternative network.	√	•
		Subfield 1 - Overview		
		Subfield 2 - Test Results		
	30 - Token Transaction Identifier	If applicable	√	•
	33 - PAN Mapping File Information	Conditional in issuer responses. Issuers should provide the entire subelement 33 on transit transactions when using third-party tokenization services.		√

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 1 - Account Number Indicator. Value: H = MDES Cloud-based Payments Token	√	*
		Subfield 2 - Account Number: Token	√	*
		Subfield 3 - Expiration Date: Token expiration date	√	*
		Subfield 5 - Token Assurance Method: Token assurance method value (as applicable)	√	
		Subfield 6 - Token Requestor ID: Token Requestor ID (as applicable)	√	
		Subfield 8 - Storage Technology: Storage technology (as applicable)	√	
	34 - ATC Information	Conditional. Not present for decline advice messages.	√	•
	71 - On-behalf Services	MDES validation results, which are the relevant combinations of the following subfields. NOTE: This data element may include different results when there are validation failures but the issuer has chosen for Mastercard to continue processing; see Cryptographic Pre-Validation Failures.	√	•
		Subfield 1: Value 50 = MDES PAN Mapping Subfield 2: Value C = Conversion of Token to PAN completed successfully Subfield 3: Mastercard use only with a space or value		

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 1: Value 51 = MDES Chip Pre-Validation Subfield 2: Value either: • V = Valid ARQC/TC and ATC and TVR/CVR • G = Application Cryptogram is valid but not an ARQC or a TC, status of TVR/CVR unknown Subfield 3: Mastercard use only with a space or value		
		Subfield 1: Value 61 = MDES Cloud-based Payments Chip Pre-Validation Service Subfield 2: Value either: • V = Valid • P = Valid MD AC; invalid UMD AC (Invalid Mobile PIN) Subfield 3: Mastercard use only with a space or value		
	79 - Chip CVR/TVR Bit Error Results	Not present when MDES performs chip pre-validation	•	•
55	Integrated Circuit Card (ICC) System-Related Data	Not present when MDES validates the chip data on behalf of the issuer	•	•
NOTE: Issuers may opt in through MDES Manager to receive DE 55 data				

DE	Name/Subelement	Comments/Subfield	Req	Res
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR): - Authorization Request/0100 and Financial Transaction Request/0200 messages: Mastercard provides the PAR when Mastercard is the BIN Controller, and the AMS stores a PAR for the PAN. - Responses: If Mastercard is not the BIN Controller, the issuer may include DE 56 (Payment Account Data) in the response message. If Mastercard is the BIN Controller, the Dual Message System (Authorization) or Single Message System will ignore DE 56 received from the issuer and will set the PAR value retrieved from AMS in DE 56.	√	√
105	Multi-Use Transaction Identification Data	If applicable	√	•
119	MDES Service Indicators, subelement 003: MDES Service Indicators for issuers	If applicable. Refer to the MDES service indicators for more details. NOTE: Issuers can opt-in to receive this element through MDES Manager.	√	•

* Issuers who issue contactless cards or devices that provide a PAN within a transaction different from what is embossed on the card, must ensure that the following is provided in DE 48, subelement 33 of Authorization Request Response/0110 messages when responding to MCC 4111, 4131, 4784, 7523 transactions:

- Subfield 1 - Account Number Indicator: Value E
- Subfield 2 - Account Number: The embossed number
- Subfield 3 - Expiration Date: The card expiration date of the embossed number

250-byte Batch Data File

When transactions are processed by MDES, Single Message System issuers receive the following in the 250-byte Batch Data File:

- The Financial (FREC)/Non-financial (NREC) and Financial/Non-financial Exception (EREC) records will contain the cardholder's PAN in the PAN field (positions 33-51).

- The Financial/Non-financial Additional Data Addendum (ADDR) will contain the token in the Mapped PAN Account field (positions 120-138).
- The Financial/Non-financial Additional Data Addendum (ADDR) will contain value C (the PAN in the Mapped PAN Account field contains the MDES token) in the Mapped PAN Indicator field (position 139).

Token Mapping Failures, Suspended/Deactivated Tokens, and Pre-Validation Failures

Mastercard might send decline advice messages to issuers when MDES cannot successfully complete token to PAN mapping or when Mastercard declines the transaction on behalf of the issuer (due to a cryptographic validation failure).

Decline advice messages are sent as either:

- Authorization Advice/0120 - All Dual Message System (Authorization) issuers will receive the declined advice messages.
- Financial Transaction Advice/0220 - Only those Single Message System issuers that have opted to receive declined advice messages will be sent online financial transaction advice messages. However, all Single Message System issuers will receive declined advices in the Non-Financial Record (NREC) of the 250-byte Batch Data File.

Token to PAN Mapping Failures

When MDES cannot successfully complete token to PAN mapping (for example, if the token to PAN mapping is not found, or the token is suspended or deactivated), Mastercard will send a decline response to the acquirer and an advice message to the issuer. Since the PAN mapping failed, DE 2 may contain the token PAN in the advice notification to the issuer. The following table shows data element values that are different from a transaction message that does not have mapping errors or cryptographic errors. The values can be present for these message types:

- Authorization Advice/0120
- Authorization Advice Response/0130
- Financial Transaction Advice/0220
- Financial Transaction Advice Response/0230

Table 41: Data element values without errors

DE	Name/Subelement	Comments/ Subfield	Req	Res
2	PAN	Cardholder's token	√	√
14	Date, Expiration	Token expiration date (if applicable)	√	•

DE	Name/Subelement	Comments/ Subfield	Req	Res
39	Response Code	One of the applicable values that the Authorization Platform/Single Message System sent to the acquirer: 14 = Invalid card number (unable to map the token to the PAN) 30 = Format error 62 = Restricted card (for suspended or deactivated token) 96 = System error (mapping failed because of technical reasons)	√	√
48	Additional Data - Private Use			
	33 - PAN Mapping File Information	Not present, unless the mapping failed because the token is suspended or deactivated	•	•
	34 - ATC Information	Not present	•	•
	71 - On-behalf Services	Subfield 1 - On-behalf Service: Value 50 = MDES PAN Mapping	√	•

DE	Name/Subelement	Comments/ Subfield	Req	Res
		Subfield 2 - On-behalf Result 1: Value either: • I = Invalid Token • F = Format Error • U = Unable to process	√	•
		Subfield 3 - On-behalf Result 2: Mastercard use only with a space or value	√	•
	79 - Chip CVR/TVR Bit Error Results	Not present	•	•
60	Advice Reason Code	Subfield 1 - Advice Reason Code: Value 141 = MDES Advice to Issuer	√	•
		Subfield 2 - Advice Detail Code: One of the following values: • 0201 = Reject: Invalid Token–PAN mapping relationship • 0202 = Reject: Token in suspended status • 0203 = Reject: Token deactivated	√	•
		Subfield 3 - Advice Detail Text: Textual information relating to the value in DE 60, subfield 2.	√	•

Reports

The Global Parameters section of the Authorization Parameter Summary Report (SI737010-AA) displays the service. A sample of this report can be found in the Authorization Manual.

The Daily Account File Activity Report for MDES (AM700010-CC) shows file maintenance activity on tokens (add, update, deactivate, delete, suspend, resume).

The Detail Account Report (AM730010-AA) includes a section for MDES to show tokens on file with Mastercard and their associated status (active, suspended, deactivated).

Samples of the Daily Account File Activity Report and Detail Account Report can be found in the Account Management System User Manual. The following Single Message System reports supporting MDES reflect the tokenization request/notification messages if there is associated activity:

- Daily Control Report (SWCHD363)
- Monthly Control Report (SWCHM363)
- Transaction Interchange Month-to-Date Report (SWCHM03)

Dual Message System (Clearing)

Issuers will receive the following data elements and values in First and Second Presentment/1240 messages when MDES has processed a contactless transaction successfully.

Table 42: Data elements and values in First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's PAN
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value M = PAN auto-entry via contactless M/Chip
23	Card Sequence Number	As applicable
55	Integrated Circuit Card System-Related Data	If provided by the acquirer NOTE: Even though DE 55 may not have been passed in Authorization, DE 55 may be passed in Clearing to the issuer if it is present and has been provided.
105	Multi-Use Transaction Identification Data	If applicable

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: One of the following values: • C (value C followed by a space) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • CC (value CC) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. • H (value H followed by a space) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • HC (value HC) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log.
Subfield 2 - Account Number: Token		
PDS 0058	Token Assurance Method	If applicable
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	If applicable
PDS 0207	Wallet Identifier	If applicable

Issuers must format subsequent retrieval request and chargeback messages to the Clearing Platform with the following:

- DE 2 with the cardholder's PAN
- DE 22, subfield 7, value M

Transaction Research Tool

Values for PDS 0001 (Mastercard Mapping Service Account Number) will be displayed on the Clearing Transaction Research Tool.

E-commerce transactions with Digital Secure Remote Payment (DSRP) Crypto

Tokenized e-commerce transactions enable secure transactions for remote payments through the submission of Digital Secure Remote Payment (DSRP) cryptograms generated by a payment application.

Payment applications use EMV based cryptography to generate cryptograms. These cryptograms in the transactions provide evidence that the cardholder verification is performed.

NOTE: For information relating to partial shipments and recurring payments, see [Merchant-initiated Transactions \(MITs\)](#).

Dual Message System (Authorization) and Single Message System EMV Data

This section describes the messages that issuers must be prepared to receive when supporting merchants in DSRP (full EMV).

The *Data element and values for Authorization messages* table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Authorization Advice/0120
- Authorization Advice Response/0130
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210
- Financial Transaction Advice/0220
- Financial Transaction Advice Response/0230
- Reversal Request/0400
- Reversal Advice/0420

NOTE: If a token transaction fails MDES cryptographic validation but the issuer has chosen for Mastercard to continue processing, the issuer will receive a request message with similar data elements and values to those in the following table. Mastercard will add the validation failure results (to DE 48, subelements 27 and/or 71) and the ATC Information (DE 48, subelement 34). For information about cryptographic validation failures, see [Chip and UCAF Pre-Validation Failures](#).

Table 43: Data element and values for Authorization messages

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's PAN	√	√

DE	Name/Subelement	Comments/Subfield	Req	Res
14	Date, Expiration	PAN expiration date	√	•
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: Value 09 = PAN/Token entry via electronic commerce containing DSRP cryptogram in DE 55 (Integrated Circuit Card [ICC] System-Related Data)	√	•
23	Card Sequence Number	As applicable	√	•
35	Track 2 Data	Not present when MDES validates the token chip data on behalf of the issuer NOTE: Issuers may opt in through MDES Manager to receive track data in DE 35 (Track 2 Data).	•	•
45	Track 1 Data	Not present when MDES validates the token chip data on behalf of the issuer NOTE: Issuers may opt in through MDES Manager to receive track data in DE 45 (Track 1 Data).	•	•

DE	Name/Subelement	Comments/Subfield	Req	Res
48	Additional Data - Private Use		√	
	22 Multi-Purpose Merchant Indicator	Subfield 1 Low-Risk Merchant Indicator - for Acquirers in the EEA, the United Kingdom, San Marino and Gibraltar when applicable in compliance with PSD 2 regulation.		
		Subfield 5 Cardholder/Merchant Initiated Transaction Indicator: As applicable		
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: If applicable	√	•
		NOTE: New Wallet Identifier values may be provided in this field without prior written notice.		
	27 - Transaction Analysis	As applicable to MCBP 2.0. Not present when the transaction comes from an alternative network.	√	•
		Subfield 1 - Overview		
		Subfield 2 - Test Results		
	30 - Token Transaction Identifier	If applicable	√	•
	33 - PAN Mapping File Information	Conditional in issuer responses. Issuers should provide the entire subelement 33 on transit transactions when using third-party tokenization services.	√	√
		Subfield 1 - Account Number Indicator: Value either:		
		• C = MDES Secure Element Token		
		• H = MDES Cloud-based Payments Token		
		Subfield 2 - Account Number: Token		
		Subfield 3 - Expiration Date: Token expiration date		
		Subfield 5 - Token Assurance Method: Token assurance method value (as applicable)		
		Subfield 6 - Token Requestor ID: Token Requestor ID (as applicable)		
		Subfield 8 - Storage Technology: Storage technology (as applicable)		

DE	Name/Subelement	Comments/Subfield	Req	Res
34	ATC Information	Conditional. Not present for decline advice messages.	√	•
43	Universal Cardholder Authentication Field [UCAF]	UCAF with the encoded AAV value if provided by the merchant.		
71	On-behalf Services	MDES validation results, which are the relevant combinations of the following subfields. NOTE: This data element may include different results when there are validation failures but the issuer has chosen for Mastercard to continue processing; see Chip and UCAF Pre-Validation Failures. Subfield 1: Value 50 = MDES PAN Mapping Subfield 2: Value C = Conversion of Token to PAN completed successfully Subfield 3: Mastercard use only with a space or value Subfield 1: Value 51 = MDES Chip Pre-Validation Subfield 2: Value either: • V = Valid ARQC/TC and ATC and TVR/CVR • G = Application Cryptogram is valid but not an ARQC or a TC, status of TVR/CVR unknown Subfield 3: Mastercard use only with a space or value Subfield 1: Value 61 = MDES Cloud-based Payments Chip Pre-Validation Service Subfield 2: Value either: • V = Valid • P = Valid MD AC; invalid UMD AC (Invalid Mobile PIN) Subfield 3: Mastercard use only with a space or value	√	•
79	Chip CVR/TVR Bit Error Results	Not present when MDES validates the token chip data on behalf of the issuer	•	•

DE	Name/Subelement	Comments/Subfield	Req	Res
55	Integrated Circuit Card (ICC) System-Related Data	Not present when MDES validates the token chip data on behalf of the issuer NOTE: Issuers may opt in through MDES Manager to receive DE 55 data.	•	•
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR): - Authorization Request/0100 and Financial Transaction Request/0200 messages: Mastercard provides the PAR when Mastercard is the BIN Controller, and the AMS stores a PAR for the PAN. - Responses: If Mastercard is not the BIN Controller, the issuer may include DE 56 (Payment Account Data) in the response message. If Mastercard is the BIN Controller, the Dual Message System (Authorization) or Single Message System will ignore DE 56 received from the issuer and will set the PAR value retrieved from AMS in DE 56.	√	√
104	Digital Payment Data, subelement 005 - Digital Authentication Data	As applicable NOTE: Issuers can opt-in to receive DE 104 (Digital Payment Data) subelements through MDES Manager.	√	•
105	Multi-Use Transaction Identification Data	If applicable	√	•
119	MDES Service Indicators, subelement 003: MDES Service Indicators for issuers	If applicable. Refer to the MDES service indicators for more details. NOTE: Issuers can opt-in to receive this element through MDES Manager.	√	•

250-byte batch data file

Single Message System issuers will receive the following in the 250-byte Batch Data File after MDES has been performed.

- Financial (FREC)/Non-financial (NREC) and Financial/Non-financial Exception (EREC) records will contain the cardholder's PAN in the PAN (PAN) field (positions 33-51).

- Financial/Non-financial Additional Data Addendum (ADDR) will contain the token in the Mapped PAN Account field (positions 120-138).
- Financial/Non-financial Additional Data Addendum (ADDR) will contain value C (The PAN in the Mapped PAN Account field contains the MDES token) in the Mapped PAN Indicator field (position 139).

Single Message Transaction Manager

Exception item processing can be accessed via Single Message Transaction Manager using the cardholder's PAN or token.

Dual Message System (Clearing) - EMV Data

When MDES has performed cryptography validation and mapping on a DSRP (Full EMV) transaction, issuers will receive the following data elements and values in First and Second Presentment/1240 messages.

Table 44: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's PAN
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value R = PAN entry via electronic commerce, including remote chip
23	Card Sequence Number	As applicable. Not always present, unless issuer opted-in to receive this value.
55	Integrated Circuit Card System-Related Data	If provided by the acquirer NOTE: Even though DE 55 may not have been passed in Authorization, DE 55 may be passed in Clearing to the issuer if it is present and has been provided.
105	Multi-Use Transaction Identification Data	If applicable

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: One of the following values: • C (value C followed by a space) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • CC (value CC) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. • H (value H followed by a space) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • HC (value HC) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. • O (value O followed by a space) = Subfield 2 contains the Static token submitted by the acquirer. The account range is owned by Mastercard. Mapping was based on the authorization log. • OC (value OC) = Subfield 2 contains the Static token submitted by the acquirer. The account range is owned by Mastercard. Mapping was based on a source other than the authorization log.
Subfield 2 - Account Number: Token		
PDS 0058	Token Assurance Method	If applicable
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	If applicable (first presentment only)
PDS 0207	Wallet Identifier	If applicable

Issuers must format subsequent retrieval request and chargeback messages to the Clearing Platform with the following:

- DE 2, containing the cardholder's PAN
- DE 22, subfield 7 value R
- PDS 0207, if applicable

Transaction Research Tool

Values for PDS 0001 (Mastercard Mapping Service Account Number), subfield 1 (Account Number Type) will be displayed on the Clearing Transaction Research Tool.

Dual Message System (Authorization) and Single Message System - UCAF data

This section describes the messages that issuers must be prepared to receive when an acquirer submits a DSRP transaction with POS Entry Mode 81 with UCAF.

The *Data element and values for Authorization messages* shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Authorization Advice/0120
- Authorization Advice Response/0130
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210
- Financial Transaction Advice/0220
- Financial Transaction Advice Response/0230
- Reversal Request/0400
- Reversal Advice/0420

Issuers must be prepared to receive DE 48, subelement 42 in Authorization request/0100 and Financial Transaction Request/0200 messages. However, Mastercard will remove DE 48, subelement 43 before sending these messages to issuers.

NOTE: If a token transaction fails MDES cryptographic validation but the issuer has chosen for Mastercard to continue processing, the issuer will receive a request message with similar data elements and values to those in the following table. Mastercard will add the validation failure results (to DE 48, subelements 27 and/or 71) and the ATC Information (DE 48, subelement 34). For information about cryptographic validation failures, refer to *Chip and UCAF Pre-Validation Failures*.

All mandatory Authorization Request/0100 and Financial Transaction Request/0200 data elements apply.

Table 45: Data elements and values for Authorization messages

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's PAN	√	√
14	Date, Expiration	PAN expiration date	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: Value 81 = PAN/Token entry via electronic commerce with optional <i>SecureCodeAAV</i> or DSRP cryptogram in UCAF or value 10 = Credential on File.	√	•
48	Additional Data - Private Use			
	22 - Multi-Purpose Merchant Indicator	Subfield 1 Low-Risk Merchant Indicator - for Acquirers in the EEA, the United Kingdom, San Marino and Gibraltar when applicable in compliance with PSD 2 regulation. Subfield 5 Cardholder/Merchant Initiated Transaction Indicator: As applicable		
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: Wallet ID (if applicable) Provides information about transactions initiated through the Masterpass Online platform or through MDES and indicates who the Wallet Provider was for the transaction. NOTE: New Wallet Identifier values may be provided in this field without prior written notice.	√	•
	27 - Transaction Analysis	As applicable to MCBP 2.0. Not present when the transaction comes from an alternative network. Subfield 1 - Overview Subfield 2 - Test Results	√	•
	30 - Token Transaction Identifier	If applicable	√	•
	32 - Mastercard Assigned ID	When available NOTE: Subfield 1 - in Financial Transaction Messages	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
	33 - PAN Mapping File Information	Conditional in issuer responses. Issuers should provide the entire subelement 33 on transit transactions when using third-party tokenization services.	√	√
		Subfield 1 - Account Number Indicator: One of the following values: • C = MDES Secure Element Token • F = MDES Static Token • H = MDES Cloud-based Payments Token		
		Subfield 2 - Account Number: Token		
		Subfield 3 - Expiration Date: Token expiration date		
		Subfield 5 - Token Assurance Method values: • 10 = Card Issuer Account Verification (Successful TAR, AS, ASI response with no authentication) • 11 = Card Issuer Interactive Cardholder Authentication – 1 Factor (Token authenticated using MDES Authentication Service (OTP)) • 14 = Card Issuer Asserted Authentication (Push Provisioning into without additional authentication beyond Issuer's own application) • 20 = Push Provisioned token with MDES Authentication service following token creation • 21 = Push Provisioned token with Identity Check™ authentication following token creation • 22 = Token authenticated via Identity Check™ Risk-based authentication (no cardholder challenge) • 23 = Token authenticated via Identity Check™ Interactive step-up (cardholder challenge) • 24 = Token with device-asserted authentication		
		Subfield 6 - Token Requestor ID: Token Requestor ID (as applicable)		

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 8 - Storage Technology: Storage technology (as applicable)		
	34 - ATC Information	Conditional. Not present for decline advice messages.	√	•
	42 - Electronic Commerce Indicators	Subfield 1 - Electronic Commerce Security Level Indicator and UCAF Collection Indicator. NOTE: For more information about the SLI values, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> Data Element 48 (Additional Data), subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator).	√	•
	43 - Universal Cardholder Authentication Field (UCAF)	with the encoded AAV value if provided by the merchant.	•	•

DE	Name/Subelement	Comments/Subfield	Req	Res
	48 - Digital Commerce Solution Indicator	These subfields might be present in transactions associated to specific programs such as Click to Pay, SQR, Pay By Link. Subfield 1=Digital Commerce Domain Identifier • 0 = Multi-Domain • 1 = Mastercard • 2 = Acquirer • 3 = Issuer • 4-9 = Reserved for future use Subfield 2=Digital Commerce Program Type • 0 = Multi-Program • 1 = SQR • 2 = C2P • 3 = In-store Biometric • 4 = Next Gen POI • 5 = Mastercard QR: Pay by Link • 6 = Mastercard QR: X QR • 7 = Tap To Own (to be assigned) • 8 = Agentic Payment (to be assigned) • 9 = EV Charging (to be assigned) Subfield 3=Digital Commerce Program Details • 10 = SQR • 20 = C2P • 30 = Biometrics • 40= Next Gen POI • 51= Merchant Presented PBL • 52= Consumer Presented PBL • 61 = Merchant Presented QR • 62= Consumer Presented QR • 70 = Tap To Own • 80 = Agentic Payment • 91 = EV Contactless • 92 = EV COF Subfield 4=Digital Commerce Check-In Details. Value: Open Text		

DE	Name/Subelement	Comments/Subfield	Req	Res
71	On-behalf Services	MDES validation results, which are the relevant combinations of the following subfields. NOTE: This data element may include different results when there are validation failures but the issuer has chosen for Mastercard to continue processing; refer to <i>Chip and UCAF Pre-Validation Failures</i>.	√	•
		Subfield 1 - Value 50 = MDES PAN Mapping Subfield 2 - Value C = Conversion of Token to PAN completed successfully Subfield 3 - Mastercard use only with a space or value		
		Subfield 1 - Value 51 = MDES Chip Pre-Validation Subfield 2 - Value either: • V = Valid ARQC/TC and ATC and TVR/CVR Subfield 3 - Mastercard use only with a space or value		
		Subfield 1 - Value 61 = MDES Cloud-based Payments Chip Pre-Validation Service Subfield 2 - Value V = Valid Subfield 3 - Mastercard use only with a space or value		

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 1 - Value 54 = Mastercard Digital Enablement Service Digital Payment Data Validation Subfield 2 - Estimated Amount Validation. Value either: • A = If the transaction amount in DE 4 is less than or equal to the approximate amount in the cryptogram • B = If the transaction amount in DE 4 is greater than the approximate amount in the cryptogram by greater than 0% to 19.99% • C = If the transaction amount in DE 4 is greater than the approximate amount in the cryptogram by 20% or more • D = If the transaction amount in DE 4 is not zero and the approximate amount in the cryptogram is zero Subfield 3 - RCAI Identifier Validation. Values either: In Europe • M = Transaction authenticated with Mastercard qualified multifactor authentication method as Strong Customer Authentication, and valid merchant dynamic linking • N = Merchant dynamic linking was not performed or was invalid Outside of Europe • M = Transaction authenticated with Mastercard qualified multifactor authentication method, and valid merchant dynamic linking • N = Merchant dynamic linking was not performed or was invalid NOTE: Issuers can opt-in to receive DE 104 (Digital Payment Data) subelements through MDES Manager. NOTE: If the cryptography cannot be validated, MDES will decline the transaction.		

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 1 - Value 55 = Merchant Validation Service Subfield 2 - Merchant Validation Results. Value either: • M = The submitted merchant data is a match to Mastercard's merchant data • N = The submitted merchant data is not a match to Mastercard's merchant data NOTE: Service results will only be provided if there was sufficient data on file for Mastercard to perform the merchant validation.		
	87 - Card Validation Code Result	As applicable. Issuer must validate the incoming DE48.92 CVC 2 value when present against the Funding PAN and provide validation result.	•	√
	92 - CVC 2	Conditionally present when submitted by the acquirer and the value is not recognized as a valid DTVC, applicable only for select programs. NOTE: Issuers must validate the incoming CVC 2 value when present against the Funding PAN and decline when the validation fails.	•	•
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR): • Authorization Request/0100 and Financial Transaction Request/0200 messages: Mastercard provides the PAR when Mastercard is the BIN Controller, and the AMS stores a PAR for the PAN. • Responses: If Mastercard is not the BIN Controller, the issuer may include DE 56 (Payment Account Data) in the response message. If Mastercard is the BIN Controller, the Dual Message System (Authorization) or Single Message System will ignore DE 56 received from the issuer and will set the PAR value retrieved from AMS in DE 56.	√	√

DE	Name/Subelement	Comments/Subfield	Req	Res
61	Point of Service Data	Subfield 3 - POS Terminal Location: Value either: • 2 = Off premises of card acceptor facility (cardholder terminal including home PC, mobile phone, PDA) • 4 = On premises of card acceptor facility (cardholder terminal including home PC, mobile phone, PDA)	√	•
		Subfield 4 - POS Cardholder Presence: Whether the cardholder is present at the point of service and the condition (if the cardholder is not present). Value either: • 2 = Card not present (mail/facsimile order) • 3 = Cardholder not present (phone or Automated Response Unit [ARU]) • 4 = Recurring transactions • 5 = Cardholder not present (Electronic order [home PC, Internet, mobile phone, PDA])	√	•
		Subfield 7 POS Transaction Status with value 0 (Normal request), 1 (Deferred Authorization), 4 (Preauthorization request) or 8 (Account Status Inquiry).		
		Subfield 10 - Cardholder-Activated Terminal Level: Whether the cardholder activated the terminal with the use of the card and the CAT security level; value 6 = Authorized Level 6 CAT: Electronic commerce.	√	•
104	Digital Payment Data	Subelement 001 - Digital Payment Cryptogram (not present in request or response)	•	•

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subelement 002 - Estimated Amount: contains the estimated purchase amount of the transaction when the payment request is presented to the consumer during checkout. This amount may not always be the final amount. The estimated amount is encoded and included within the cryptogram. NOTE: The estimated amount is in the transaction currency defined in DE 49 (Currency Code, Transaction).	√	•
		Subelement 003 - Remote Commerce Acceptor Identifier: Contains a merchant identifier such as the merchant business website URL or reverse domain name as presented to the consumer during checkout. The merchant identifier is provided encoded as Base64. NOTE: Issuers can opt-in to receive DE 104 (Digital Payment Data) subelements through MDES Manager.	√	•
		Subelement 004 - Digital Service Provider: As applicable NOTE: Issuers can opt-in to receive DE 104 (Digital Payment Data) subelements through MDES Manager.	√	•
		Subelement 005 - Digital Authentication Data: As applicable NOTE: Issuers can opt-in to receive DE 104 (Digital Payment Data) subelements through MDES Manager.	√	•
105	Multi-Use Transaction Identification Data	If applicable	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
119	MDES Service Indicators, subelement 003: MDES Service Indicators for issuers	If applicable. Refer to the MDES service indicators for more details. NOTE: Issuers can opt-in to receive this element through MDES Manager.	√	•

Dual Message System (Clearing) - UCAF data

When MDES has performed cryptography validation and mapping on a DSRP transaction containing UCAF, issuers will receive the following data elements and values in First and Second Presentment/1240 messages.

Table 46: Data elements and values in First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's PAN
22	Point of Service Data Code	Subfield 7- Card Data - Input Mode: Value S = Electronic commerce
105	Multi-Use Transaction Identification Data	If applicable

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: One of the following values: • C (value C followed by a space) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • CC (value CC) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. • H (value H followed by a space) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • HC (value HC) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. • O (value O followed by a space) = Subfield 2 contains the Static token submitted by the acquirer. The account range is owned by Mastercard. Mapping was based on the authorization log. • OC (value OC) = Subfield 2 contains the Static token submitted by the acquirer. The account range is owned by Mastercard. Mapping was based on a source other than the authorization log.
		Subfield 2 - Account Number: Token
PDS 0016	Digital Service Provider	If applicable

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0020	Digital Commerce Solution Indicator	These subfields might be present in transactions associated to specific programs such as Click to Pay, SQR, Pay By Link. Subfield 1=Digital Commerce Domain Identifier • 0 = Multi-Domain • 1 = Mastercard • 2 = Acquirer • 3 = Issuer • 4-9 = Reserved for future use Subfield 2=Digital Commerce Program Type • 0 = Multi-Program • 1 = SQR • 2 = C2P • 3 = In-store Biometric • 4 = Next Gen POI • 5 = Mastercard QR: Pay by Link • 6 = Mastercard QR: X QR • 7 = Tap To Own (to be assigned) • 8 = Agentic Payment (to be assigned) • 9 =EV Charging (to be assigned) Subfield 3=Digital Commerce Program Details • 10 = SQR • 20 = C2P • 30 = Biometrics • 40= Next Gen POI • 51= Merchant Presented PBL • 52= Consumer Presented PBL • 61 = Merchant Presented QR • 62= Consumer Presented QR • 70 = Tap To Own • 80 = Agentic Payment • 91 = EV Contactless • 92 = EV COF Subfield 4=Digital Commerce Check-In Details. Value: Open Text
PDS 0052	Electronic Commerce Security Level Indicator	Refer to Authorization Request Response/0110 or Financial Transaction Request Response/0210

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0058	Token Assurance Method	Values: • 10 = Card Issuer Account Verification (Successful TAR, AS, ASI response with no authentication) • 11 = Card Issuer Interactive Cardholder Authentication – 1 Factor (Token authenticated using MDES Authentication Service (OTP)) • 14 = Card Issuer Asserted Authentication (Push Provisioning into without additional authentication beyond Issuer's own application) • 20 = Push Provisioned token with MDES Authentication service following token creation • 21 = Push Provisioned token with Identity Check™ authentication following token creation • 22 = Token authenticated via Identity Check™ Risk-based authentication (no cardholder challenge) • 23 = Token authenticated via Identity Check™ Interactive step-up (cardholder challenge) • 24 = Token with device-asserted authentication
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	If applicable
PDS 0207	Wallet Identifier	If applicable
PDS 0218	Cardholder/Merchant Initiated Transaction Indicator	As applicable

Token Mapping Failures, Suspended/Deactivated Tokens, and Pre-validation Failures

Mastercard might send decline advice messages to issuers when MDES cannot successfully complete token to PAN mapping or when Mastercard declines the transaction on behalf of the issuer (due to a cryptographic validation failure).

Decline advice messages are sent as either:

- Authorization Advice/0120 - All Dual Message System (Authorization) issuers will receive the declined advice messages.
- Financial Transaction Advice/0220 - Only those Single Message System issuers that have opted to receive declined advice messages will be sent online financial transaction advice messages. However, all Single Message System issuers will receive declined advices in the Non-Financial Record (NREC) of the 250-byte Batch Data File.

Token to PAN-Mapping Failures

When MDES cannot successfully complete token to PAN mapping (for example, if the token to PAN mapping is not found, or the token is suspended or deactivated), Mastercard will send a decline response to the acquirer and an advice message to the issuer. Since the PAN mapping failed, DE 2 may contain the token PAN in the advice notification to the Issuer.

The *Data element values without errors* table shows data element values that are different from a transaction message that does not have mapping errors or cryptographic errors. The values can be present for these message types:

- Authorization Advice/0120
- Authorization Advice Response/0130
- Financial Transaction Advice Response/0230

Table 47: Data element values without errors

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's token	√	√
14	Date, Expiration	PAN expiration date (if applicable)	√	•
39	Response Code	One of the applicable values that the Authorization Platform/Single Message System sent to the acquirer: • 14 = Invalid card number (unable to map the token to the PAN) • 30 = Format error • 62 = Restricted card (for suspended or deactivated token) • 96 = System error (mapping failed because of technical reasons)	√	√
48	Additional Data - Private Use			
	33 - PAN Mapping File Information	Not present, unless the mapping failed because the token is suspended or deactivated	•	•
	34 - ATC Information	Not present	•	•
	42 - Electronic Commerce Indicators	Not present	•	•
	43 - Universal Cardholder Authentication Field (UCAF)	Not present	•	•
	71 - On-behalf Services	Subfield 1 - On-behalf Service: Value 50 = MDES PAN Mapping	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 2 - On-behalf Result 1: Value either: • I = Invalid Token • U = Unable to process	√	•
		Subfield 3 - On-behalf Result 2: Mastercard use only with a space or value	√	•
	79 - Chip CVR/TVR Bit Error Results	Not present	•	•
60	Advice Reason Code	Subfield 1 - Advice Reason Code: Value 141 = MDES Advice to Issuer	√	•
		Subfield 2 - Advice Detail Code: One of the following values: • 0201 = Reject: Invalid Token-PAN mapping relationship • 0202 = Reject: Token in suspended status • 0203 = Reject: Token deactivated	√	•
		Subfield 3 - Advice Detail Text: Textual information relating to the value in DE 60, subfield 2.	√	•
105	Multi-Use Transaction Identification Data	Not applicable	√	•
119	MDES Service Indicators, subelement 003: MDES Service Indicators for issuers	If applicable. Refer to the MDES service indicators for more details. NOTE: Issuers can opt-in to receive this element through MDES Manager.	√	•

Chip and UCAF Pre-Validation Failures

When a token transaction fails MDES cryptographic validation, the issuer decision matrix or MDES Transaction Analysis service (for MCBP 2.0 tokens) can determine whether Mastercard declines the transaction or continues processing. For more information, see MDES Transaction Validation and Decision Matrix.

If Mastercard declines on behalf of the issuer, a decline advice message is sent to the issuer, and a decline response message is sent to the acquirer. Otherwise, if processing continues, the validation failure results are provided in the authorization request message sent to the issuer.

NOTE: The authorization request message contains similar data elements to those present for a transaction without cryptographic errors; see the appropriate Dual Message System (Authorization) and Single Message System section (for EMV Data or UCAF).

The *Data element values without errors* table shows data element values that are different from a transaction message that does not have mapping errors or cryptographic errors. The values can be present for these decline advice messages:

- Authorization Advice/0120
- Authorization Advice Response/0130
- Financial Transaction Advice Response/0230

Table 48: Data element values without errors

DE	Name/Subelement	Comments/Subfield	Req	Res
39	Response Code	Value 05 (Do not honor) that the Authorization Platform/Single Message System sent to the acquirer	√	√
48	Additional Data - Private Use			
	27 - Transaction Analysis	As applicable to MCBP 2.0. Not present when the transaction comes from an alternative network.	√	•
		Subfield 1 - Overview		
		Subfield 2 - Test Results		
	34 - ATC Information	Not present	•	•
	Failure values in subelement 71 - On-behalf Services	Subfield 1 - On-behalf Service: One of the following values: • 51 = MDES Chip Pre-Validation • 61 = MDES Cloud-based Payments Chip Pre-Validation Service	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 2 - On-behalf Result 1: One of the following values: MDES chip (and UCAF) pre-validation failure: • A = ATC outside allowed range (applicable when ATC value is dynamic [varying] value) • E = ATC Replay • I = Invalid Cryptogram • K = No matching key file for this PAN, PAN expiry date and KDI combination • T = Valid ARQC/TC and ATC; TVR/CVR invalid • U = Unable to process MDES cloud-based payments chip (MCBP chip and UCAF) pre-validation failure: • K = No matching key file for this PAN, PAN expiry date and KDI combination • P = Valid MD AC; invalid UMD AC (Invalid Mobile PIN)	√	•
		Subfield 3 - On-behalf Result 2: Mastercard use only with a space or value	√	•
	79 - Chip CVR/TVR Bit Error Results	As applicable	√	•
60	Advice Reason Code	Subfield 1 - Advice Reason Code: Value 141 = MDES Advice to Issuer	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 2 - Advice Detail Code: One of the following values: MDES chip (and UCAF) pre-validation failure: • 0032 = Reject: Chip Data Processing Error • 0034 = Reject: Chip validation failed • 0035 = Reject: TVR/CVR validation failed • 0037 = Reject: No matching key File for this PAN, PAN expiry date and KDI combination - Validation of ARQC and CVR/TVR not performed, status unknown • 0043 = Reject: ATC outside allowed range • 0046 = Reject: ATC Replay MDES cloud-based payments chip (MCBP chip and UCAF) pre-validation failure: • 0204 = Reject: ATC Invalid - Not in List of Currently Active Single Use Keys • 0205 = Reject: ATC Replay • 0206 = Reject: Invalid MD AC and UMD AC (Invalid Mobile PIN) • 0207 = Reject: Valid MD AC; Invalid UMD AC (Mobile PIN Try Counter Max Limit not Reached, Token not Suspended) • 0208 = Reject: Invalid MD AC; Valid UMD AC • 0209 = Reject: Valid MD AC; Invalid UMD AC (Mobile PIN Try Counter Max Limit Reached, Token Suspended) • 0211 = Reject: TVR/CVR validation failed • 0212 = Reject: Unable to Process	√	•
		Subfield 3 - Advice Detail Text: Textual information relating to the value in DE 60, subfield 2.	√	•
105	Multi-Use Transaction Identification Data	If applicable	√	•

250-byte Batch Data File

When MDES cannot successfully complete token to PAN mapping (for example, if the token to PAN mapping is not found, or the token is suspended or deactivated):

- The Non-financial (NREC) record will contain the token in the PAN field (positions 33-51) and the applicable advice reason code in the Advice Reason Code field (positions 108-114).
- The Non-financial Additional Data Addendum (ADDR) will contain blanks in the Mapped PAN Account field (positions 120-138) and a blank in the Mapped PAN Indicator field (position 139).

When a token transaction fails cryptography validation within MDES and Mastercard sends a decline response to the acquirer:

- The Non-financial (NREC) record will contain the cardholder's PAN in the PAN (PAN) field (positions 33-51) and the applicable advice reason code in the Advice Reason Code field (positions 108-114).
- The Non-financial Additional Data Addendum (ADDR) will contain the token in the Mapped PAN Account field (positions 120-138).
- The Non-financial Additional Data Addendum (ADDR) will contain value C (The PAN in the Mapped PAN Account field contains the MDES token) in the Mapped PAN Indicator field (position 139).

E-Commerce Transactions with Dynamic Token Verification Code (DTVC)

Mastercard has enhanced MDES processing to allow merchants and acquirers with legacy transaction processing systems that are not yet upgraded to support tokenized ecommerce transactions with DSRP UCAF data or DSRP full EMV data, to submit static token transactions without the presence of DSRP cryptogram in either UCAF or DPD field.

MDES generate static tokens (previously known as Card on File tokens) for use by acquirers supporting e-commerce payments. MDES maps the static token, in the e-commerce transaction, to the cardholder's PAN before forwarding the request to the issuer.

To further strengthen the security of the static token transaction, Mastercard introduced support for dynamic token verification code (DTVC), an alternative to 3-digit static CVC2 that allows merchants to process tokenized transactions. Mastercard during the authorization processing will recognize and validate the DTVC before sending the request to the issuer for approval.

NOTE: For information relating to partial shipments and recurring payments, see Partial Shipments and Recurring Payment Transactions.

Dual Message System (Authorization) and Single Message System

This section describes the messages that issuers receive and send when MDES has successfully processed an e-commerce transaction (POS Entry Mode 10, 81 or 82) containing no UCAF data (DE 48, subelement 43 or DE 104, subelement 001).

The *Data elements and values for e-commerce transaction* table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Authorization Advice/0120
- Authorization Advice Response/0130
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210
- Financial Transaction Advice/0220
- Financial Transaction Advice Response/0230
- Reversal Request/0400
- Reversal Advice/0420

Table 49: Data elements and values for e-commerce transaction

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's PAN	√	√
14	Date, Expiration	PAN expiration date	√	•
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: • Value 81 = PAN/Token entry through electronic commerce with optional <i>SecureCode</i>-AAV or DSRP cryptogram in UCAF • Value 82 = Electronic Commerce PAN Auto Entry through Server (issuer, acquirer, or third-party vendor system).	√	•
48	Additional Data - Private Use			
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: Wallet ID (if applicable)	√	•
NOTE: New Wallet Identifier values may be provided in this field without prior written notice.				

DE	Name/Subelement	Comments/Subfield	Req	Res
	30 - Token Transaction Identifier	If applicable	√	•
	32 - Mastercard Assigned ID	When available	√	•
	NOTE: Subfield 1 - in Financial Transaction Messages.			
	33 - PAN Mapping File Information	Conditional in issuer responses. Issuers should provide the entire subelement 33 on transit transactions when using third-party tokenization services.	√	√
		Subfield 1- Account Number Indicator: Value either: • C = MDES Secure Element Token • H = MDES Cloud-based Payments Token • F = MDES Static Token		
		Subfield 2 - Account Number: Token		
		Subfield 3 - Expiration Date: Token expiration date		

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 5 - Token Assurance Method: Assigned token assurance method values (if applicable): • 10 = Card Issuer Account Verification (Successful TAR, AS, ASI response with no authentication) • 11 = Card Issuer Interactive Cardholder Authentication – 1 Factor (Token authenticated using MDES Authentication Service (OTP)) • 14 = Card Issuer Asserted Authentication (Push Provisioning into without additional authentication beyond Issuer's own application) • 20 = Push Provisioned token with MDES Authentication service following token creation • 21 = Push Provisioned token with Identity Check™ authentication following token creation • 22 = Token authenticated via Identity Check™ Risk-based authentication (no cardholder challenge) • 23 = Token authenticated through Identity Check™ Interactive step-up (cardholder challenge) • 24 = Token with device-asserted authentication		
		Subfield 6 - Token Requestor ID: Token Requestor ID (if applicable)		
		Subfield 8 - Storage Technology: Storage technology (if applicable)		

DE	Name/Subelement	Comments/Subfield	Req	Res
	42 - Electronic Commerce Indicators	Subfield 1 - Electronic Commerce Security Level Indicator and UCAF Collection Indicator. Issuers must be prepared to echo the value in Authorization Request Response/0110 and Financial Transaction Request Response/0210 messages. NOTE: For more information about the SLI values, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide Data Element 48 (Additional Data)</i>, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator).	√	√
	43 - Universal Cardholder Authentication Field (UCAF)	Not present	.	.
	48 - Digital Commerce Solution Indicator	These subfields might be present in transactions associated to specific programs such as Click to Pay, SQR, Pay By Link. • Subfield 1=Digital Commerce Domain Identifier • Subfield 2=Digital Commerce Program Type • Subfield 3=Digital Commerce Program Details • Subfield 4=Digital Commerce Check-In Details For full list of applicable values, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or <i>Mastercard Network Processing Single Message System Guide</i>.		
	71 - On-behalf Services	MDES validation results, which are the relevant combinations of the following subfields. Subfield 1: Value 50 = MDES PAN Mapping Subfield 2: Value C = Conversion of Token to PAN completed successfully Subfield 3: Mastercard use only with a space or value	√	.

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 1 - Value 55 = Merchant Validation Service Subfield 2 - Merchant Validation Results. Value either: • M = The submitted merchant data is a match to Mastercard's merchant data • N = The submitted merchant data is not a match to Mastercard's merchant data NOTE: Service results will only be provided if there was sufficient data on file for Mastercard to perform the merchant validation.		
		Subfield 1: Value 61 = MDES Cloud-based Payments Chip Pre-Validation Service Subfield 2: Value V = Valid or value I = Invalid Subfield 3: No value, or value T = Dynamic Token Verification Code (DTVC) verified		
	87 - Card Validation Code Result	As applicable. Issuer must validate the incoming DE48.92 CVC 2 value when present against the Funding PAN and provide validation result.	√	•
	92 - CVC 2	CVC 2 (when applicable)	√	•
		NOTE: Issuers must validate the incoming CVC2 value when present against the Funding PAN and decline when the validation fails.		

DE	Name/Subelement	Comments/Subfield	Req	Res
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR): • Authorization Request/0100 and Financial Transaction Request/0200 messages: Mastercard provides the PAR when Mastercard is the BIN Controller, and the AMS stores a PAR for the PAN. • Responses: If Mastercard is not the BIN Controller, the issuer may include DE 56 (Payment Account Data) in the response message. If Mastercard is the BIN Controller, the Dual Message System (Authorization) or Single Message System will ignore DE 56 received from the issuer and will set the PAR value retrieved from AMS in DE 56.	√	√
104	Digital Payment Data	• Subelement 004: Digital Service Provider: As applicable • Subelement 005: Digital Authentication Data: As applicable NOTE: Issuers can opt-in to receive DE 104 (Digital Payment Data) subelements through MDES Manager.	√	•
105	Multi-Use Transaction Identification Data	If applicable	√	•
119	MDES Service Indicators, subelement 003: MDES Service Indicators for issuers	If applicable. Refer to the MDES service indicators for more details. NOTE: Issuers can opt-in to receive this element through MDES Manager.	√	•

Token Mapping Failures, Suspended/Deactivated Tokens, and Pre-Validation Failures

Mastercard might send decline advice messages to issuers when MDES cannot successfully complete token to PAN mapping or when Mastercard declines the transaction on behalf of the issuer (due to a cryptographic validation failure). When PAN mapping fails DE 2 may contain the token PAN in the advice notification to the Issuer.

The scenarios and data element values are similar to those for DSRP transactions with UCAF data. Refer to 'Token Mapping Failures, Suspended/Deactivated Tokens, and Pre-Validation Failures' in the DSRP Transactions section.

Dual Message System (Clearing)

When Mastercard has processed an e-commerce transaction (POS Entry Mode 81) containing no UCAF data (DE 48, subelement 43), issuers will receive the following data elements and values in First and Second Presentment/1240 messages.

Table 50: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's PAN
PDS0020	Digital Commerce Solution Indicator	These subfields might be present in transactions associated to specific programs such as Click to Pay, SQR, Pay By Link. • Subfield 1=Digital Commerce Domain Identifier • Subfield 2=Digital Commerce Program Type • Subfield 3=Digital Commerce Program Details • Subfield 4=Digital Commerce Check-In Details For full list of applicable values, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i>.
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value S = Electronic commerce
105	Multi-Use Transaction Identification Data	If applicable

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: One of the following values: • C (value C followed by a space) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • CC (value CC) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. • H (value H followed by a space) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • HC (value HC) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. • O (value O followed by a space) = Subfield 2 contains the Static token submitted by the acquirer. The account range is owned by Mastercard. Mapping was based on the authorization log. • OC (value OC) = Subfield 2 contains the Static token submitted by the acquirer. The account range is owned by Mastercard. Mapping was based on a source other than the authorization log.
		Subfield 2 - Account Number: Token
PDS 0052	Electronic Commerce Security Level Indicator	Refer to Authorization Request Response/0110 or Financial Transaction Request Response/0210

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0058	Token Assurance Method	Values: • 10 = Card Issuer Account Verification (Successful TAR, AS, ASI response with no authentication) • 11 = Card Issuer Interactive Cardholder Authentication – 1 Factor (Token authenticated using MDES Authentication Service (OTP)) • 14 = Card Issuer Asserted Authentication (Push Provisioning into without additional authentication beyond Issuer's own application) • 20 = Push Provisioned token with MDES Authentication service following token creation • 21 = Push Provisioned token with Identity Check™ authentication following token creation • 22 = Token authenticated through Identity Check™ Risk-based authentication (no cardholder challenge) • 23 = Token authenticated through Identity Check™ Interactive step-up (cardholder challenge) • 24 = Token with device-asserted authentication
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	If applicable
PDS 0207	Wallet Identifier	If applicable

Dynamic magnetic stripe transactions

This section describes the messages that issuers receive and send when MDES has successfully processed a Dynamic Magnetic Stripe Data token transaction.

Dual Message System (Authorization) and Single Message System

The *Data elements and values* table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Authorization Advice/0120
- Authorization Advice Response/0130
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210
- Financial Transaction Advice/0220
- Financial Transaction Advice Response/0230

- Reversal Request/0400
- Reversal Advice/0420

Table 51: Data elements and values

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's PAN	√	√
14	Date, Expiration	PAN expiration date	√	•
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: Value 90 = PAN auto-entry through magnetic stripe	√	•
35	Track 2 Data	Not present	•	•
		NOTE: Issuers may opt in through MDES Manager to receive track data in DE 35 (Track 2 Data).		
45	Track 1 Data	Not present	•	•
		NOTE: Issuers may opt in through MDES Manager to receive track data in DE 45 (Track 1 Data).		
48	Additional Data - Private Use			
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: If applicable	√	•
		NOTE: New Wallet Identifier values may be provided in this field without prior written notice.		
	27 - Transaction Analysis	As applicable to MCBP 2.0. Not present when the transaction comes from an alternative network.	√	•
		Subfield 1 - Overview		
		Subfield 2 - Test Results		
	33 - PAN Mapping File Information	Conditional in issuer responses. Issuers should provide the entire subelement 33 on transit transactions when using third-party tokenization services.	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 1 - Account Number Indicator: Value either: • C = MDES Secure Element Token • H = MDES Cloud-based Payments Token	√	√
		Subfield 2 - Account Number: Token		
		Subfield 3 - Expiration Date: Token expiration date		
		Subfield 5 - Token Assurance Method: Token assurance method value (as applicable)		
		Subfield 6 - Token Requestor ID: Token Requestor ID (as applicable)		
		Subfield 8 - Storage Technology: Storage technology (as applicable)		
49 - Time Validation Information		Subfield 1 - Time Value: The time data derived from the acquirer transaction to be used in the time validation	√	•
		Subfield 2 - Time Discrepancy Value: A positive value representing the differential in minutes between the transaction time data and service-calculated time	√	•
		Subfield 3 - Time Discrepancy Indicator: A value that indicates if the time discrepancy value is below, above, or within the minimum and maximum values for the time validation window or that indicates time validation was not performed	√	•
71 - On-behalf Services		MDES validation results, which are the following combinations of subfields.	√	•
		Subfield 1: Value 50 = MDES PAN Mapping		
		Subfield 2: Value C = Conversion of Token to PAN completed successfully		
		Subfield 3: Mastercard use only with a space or value		

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 1: Value 52 = MDES CVC 3 Pre-Validation Subfield 2: Value V = Valid Subfield 3: Mastercard use only with a space or a value		
		Subfield 1: Value 62 = MDES Cloud-based Payments Magnetic Stripe Pre-Validation Service Subfield 2: Value V = Valid Subfield 3: Mastercard use only with a space or a value		
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR): - Authorization Request/0100 and Financial Transaction Request/0200 messages: Mastercard provides the PAR when Mastercard is the BIN Controller, and the AMS stores a PAR for the PAN. - Responses: If Mastercard is not the BIN Controller, the issuer may include DE 56 (Payment Account Data) in the response message. If Mastercard is the BIN Controller, the Dual Message System (Authorization) or Single Message System will ignore DE 56 received from the issuer and will set the PAR value retrieved from AMS in DE 56.	√	√
105	Multi-Use Transaction Identification Data	If applicable	√	•
119	MDES Service Indicators, subelement 003: MDES Service Indicators for issuers	If applicable. Refer to the MDES service indicators for more details. NOTE: Issuers can opt-in to receive this element through MDES Manager.	√	•

Decision Matrix for MDES Dynamic Magnetic Stripe Data

MDES supports timestamp validation of Dynamic Magnetic Stripe Data token transactions to help ensure transactions are not inappropriately delayed. The time validation information is in DE 48, subelement 49.

During the issuer enablement process, issuers can set up a decision matrix in MDES to designate the action Mastercard should take, on behalf of the issuer, if the time validation is outside the time validation window. For more information about the decision matrix, see MDES Transaction Validation and Decision Matrix.

Validations and Advice Messages

Mastercard might send decline Authorization Advice/0120 or Financial Transaction Advice/0220 messages to issuers in the following situations:

- MDES cannot successfully complete token to PAN mapping (for example, if the token to PAN mapping is not found, or the token is suspended or deactivated)
- Mastercard declines the transaction on behalf of the issuer, because of a cryptographic validation failure or the time validation is outside the time validation window

The "Validations and Advice Messages" table shows data element values that are different from a transaction message that does not have mapping errors or cryptographic errors:

Table 52: Validations and Advice Messages

DE	Name/Subelement	Comments/Subfield	Req	Res
39	Response Code	One of the following values, when the token to PAN mapping or pre-validation is not successful: • 05 = Do not honor; returned when the issuer has chosen for Mastercard to decline on their behalf and dynamic CVC 3 pre-validation failed • 14 = Invalid card number; returned when MDES is unable to map the token to the PAN • 30 = Format error • 62 = Restricted card; returned when a token has been suspended or deactivated from use • 96 = System error; returned when the mapping failed because of technical reasons	√	•
48	Additional Data - Private Use			

DE	Name/Subelement	Comments/Subfield	Req	Res
	33 - PAN Mapping File Information	Might not be present		
	87 - Card Validation Code Result	Authorization Request Response/0110 and Financial Transaction Request Response/0210 messages: When the MDES dynamic CVC 3 pre-validation is not successful, contains one of the following values: • E = Length of unpredictable number was not a valid length • P = Unable to process • Y = Invalid	√	•
60	Advice Reason Code	Subfield 1 - Advice Reason Code: Value 141 = MDES Advice to Issuer	√	•
		Subfield 2 - Advice Detail Code: One of the following values: • 0009 = Reject: Invalid Time Validation • 0201 = Reject: Invalid Token - PAN mapping relationship • 0202 = Reject: Token in suspended status • 0203 = Reject: Token deactivated	√	•
		Subfield 3 - Advice Detail Text: Textual information relating to the value in DE 60, subfield 2.	√	•

DE 48, subelement 49 will not be present in Financial Transaction Advice/0220 Debit Mastercard (Force Post).

Issuers may also receive an Authorization Advice/0120 - System-generated or Financial Transaction Advice/0220 - System-generated message if a contactless magnetic stripe transaction is submitted with DE 22, subfield 1, value 90 instead of value 91 (PAN auto-entry through contactless magnetic stripe). Mastercard will reject these types of transactions with DE 39 (Response Code), value 05 (Do not honor).

Merchant-initiated Transactions (MITs)

This section describes the messages that issuers receive and send when MDES has processed a partial shipment or recurring payment authorization transaction with POS Entry Mode 81.

These transactions apply to DSRP and Tokenized E-Commerce with Dynamic Payment Credentials.

NOTE: Issuers should continue to review transactions that do not have cryptography validation and confirm partial shipments or recurring payments correspond to an original transaction detail.

Dual Message System (Authorization)

The *Data elements and values* table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110

All mandatory request and response data elements apply, even if not specifically listed here in this layout.

Table 53: Data elements and values

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's PAN	√	√
4	Amount, Transaction		√	√
14	Date, Expiration	PAN expiration date	√	•
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: • 81 = PAN/Token entry through electronic commerce with optional. The amount of funds the cardholder requested in the local currency of the acquirer or source location of the transaction. The amount might be a proportion of the original transaction amount, reflecting the value of part of the shipment.<i>SecureCode</i> • 10: Credential on file-AAV or DSRP cryptogram in UCAF • 82 = PAN Auto Entry through Server (issuer, acquirer, or third-party vendor system).	√	•
48	Additional Data - Private Use			
	22 - Multi-Purpose Merchant Indicator	Subfield 05 - Cardholder/Merchant Initiated Transaction Indicator - As applicable	√	√

DE	Name/Subelement	Comments/Subfield	Req	Res
	33 - PAN Mapping File Information	AAV or DSRP The amount of funds the cardholder requested. Supports the mapping between the virtual account data and actual account data. Conditional in issuer responses. Issuers should provide the entire subelement 33 on transit transactions when using third-party tokenization services.	√	√
		Subfield 1 - Account Number Indicator: Value either: • C = MDES Secure Element Token • H = MDES Cloud-based Payments Token		
		Subfield 2 - Account Number: Token		
		Subfield 3 - Expiration Date: Token expiration date		
		Subfield 5 - Token Assurance Method: Token assurance method values: • 10 = Card Issuer Account Verification (Successful TAR, AS, ASI response with no authentication) • 11 = Card Issuer Interactive Cardholder Authentication – 1 Factor (Token authenticated using MDES Authentication Service (OTP)) • 14 = Card Issuer Asserted Authentication (Push Provisioning into without additional authentication beyond Issuer's own application) • 20 = Push Provisioned token with MDES Authentication service following token creation • 21 = Push Provisioned token with Identity Check™ authentication following token creation • 22 = Token authenticated through Identity Check™ Risk-based authentication (no cardholder challenge) • 23 = Token authenticated through Identity Check™ Interactive step-up (cardholder challenge) • 24 = Token with device-asserted authentication		

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 6 - Token Requestor ID: Token Requestor ID (as applicable)		
		Subfield 8 - Storage Technology: Storage technology (as applicable)		
	42 - Electronic Commerce Indicators	Subfield 1 - Electronic Commerce Security Level Indicator and UCAF Collection Indicator. NOTE: For more information about the SLI values, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide Data Element 48 (Additional Data)</i>, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator).	√	•
	43 - Universal Cardholder Authentication Field (UCAF)	As applicable	•	•
	71 - On-behalf Services	Subfield 1 - On-behalf Service: Value 50 = MDES PAN Mapping	√	•
		Subfield 2 - On-behalf Result 1: Value C = Conversion of Token to PAN completed successfully	√	•
		Subfield 3 - On-behalf Result 2: Mastercard use only with a space or value	√	•
61	POS Data	Subfield 3 - POS Terminal Location: Value either: • 2 = Off premises of card acceptor facility (cardholder terminal including home PC, mobile phone, PDA) • 4 = On premises of card acceptor facility (cardholder terminal including home PC, mobile phone, PDA)	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 4 - POS Cardholder Presence: Whether the cardholder is present at the point of service and the condition (if the cardholder is not present); value either: Subfield 10 - Cardholder-Activated Terminal Level: Whether the cardholder activated the terminal with the use of the card and the CAT security level; value 6 = Authorized Level 6 CAT: Electronic commerce 4 = Standing order/recurring transactions 5 = Electronic order (home PC, Internet, mobile phone, PDA)	√	•
104	Digital Payment Data	- Subelement 004 - Digital Service Provider: As applicable - Subelement 005 - Digital Authentication Data: As applicable NOTE: Issuers can opt-in to receive DE 104 (Digital Payment Data) subelements through MDES Manager.	√	•
105	Multi-Use Transaction Identification Data	If applicable	√	•
119	MDES Service Indicators, subelement 003: MDES Service Indicators for issuers	If applicable. Refer to the MDES service indicators for more details. NOTE: Issuers can opt-in to receive this element through MDES Manager.	√	•

Dual Message System (Clearing)

When MDES has performed mapping on a partial shipment or recurring payment clearing transaction, issuers will receive a presentment message with the following data elements and values.

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's PAN

DE/PDS	Name/Subelement	Comments/Subfield
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value S = Electronic commerce Subfield 7: Value 7 = Credential on file
Subfield 10 - Cardholder-Activated TerminalPD S 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: One of the following values: • C (value C followed by a space) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • CC (value CC) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. • H (value H followed by a space) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • HC (value HC) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. • O (value O followed by a space) = Subfield 2 contains the Static token submitted by the acquirer. The account range is owned by Mastercard. Mapping was based on the authorization log. • OC (value OC) = Subfield 2 contains the Static token submitted by the acquirer. The account range is owned by Mastercard. Mapping was based on a source other than the authorization log.
		Subfield 2 - Account Number: Token
105	Multi-Use Transaction Identification Data	If applicable
PDS 0052	Electronic Commerce Security Level Indicator	Refer to Authorization Request Response/0110 or Financial Transaction Request Response/0210

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0058	Token Assurance Method	If applicable Values: • 10 = Card Issuer Account Verification (Successful TAR, AS, ASI response with no authentication) • 11 = Card Issuer Interactive Cardholder Authentication - 1 Factor (Token authenticated using MDES Authentication Service (OTP)) • 14 = Card Issuer Asserted Authentication (Push Provisioning into without additional authentication beyond Issuer's own application) • 20 = Push Provisioned token with MDES Authentication service following token creation • 21 = Push Provisioned token with Identity Check™ authentication following token creation • 22 = Token authenticated through Identity Check™ Risk-based authentication (no cardholder challenge) • 23 = Token authenticated through Identity Check™ Interactive step-up (cardholder challenge) • 24 = Token with device-asserted authentication
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	If applicable
PDS 0207	Wallet Identifier	If applicable

Stand-In authorization

If the funding account range and transaction type participate in Stand-In authorization, then Mastercard may apply Stand-In authorization processing on the MDES token transaction using the funding account range parameters.

Mastercard Stand-In service will provide issuers with an Authorization Advice/0120 or Financial Transaction Advice/0220 message depending on whether they receive their advice messages through the Dual Message System (Authorization) or the Single Message System.

For further details on the Stand-In service, refer to the *Stand-In Authorization Reference Guide*.

The detailed specifications for an Authorization Advice/0120 or Financial Transaction Advice/0220 system-generated advice can be found within the *Mastercard Network Processing Dual*

Message Authorization System Guide or Mastercard Network Processing Single Message System Guide.

MoneySend Transactions

The MoneySend and Funding Transactions service enables person-to-person transfers, account-to-account transfers, agent cash out, credit card bill payment, business to consumer disbursement, government/non-government organization to consumer disbursement, and business money transfers by allowing consumers to use their Mastercard, Debit Mastercard, Cirrus and Maestro card on MDES enabled wallets to send and access funds.

The Mastercard® MoneySend™ and Funding Transactions service also allows the use of multiple channels to initiate transactions such as an ATM, a bank branch, a stand-alone kiosk, mobile, or over the Internet.

NOTE: An issuing country, receiving institution, or receiving institution's account range must be enabled for the MoneySend service. The network blocking service (OBS 33) is executed on all MoneySend transactions. For information about the Mastercard MoneySend service, refer to the Mastercard Network Processing Dual Message Authorization System Guide.

Dual Message System (Authorization) and Single Message System - e-Commerce MoneySend Funding Transactions

This section describes the messages that issuers must be prepared to receive when an acquirer submits a MoneySend payment transaction with POS Entry Mode 81.

The *Data elements and values for MoneySend transactions* table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Authorization Advice/0120
- Authorization Advice Response/0130
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210
- Financial Transaction Advice/0220
- Financial Transaction Advice Response/0230
- Reversal Request/0400
- Reversal Advice/0420

All mandatory Authorization Request/0100 and Financial Transaction Request/0200 data elements apply.

Table 54: Data elements and values for MoneySend transactions

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's PAN	√	√
3	Processing Code	Value 00 = Funding Transaction	√	√
14	Date, Expiration	PAN expiration date	√	•
18	Merchant Type	MCC 6538 (MoneySend Funding) MCC 6540 (Funding Transaction) MCC 4829 (Money Transfer) Refer to the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> for details.	√	•
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: Value 81 = PAN/Token entry through electronic commerce or Value 10 = Credential on file	√	•
39	Response Code	For the full list of codes, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or the <i>Mastercard Network Processing Single Message System Guide</i> .	•	√

DE	Name/Subelement	Comments/Subfield	Req	Res
48	Additional Data - Private Use			
	Transaction Category Code	R = Face-to-face retail (applicable to MCCs 6538 and 6540) T = Non-face-to-face retail (applicable to MCCs 6538 and 6540) U = Unique Transaction - Quasi Cash Disbursement (applicable to MCC 4829)	√	√
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: If applicable	√	•
		NOTE: New Wallet Identifier values may be provided in this field without prior written notice.		
	27 - Transaction Analysis As applicable to MCBP 2.0	Not present when the transaction comes from an alternative network. • Subfield 1 - Overview • Subfield 2 - Test Results	√	•
	30 - Token Transaction Identifier	If applicable	√	•
	32 - Mastercard Assigned ID	When applicable	√	•
	33 - PAN Mapping File Information	Conditional in issuer responses. Issuers should provide the entire subelement 33 on transit transactions when using third-party tokenization services. Subfield 1 - Account Number Indicator: • C = MDES Secure Element Token • F = MDES Static Token • H = MDES Cloud-based Payments Token Subfield 2 - Account Number: Token Subfield 3 - Expiration Date: Token expiration date Subfield 5 - Token Assurance Method: Token assurance method value (as applicable) Subfield 6 - Token Requestor ID: Token Requestor ID (as applicable)	√	√

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 8 - Storage Technology: Storage technology (as applicable)		
34 - ATC Information Conditional		Not present for decline advice messages	√	•
42 - Electronic Commerce Indicator		Subfield 1 - Electronic Commerce Security Level Indicator and UCAF Collection Indicator.	√	√
		NOTE: For more information about the SLI values, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide Data Element 48 (Additional Data), subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator).</i>		
71 - On-behalf Services		MDES validation results, which are the relevant combinations of the following subfields.	√	•
		NOTE: This data element may include different results when there are validation failures but the issuer has chosen for Mastercard to continue processing; refer to Cryptographic Pre-Validation Failures.		
		Subfield 1: Value 50 = MDES PAN Mapping		
		Subfield 2: Value C = Conversion of Token to PAN completed successfully		
		Subfield 3: Mastercard use only with a space or value		
		Subfield 1: Value 33 = MoneySend and Funding Blocking Service		
		Subfield 2: All values applicable for MoneySend payments		
		Subfield 3: Mastercard use only with a space or value		

DE	Name/Subelement	Comments/Subfield	Req	Res
	77 - Transaction Type Identifier	• C07 = General Person-to-Person • C52 = General Transfer to Own Account • C53 = Agent Cash Out • C54 = Payment of Own Credit Card Bill • C55 = Business Disbursement • C56 = Government/Non-profit Disbursement • C57 = Rapid Merchant Settlement • C65 = General Business to Business Transfer • F07 = General Person-to-Person Transfer • F08 • F52 = General Transfer to Own Account • F53 = Agent Cash Out • F54 = Payment of Own Credit Card Bill • F55 = Business Disbursement • F61 = Transfer to Own Staged Digital Wallet Account • F64 = Transfer to Own Debit or Prepaid Account • F65 = General Business-to-Business Transfer	√	•
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR): • Authorization Request/0100 and Financial Transaction Request/0200 messages: Mastercard provides the PAR when Mastercard is the BIN Controller, and the AMS stores a PAR for the PAN. • Responses: If Mastercard is not the BIN Controller, the issuer may include DE 56 (Payment Account Data) in the response message. If Mastercard is the BIN Controller, the Dual Message System (Authorization) or Single Message System will ignore DE 56 received from the issuer and will set the PAR value retrieved from AMS in DE 56.	√	√
61	POS Data	Subfield 04 - POS Cardholder Presence: Value 5 = Cardholder not present (Electronic order [home PC, Internet, mobile phone, PDA])	√	√
		Subfield 05 - POS Card Presence: Value 1 = Card not present	√	√
		Subfield 10 - Cardholder-Activated Terminal Level: Value 6 = Electronic commerce	√	√

DE	Name/Subelement	Comments/Subfield	Req	Res
104	Digital Payment Data	Subfield 001 - Digital Payment Cryptogram (not present in request or response)	•	•
		Subfield 002 - Estimated Amount: contains the estimated purchase amount of the transaction when the payment request is presented to the consumer during checkout. This amount may not always be the final amount. The estimated amount is encoded and included within the cryptogram.	√	•
		NOTE: The estimated amount is in the transaction currency defined in DE 49 (currency Code, Transaction).		
		Subfield 003 - Remote Commerce Acceptor Identifier: contains a merchant identifier such as the merchant business website URL or reverse domain name as presented to the consumer during checkout. The merchant identifier is provided encoded as Base64.	√	•
105	Multi-Use Transaction Identification Data	If applicable	√	•
108	Additional Transaction Reference Data	As sent by acquirer	√	√
119	MDES Service Indicators, subelement 003: MDES Service Indicators for issuers	If applicable. Refer to the MDES service indicators for more details. NOTE: Issuers can opt-in to receive this element through MDES Manager.	√	•

Token mapping failures, Suspended/Deactivated tokens

Mastercard might send decline advice messages to issuers when MDES cannot successfully complete token to PAN mapping or when Mastercard declines the transaction on behalf of the issuer (due to a cryptographic validation failure). Decline advice messages are sent as either:

- Authorization Advice/0120 - All Dual Message System (Authorization) issuers will receive the declined advice messages.
- Financial Transaction Advice/0220 - Only those Single Message System issuers that have opted to receive declined advice messages will be sent online financial transaction advice messages. However, all Single Message System issuers will receive declined advice in the Non-Financial Record (NREC) of the 250-byte batch data file.

250-byte batch data file

When transactions are processed by MDES, Single Message System issuers receive the following in the 250-byte batch data file:

- The Financial (FREC)/Non-Financial (NREC) and Financial/Non-Financial Exception (EREC) records will contain the cardholder's PAN in the PAN field (positions 33-51).
- The Financial/Non-Financial Additional Data Addendum (ADDR) will contain the token in the Mapped PAN Account field (positions 120-138).
- The Financial/Non-Financial Additional Data Addendum (ADDR) will contain value C (the PAN in the Mapped PAN Account field contains the MDES token) in the Mapped PAN Indicator field (position 139).

Dual Message System (Clearing) - e-Commerce MoneySend Funding Transactions

Issuers will receive the following data elements and values in First and Second Presentment/1240 messages when MDES has processed a contactless transaction successfully.

Table 55: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's token
3	Processing Code	Value 00 = Funding
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value S = Electronic Commerce
26	Card Acceptor Business Code (MCC)	MCC 6538 (MoneySend Funding) or MCC 6540 (Funding Transaction) or MCC 4829 (Money Transfer)
105	Multi-Use Transaction Identification Data	If applicable

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: One of the following values: • C (value C followed by a space) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • CC (value CC) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. • H (value H followed by a space) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • HC (value HC) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. Subfield 2 - Account Number: Token
PDS 0023	Terminal Type	Value CT6 = Electronic commerce transaction
PDS 0043	Transaction Type Identifier	As applicable for use cases. Refer to the definitions in the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> and the <i>Mastercard Network Processing Dual Message Authorization System Guide</i>.
PDS 0052	Electronic Commerce Security Level Indicator	Refer to Authorization Request Response/0110 or Financial Transaction Request Response/0210.
PDS 0058	Token Assurance Method	Optional
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	If applicable
PDS 0207	Wallet Identifier	If applicable
PDS 0765	MoneySend Receiver Data	If applicable

Dual Message System (Authorization) and Single Message System - e-Commerce MoneySend Payment Transactions

This section describes the messages that issuers must be prepared to receive when an acquirer submits a contactless MoneySend payment funding transaction.

The *Data elements and values for MoneySend transactions* table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Authorization Advice/0120
- Authorization Advice Response/0130
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210
- Financial Transaction Advice/0220
- Financial Transaction Advice Response/0230
- Reversal Request/0400
- Reversal Advice/0420

All mandatory Authorization Request/0100 and Financial Transaction Request/0200 data elements apply.

Table 56: Data elements and values for MoneySend transactions

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's PAN	√	√
3	Processing Code	Value 28 = Payment	√	√
14	Date, Expiration	PAN expiration date	√	•
18	Merchant Type	MCC 6536 (MoneySend Intracountry) MCC 6537 (MoneySend Intercountry) Refer to the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> for details.	√	•
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: Value 81 = PAN/Token entry via electronic Commerce or Value 10 = Credential on file	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
39	Response Code	For the full list of codes, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or the <i>Mastercard Network Processing Single Message System Guide</i> .	•	√
48	Additional Data - Private Use			
	Transaction Category Code (TCC)	P = Payment Transaction	√	√
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: (if applicable)	√	•
		NOTE: New Wallet Identifier values may be provided in this field without prior written notice.		
	27 - Transaction Analysis As applicable to MCBP 2.0	Not present when the transaction comes from an alternative network: • Subfield 1 - Overview • Subfield 2 - Test Results	√	•
	30 - Token Transaction Identifier	If applicable	√	•
	32 - Mastercard Assigned ID	When applicable	√	•
	33 - PAN Mapping File Information	Conditional in issuer responses. Issuers should provide the entire subelement 33 on transit transactions when using third-party tokenization services.	√	√
		Subfield 1 - Account Number Indicator: • C = MDES Secure Element Token • F = MDES Static Token • H = MDES Cloud-based Payments Token		
		Subfield 2 - Account Number: Token		
		Subfield 3 - Expiration Date: Token expiration date		
		Subfield 5 - Token Assurance Method: Token assurance method value (as applicable)		
		Subfield 6 - Token Requestor ID: Token Requestor ID (as applicable)		
		Subfield 8 - Storage Technology: Storage technology (as applicable)		

DE	Name/Subelement	Comments/Subfield	Req	Res
	34 - ATC Information Conditional	Not present for decline advice messages	√	•
	42 - Electronic Commerce Indicators	Subfield 1 - Electronic Commerce Security Level Indicator and UCAF Collection Indicator. NOTE: For more information about the SLI values, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide Data Element 48 (Additional Data)</i>, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator).	√	√
	71 - On-behalf Services	MDES validation results, which are the relevant combinations of the following subfields: NOTE: This data element may include different results when there are validation failures but the issuer has chosen for Mastercard to continue processing; see Cryptographic Pre-Validation Failures. Subfield 1: Value 50 = MDES PAN Mapping Subfield 2: Value C = Conversion of Token to PAN completed successfully Subfield 3: Mastercard use only with a space or value Subfield 1: Value 33 = MoneySend and Funding Blocking Service Subfield 2: All values applicable for MoneySend payments Subfield 3: Mastercard use only with a space or value	√	•
	77 - Transaction Type Identifier	• C07 = MoneySend Person-to-Person • C52 = MoneySend Account-to-Account Transfers • C53 = MoneySend Agent Cash Out • C54 = MoneySend Credit Card Bill Payment • C55 = MoneySend Business Disbursement • C56 = MoneySend Government/ Nonprofit Disbursement • C57 = MoneySend Acquirer Merchant Settlement • C58 = MoneySend Cash2ATM • C59 = MoneySend Cash2Card • C65 = MoneySend Business to Business Transfer	√	√

DE	Name/Subelement	Comments/Subfield	Req	Res
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR): • Authorization Request/0100 and Financial Transaction Request/0200 messages: Mastercard provides the PAR when Mastercard is the BIN Controller, and the AMS stores a PAR for the PAN. • Responses: If Mastercard is not the BIN Controller, the issuer may include DE 56 (Payment Account Data) in the response message. If Mastercard is the BIN Controller, the Dual Message System (Authorization) or Single Message System will ignore DE 56 received from the issuer and will set the PAR value retrieved from AMS in DE 56.	√	√
61	POS Data	Subfield 04 - POS Cardholder Presence: Value 5 = Cardholder not present (Electronic order [home PC, Internet, mobile phone, PDA])	√	√
		Subfield 05 - POS Card Presence: Value 1 = Card not present	√	√
		Subfield 10 - Cardholder-Activated Terminal Level: Value 6 = Electronic commerce	√	√
104	Digital Payment Data	Subfield 001 - Digital Payment Cryptogram (not present in request or response)	•	•
		Subfield 002 - Estimated Amount: contains the estimated purchase amount of the transaction when the payment request is presented to the consumer during checkout. This amount may not always be the final amount. The estimated amount is encoded and included within the cryptogram. NOTE: The estimated amount is in the transaction currency defined in DE 49 (currency Code, Transaction).	√	•
		Subfield 003 - Remote Commerce Acceptor Identifier: contains a merchant identifier such as the merchant business website URL or reverse domain name as presented to the consumer during checkout. The merchant identifier is provided encoded as Base64.	√	•
105		If applicable	√	•
108	Additional Transaction Reference Data	As sent by acquirer	√	√

DE	Name/Subelement	Comments/Subfield	Req	Res
119	MDES Service Indicators, subelement 003: MDES Service Indicators for issuers	If applicable. Refer to the MDES service indicators for more details. NOTE: Issuers can opt-in to receive this element through MDES Manager.	√	•

Token mapping failures, Suspended/Deactivated tokens

Mastercard might send decline advice messages to issuers when MDES cannot successfully complete token to PAN mapping or when Mastercard declines the transaction on behalf of the issuer (due to a cryptographic validation failure). Decline advice messages are sent as either:

- Authorization Advice/0120 - All Dual Message System (Authorization) issuers will receive the declined advice messages
- Financial Transaction Advice/0220 - Only those Single Message System issuers that have opted to receive declined advice messages will be sent online financial transaction advice messages. However, all Single Message System issuers will receive declined advices in the Non-Financial Record (NREC) of the 250-byte batch data file.

250-byte batch data file

When transactions are processed by MDES, Single Message System issuers receive the following in the 250-byte batch data file:

- The Financial (FREC)/Non-Financial (NREC) and Financial/Non-Financial Exception (EREC) records will contain the cardholder's PAN in the PAN field (positions 33-51).
- The Financial/Non-Financial Additional Data Addendum (ADDR) will contain the token in the Mapped PAN Account field (positions 120-138).
- The Financial/Non-Financial Additional Data Addendum (ADDR) will contain value C (the PAN in the Mapped PAN Account field contains the MDES token) in the Mapped PAN Indicator field (position 139).

Dual Message System (Clearing) - e-Commerce MoneySend Payment Transactions

Issuers will receive the following data elements and values in First and Second Presentment/1240 messages when MDES has processed a contactless transaction successfully.

Table 57: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's PAN
3	Processing Code	Value 28 = Payment

DE/PDS	Name/Subelement	Comments/Subfield
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: Value S = Electronic Commerce
26	Card Acceptor Business Code (MCC)	MCC 6536 (MoneySend Intracountry) MCC 6537 (MoneySend Intercountry)
105	Multi-Use Transaction Identification Data	If applicable
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: One of the following values: - C (value C followed by a space) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. - CC (value CC) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. - H (value H followed by a space) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. - HC (value HC) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log.
		Subfield 2 - Account Number: Token
PDS 0023	Terminal Type	Value CT6 = Electronic commerce transaction
PDS 0043	Transaction Type Identifier	As applicable for use cases. Refer to the definitions in the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> and the <i>Mastercard Network Processing Dual Message Authorization System Guide</i>.
PDS 0052	Electronic Commerce Security Level Indicator	Refer to Authorization Request Response/0110 or Financial Transaction Request Response/0210
PDS 0058	Token Assurance Method	Optional

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	If applicable
PDS 0207	Wallet Identifier	If applicable
PDS 0765	MoneySend Receiver Data	As applicable

Dual Message System (Authorization) and Single Message System - Contactless MoneySend Funding Transactions

This section describes the messages that issuers must be prepared to receive when an acquirer submits a contactless MoneySend funding transaction.

The *Data elements and values for MoneySend transaction* table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Authorization Advice/0120
- Authorization Advice Response/0130
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210
- Financial Transaction Advice/0220
- Financial Transaction Advice Response/0230
- Reversal Request/0400
- Reversal Advice/0420

All mandatory Authorization Request/0100 and Financial Transaction Request/0200 data elements apply.

Table 58: Data elements and values for MoneySend transaction

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's PAN	√	√
3	Processing Code	Value 00 = Funding Transaction	√	√
14	Date, Expiration	PAN expiration date	√	•

Issuer-specific information

Dual Message System (Authorization) and Single Message System - Contactless MoneySend
Funding Transactions

DE	Name/ Subelement	Comments/Subfield	Req	Res
18	Merchant Type	MCC 6538 (MoneySend Funding) MCC 6540 (Funding Transaction) MCC 4829 (Money Transfer) Refer to the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> for details.	√	•
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: Value 07 = PAN auto-entry through contactless M/Chip or Value 90 = PAN auto-entry through magnetic stripe	√	•
23	Card Sequence Number	As applicable	√	•
35	Track2 Data	Not present when MDES validates the chip or dynamic CVC 3 data on behalf of the issuer. NOTE: Issuers may opt in through MDES Manager to receive track data in DE 35 (Track 2 Data).	•	•
39	Response Code	For the full list of codes, refer to the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> or the <i>Mastercard Network Processing Single Message System Guide</i> .	•	√

DE	Name/ Subelement	Comments/Subfield	Req	Res
45	Track1 Data	Not present when MDES validates the chip or dynamic CVC 3 data on behalf of the issuer. NOTE: Issuers may opt in through MDES Manager to receive track data in DE 45 (Track 2 Data).	•	•
48	Additional Data - Private Use			
	Transaction Category Code	R = Face-to-face retail (applicable to MCCs 6538 and 6540) T = Non-face-to-face retail (applicable to MCCs 6538 and 6540) U = Unique Transaction - Quasi Cash Disbursement (applicable to MCC 4829)	√	√
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: If applicable NOTE: New Wallet Identifier values may be provided in this field without prior written notice.	√	•
	27 - Transaction Analysis As applicable to MCBP 2.0	Not present when the transaction comes from an alternative network: • Subfield 1 - Overview • Subfield 2 - Test Results	√	•
	30 - Token Transaction Identifier	If applicable	√	•
	32 - Mastercard Assigned ID	When applicable	√	•
	33 - PAN Mapping File Information	Conditional in issuer responses. Issuers should provide the entire subelement 33 on transit transactions when using third-party tokenization services. Subfield 1 - Account Number Indicator: • C = MDES Secure Element Token • F = MDES Static Token • H = MDES Cloud-based Payments Token Subfield 2 - Account Number: Token Subfield 3 - Expiration Date: Token expiration date Subfield 5 - Token Assurance Method: Token assurance method value (as applicable)	√	√

DE	Name/ Subelement	Comments/Subfield	Req	Res
		Subfield 6 - Token Requestor ID: Token Requestor ID (as applicable)		
		Subfield 8 - Storage Technology: Storage technology (as applicable)		
	34 - ATC Information Conditional	Not present for decline advice messages	√	•
	71 - On-behalf Services	MDES validation results, which are the relevant combinations of the following subfields: NOTE: This data element may include different results when there are validation failures but the issuer has chosen for Mastercard to continue processing; see Cryptographic Pre-Validation Failures. Subfield 1: Value 50 = MDES PAN Mapping Subfield 2: Value C = Conversion of Token to PAN completed successfully Subfield 3: Mastercard use only with a space or value Subfield 1: Value 33 = MoneySend and Funding Blocking Service Subfield 2: All values applicable for MoneySend payments Subfield 3: Mastercard use only with a space or value	√	•
	77 -Transaction Type Identifier	• C07 = General Person-to-Person • C52 = General Transfer to Own Account • C53 = Agent Cash Out • C54 = Payment of Own Credit Card Bill • C55 = Business Disbursement • C56 = Government/Non-profit Disbursement • C57 = Rapid Merchant Settlement • C65 = General Business to Business Transfer • F07 = General Person-to-Person Transfer • F08 • F52 = General Transfer to Own Account • F53 = Agent Cash Out • F54 = Payment of Own Credit Card Bill • F55 = Business Disbursement • F61 = Transfer to Own Staged Digital Wallet Account • F64 = Transfer to Own Debit or Prepaid Account • F65 = General Business-to-Business Transfer	√	•

DE	Name/ Subelement	Comments/Subfield	Req	Res
55	Integrated Circuit Card (ICC) System-Related Data	Not present when MDES validates the chip or dynamic CVC 3 data on behalf of the issuer. NOTE: Issuers may opt in through MDES Manager to receive DE 55 data.	•	•
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR): - Authorization Request/0100 and Financial Transaction Request/0200 messages: Mastercard provides the PAR when Mastercard is the BIN Controller, and the AMS stores a PAR for the PAN. - Responses: If Mastercard is not the BIN Controller, the issuer may include DE 56 (Payment Account Data) in the response message. If Mastercard is the BIN Controller, the Dual Message System (Authorization) or Single Message System will ignore DE 56 received from the issuer and will set the PAR value retrieved from AMS in DE 56.	√	√
61	POS Data	Subfield 04 - POS Cardholder Presence: Value 0 = Cardholder present	√	√
		Subfield 05 - POS Card Presence: Value 0 = Card present	√	√
		Subfield 10 - Cardholder-Activated Terminal Level: As applicable	√	√
105	Multi-Use Transaction Identification Data	If applicable	√	•
108	Additional Transaction Reference Data	As sent by acquirer	√	√
119	MDES Service Indicators, subelement 003: MDES Service Indicators for issuers	If applicable. Refer to the MDES service indicators for more details. NOTE: Issuers can opt-in to receive this element through MDES Manager.	√	•

Token mapping failures, Suspended/Deactivated tokens

Mastercard might send decline advice messages to issuers when MDES cannot successfully complete token to PAN mapping or when Mastercard declines the transaction on behalf of the issuer (due to a cryptographic validation failure). Decline advice messages are sent as either:

- Authorization Advice/0120 - All Dual Message System (Authorization) issuers will receive the declined advice messages.
- Financial Transaction Advice/0220 - Only those Single Message System issuers that have opted to receive declined advice messages will be sent online financial transaction advice messages. However, all Single Message System issuers will receive declined advice in the Non-Financial Record (NREC) of the 250-byte batch data file.

250-byte batch data file

When transactions are processed by MDES, Single Message System issuers receive the following in the 250-byte batch data file:

- The Financial (FREC)/Non-Financial (NREC) and Financial/Non-Financial Exception (EREC) records will contain the cardholder's PAN in the PAN field (positions 33-51).
- The Financial/Non-Financial Additional Data Addendum (ADDR) will contain the token in the Mapped PAN Account field (positions 120-138).
- The Financial/Non-Financial Additional Data Addendum (ADDR) will contain value C (the PAN in the Mapped PAN Account field contains the MDES token) in the Mapped PAN Indicator field (position 139).

Dual Message System (Clearing) - Contactless MoneySend Funding Transactions

Issuers will receive the following data elements and values in First and Second Presentment/1240 messages when MDES has processed a contactless transaction successfully.

Table 59: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's PAN
3	Processing Code	Value 00 = Funding
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: As applicable
23	Card Sequence Number	As applicable
26	Card Acceptor Business Code (MCC)	MCC 6538 (MoneySend Funding) or MCC 6540 (Funding Transaction) or MCC 4829 (Money Transfer)
55	Integrated Circuit Card System-Related Data	If provided by the acquirer NOTE: Even though DE 55 may not have been passed in Authorization, DE 55 may be passed in Clearing to the issuer if it is present and has been provided.

DE/PDS	Name/Subelement	Comments/Subfield
105	Multi-Use Transaction Identification Data	Not applicable
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: One of the following values: • C (value C followed by a space) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • CC (value CC) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. • H (value H followed by a space) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • HC (value HC) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. Subfield 2 - Account Number: Token
PDS 0023	Terminal Type	As applicable
PDS 0043	Transaction Type Identifier	As applicable for use cases. Refer to the definitions in the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> and the <i>Mastercard Network Processing Dual Message Authorization System Guide</i>.
PDS 0058	Token Assurance Method	Optional
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	If applicable
PDS 0207	Wallet Identifier	If applicable
PDS 0765	MoneySend Receiver Data	If applicable

Dual Message System (Authorization) and Single Message System - Contactless MoneySend Payment Transactions

This section describes the messages that issuers must be prepared to receive when an acquirer submits a contactless MoneySend funding transaction.

The *Data elements and values for MoneySend transactions* table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Authorization Advice/0120
- Authorization Advice Response/0130
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210
- Financial Transaction Advice/0220
- Financial Transaction Advice Response/0230
- Reversal Request/0400
- Reversal Advice/0420

All mandatory Authorization Request/0100 and Financial Transaction Request/0200 data elements apply.

Table 60: Data elements and values for MoneySend transactions

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's PAN	√	√
3	Processing Code	Value 28 = Payment	√	√
14	Date, Expiration	PAN expiration date	√	•
18	Merchant Type	MCC 6536 (MoneySend Intracountry) MCC 6537 (MoneySend Intercountry) Refer to the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> for details.	√	•
22	POS Entry Mode	Value 07 = PAN auto-entry through Contactless M/Chip or Value 90 = PAN auto-entry through magnetic stripe	√	•
23	Card Sequence Number	As applicable	√	•

DE	Name/Subelement	Comments/Subfield	Req	Res
35	Track2 Data	Not present when MDES validates the chip or dynamic CVC 3 data on behalf of the issuer.	•	•
NOTE: Issuers may opt in through MDES Manager to receive track data in DE 35 (Track 2 Data).				

DE	Name/Subelement	Comments/Subfield	Req	Res
45	Track1 Data	Not present when MDES validates the chip or dynamic CVC 3 data on behalf of the issuer.	•	•
NOTE: Issuers may opt in through MDES Manager to receive track data in DE 45 (Track 1 Data).				
48	Additional Data - Private Use			
	Transaction Category Code (TCC)	P = Payment Transaction	√	√
	23 - Payment Initiation Channel	Subfield 1 - Device Type: If applicable	√	•
	26 - Wallet Program Data	Subfield 1 - Wallet Identifier: If applicable	√	•
NOTE: New Wallet Identifier values may be provided in this field without prior written notice.				
	27 - Transaction Analysis As applicable to MCBP 2.0	Not present when the transaction comes from an alternative network: • Subfield 1 - Overview • Subfield 2 - Test Results	√	•
	30 - Token Transaction Identifier	If applicable	√	•
	32 - Mastercard Assigned ID	When applicable	√	•
	33 - PAN Mapping File Information	Conditional in issuer responses. Issuers should provide the entire subelement 33 on transit transactions when using third-party tokenization services.	√	√
		Subfield 1 - Account Number Indicator: • C = MDES Secure Element Token • F = MDES Static Token • H = MDES Cloud-based Payments Token		
		Subfield 2 - Account Number: Token		
		Subfield 3 - Expiration Date: Token expiration date		
		Subfield 5 - Token Assurance Method: Token assurance method value (as applicable)		
		Subfield 6 - Token Requestor ID: Token Requestor ID (as applicable)		

DE	Name/Subelement	Comments/Subfield	Req	Res
		Subfield 8 - Storage Technology: Storage technology (as applicable)		
	34 - ATC Information Conditional	Not present for decline advice messages	√	•
	71 - On-behalf Services	MDES validation results, which are the relevant combinations of the following subfields: NOTE: This data element may include different results when there are validation failures but the issuer has chosen for Mastercard to continue processing; see Cryptographic Pre-Validation Failures. Subfield 1: Value 50 = MDES PAN Mapping Subfield 2: Value C = Conversion of Token to PAN completed successfully Subfield 3: Mastercard use only with a space or value Subfield 1: Value 33 = MoneySend and Funding Blocking Service Subfield 2: All values applicable for MoneySend payments Subfield 3: Mastercard use only with a space or value	√	•
	77 - Transaction Type Identifier	• C07 = MoneySend Person-to-Person • C52 = MoneySend Account-to-Account Transfers • C53 = MoneySend Agent Cash Out • C54 = MoneySend Credit Card Bill Payment • C55 = MoneySend Business Disbursement • C56 = MoneySend Government/ Nonprofit Disbursement • C57 = MoneySend Acquirer Merchant Settlement • C58 = MoneySend Cash2ATM • C59 = MoneySend Cash2Card • C65 = MoneySend Business to Business Transfer	√	√
55	Integrated Circuit Card (ICC) System-Related Data	Not present when MDES validates the chip or dynamic CVC 3 data on behalf of the issuer. NOTE: Issuers may opt in through MDES Manager to receive DE 55 data.	•	•

DE	Name/Subelement	Comments/Subfield	Req	Res
56	Payment Account Data, subelement 01 - Payment Account Data	Subfield 01 - Payment Account Reference (PAR): • Authorization Request/0100 and Financial Transaction Request/0200 messages: Mastercard provides the PAR when Mastercard is the BIN Controller, and the AMS stores a PAR for the PAN. • Responses: If Mastercard is not the BIN Controller, the issuer may include DE 56 (Payment Account Data) in the response message. If Mastercard is the BIN Controller, the Dual Message System (Authorization) or Single Message System will ignore DE 56 received from the issuer and will set the PAR value retrieved from AMS in DE 56.	√	√
61	POS Data	Subfield 04 - POS Cardholder Presence: Value 0 = Cardholder present	√	√
		Subfield 05 - POS Card Presence: Value 0 = Card present	√	√
		Subfield 10 - Cardholder-Activated Terminal Level: As applicable	√	√
105	Multi-Use Transaction Identification Data	If applicable	√	•
108	Additional Transaction Reference Data	As sent by acquirer	√	√
119	MDES Service Indicators, subelement 003: MDES Service Indicators for issuers	If applicable. Refer to the MDES service indicators for more details. NOTE: Issuers can opt-in to receive this element through MDES Manager.	√	•

Token mapping failures, Suspended/Deactivated tokens

Mastercard might send decline advice messages to issuers when MDES cannot successfully complete token to PAN mapping or when Mastercard declines the transaction on behalf of the issuer (due to a cryptographic validation failure). Decline advice messages are sent as either:

- Authorization Advice/0120 - All Dual Message System (Authorization) issuers will receive the declined advice messages.
- Financial Transaction Advice/0220 - Only those Single Message System issuers that have opted to receive declined advice messages will be sent online financial transaction advice

messages. However, all Single Message System issuers will receive declined advice in the Non-Financial Record (NREC) of the 250-byte batch data file.

250-byte batch data file

When transactions are processed by MDES, Single Message System issuers receive the following in the 250-byte batch data file:

- The Financial (FREC)/Non-Financial (NREC) and Financial/Non-Financial Exception (EREC) records will contain the cardholder's PAN in the PAN field (positions 33-51).
- The Financial/Non-Financial Additional Data Addendum (ADDR) will contain the token in the Mapped PAN Account field (positions 120-138).
- The Financial/Non-Financial Additional Data Addendum (ADDR) will contain value C (the PAN in the Mapped PAN Account field contains the MDES token) in the Mapped PAN Indicator field (position 139).

Dual Message System (Clearing) - Contactless MoneySend Payment Transactions

Issuers will receive the following data elements and values in First and Second Presentment/1240 messages when MDES has processed a contactless transaction successfully.

Table 61: First and Second Presentment/1240 messages

DE/PDS	Name/Subelement	Comments/Subfield
2	PAN	Cardholder's PAN
3	Processing Code	Value 28 = Payment
22	Point of Service Data Code	Subfield 7 - Card Data - Input Mode: As applicable
23	Card Sequence Number	As applicable
26	Card Acceptor Business Code (MCC)	MCC 6536 (MoneySend Intracountry) MCC 6537 (MoneySend Intercountry)
55	Integrated Circuit Card System-Related Data	If provided by the acquirer NOTE: Even though DE 55 may not have been passed in Authorization, DE 55 may be passed in Clearing to the issuer if it is present and has been provided.
105	Multi-Use Transaction Identification Data	If applicable

DE/PDS	Name/Subelement	Comments/Subfield
PDS 0001	Mastercard Mapping Service Account Number	Subfield 1 - Account Number Type: One of the following values: • C (value C followed by a space) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • CC (value CC) = Subfield 2 contains the MDES secure element device token used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. • H (value H followed by a space) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on the authorization log. • HC (value HC) = Subfield 2 contains the MDES Cloud-Based Payments token device account number used at the point of sale. The account range is owned by an issuer. Mapping was based on a source other than the authorization log. Subfield 2 - Account Number: Token
PDS 0023	Terminal Type	As applicable
PDS 0043	Transaction Type Identifier	As applicable for use cases. Refer to the definitions in the <i>Mastercard MoneySend and Funding Transactions Program Standards</i> and the <i>Mastercard Network Processing Dual Message Authorization System Guide</i>.
PDS 0058	Token Assurance Method	Optional
PDS 0059	Token Requestor ID	If applicable
PDS 0198	Device Type	If applicable
PDS 0207	Wallet Identifier	If applicable
PDS 0765	MoneySend Receiver Data	As applicable

Purchase mail order/telephone order transactions

This section describes the data that issuers receive and send when MDES has processed a tokenized purchase mail order/telephone order (MO/TO) transaction.

The data elements (DE) for Tokenized Purchase MO/TO transactions table shows the data elements and values that can be present for these message types:

- Authorization Request/0100
- Authorization Request Response/0110
- Financial Transaction Request/0200
- Financial Transaction Request Response/0210

All mandatory data elements apply, even if not specifically shown in the table.

Table 62: DE for Tokenized Purchase MO/TO transactions

DE	Name/Subelement	Comments/Subfield	Req	Res
2	PAN	Cardholder's token	✓	✓
3	Processing Code	Subfield 1 - Cardholder Transaction Type Code: 00 = Purchase	✓	✓
4	Amount, Transaction	The amount of funds the cardholder requested in the local currency of the acquirer or source location of the transaction.	✓	✓
14	Date, Expiration	Token expiration date	✓	

DE	Name/Subelement	Comments/Subfield	Req	Res
22	POS Entry Mode	Subfield 1 - POS Terminal PAN Entry Mode: 10 = Credential on File. 81 = PAN/Token entry via electronic commerce with optional SecureCode-AAV or DSRP cryptogram in UCAF.	✓	
48	Additional Data - Private Use			
	33 - PAN Mapping File Information	Subfield 1 - Account Number Indicator: - C = MDES Secure Element Token - F = MDES Static Token - H = MDES Cloud-based Payments Token	✓	✓
		Subfield 2 – Account Number: Token	✓	✓
		Subfield 3 – Expiration Date: Token expiration date	✓	✓
		Subfield 5 – Token Assurance Method (as applicable)	✓	✓
		Subfield 6 – Token Requestor ID (as applicable)	✓	✓
		Subfield 8 – Storage Technology (as applicable)	✓	✓
	42 - Electronic Commerce Indicators	Subfield 1 - Electronic Commerce Security Level Indicator and UCAF Collection Indicator. CITs with Static token: 210 = Present in unauthenticated transactions, or in transactions where an Identity Check merchant has chosen not to undertake Identity Check, or where Identity Check failed authentication and the merchant desires to proceed with the transaction. Also present in transactions with a DTVC. CITs with Secure Element or Cloud token: 246 = Tokenized transaction with DSRP cryptogram. MITs: 247 = Tokenized Merchant Initiated Transaction. NOTE: For more information about the SLI values, see the <i>Mastercard Network Processing Dual Message Authorization System Guide</i> Data Element 48 (Additional Data), subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator).	✓	✓

DE	Name/Subelement	Comments/Subfield	Req	Res
61	Point of Service Data	Subfield 4 - POS Cardholder Presence: CITs: • 2 = Cardholder not present (mail/facsimile order) • 3 = Cardholder not present (phone or Automated Response Unit [ARU]) MITs: • 4 = Standing order/recurring transactions • 5 = Cardholder not present (Electronic order [home PC, Internet, mobile phone, PDA])	✓	
		Subfield 7 - POS Transaction Status: 0 = Normal request (original presentment)	✓	
105	Multi-Use Transaction Identification Data	If applicable	✓	•
119	MDES Service Indicators, subelement 003: MDES Service Indicators for issuers	If applicable. Refer to the MDES service indicators for more details. NOTE: Issuers can opt-in to receive this element through MDES Manager.	✓	•

Appendix A Guidance for acquirers on typical e-Commerce purchase submissions

Table 63: Legend

*	Mandatory for purchase e-Commerce consumer initiated transactions DSRP cryptograms are optional for e-Commerce MITs, partial shipments, ASIs, incremental pre-authorizations, and refunds.
**	This guidance assumes no 3DS authentication; if 3DS is used, corresponding SLIs apply.

Table 64: e-Commerce purchase submissions: guidance for acquirers - Use case set 1

Transaction identifiers	Data Element	Populated by	Click to Pay (Cloud Token)	Click to Pay (Static Token)	Click to Pay (Cloud Token) & Passkey	Tokenized Card on File (SCOF)	Tokenized Card on File (SCOF) & Passkey	Tokenized Merchant Initiated Transactions (MIT)
PAN	DE 2	Acquirer	Token	Token	Token	Token	Token	Token
Date, Expiration	DE 14	Acquirer	Token Expiration Date	Token Expiration Date	Token Expiration Date	Token Expiration Date	Token Expiration Date	Token Expiration Date
POS Entry Method	DE 22, SF 1	Acquirer	81	81	81	10	10	10, 81
CIT/MIT indicator	DE 48, SE 22, SF 5	Acquirer	As needed	As needed	As needed	Mandatory	Mandatory	As needed

Transaction identifiers	Data Element	Populated by	Click to Pay (Cloud Token)	Click to Pay (Static Token)	Click to Pay (Cloud Token) & Passkey	Tokenized Card on File (SCOF)	Tokenized Card on File (SCOF) & Passkey	Tokenized Merchant Initiated Transactions (MIT)
Electronic Commerce Security Level Indicator**	DE 48, SE 42, SF 1	Acquirer	246	210	242	246	242	247
Digital Commerce Domain Identifier	DE 48, SE 48, SF 001	Acquirer/Mastercard	1	1	1	As needed	As needed	N/A
Digital Commerce Program Type	DE 48, SE 48, SF 002	Acquirer/Mastercard	2	2	2	As needed	As needed	N/A
Digital Commerce Check-In Entity Type and Name	DE 48, SE 48, SF 003	Acquirer/Mastercard	20	20	20	As needed	As needed	N/A
Digital Commerce Check-In Method	DE 48, SE 48, SF 004	Acquirer/Mastercard	N/A	N/A	N/A	As needed	As needed	N/A
CVC 2	DE 48, SE 92	Acquirer	Required	Optional	Required	Required	Required	Not required
POS Terminal Attendance	DE 61, SF 1	Acquirer	1	1	1	1	1	2
POS Terminal Location	DE 61, SF 3	Acquirer	2	2	2	2	2	3

Transaction identifiers	Data Element	Populated by	Click to Pay (Cloud Token)	Click to Pay (Static Token)	Click to Pay (Cloud Token) & Passkey	Tokenized Card on File (SCOF)	Tokenized Card on File (SCOF) & Passkey	Tokenized Merchant Initiated Transactions (MIT)
POS Cardholder Presence	DE 61, SF 4	Acquirer	5	5	5	5	5	4
POS Card Presence	DE 61, SF 5	Acquirer	1	1	1	1	1	1
Cardholder-Activated Terminal Level	DE 61, SF 10	Acquirer	6	6	6	6	6	0
Digital Payment Cryptogram	DE 104, SE 001	Acquirer	Conditional*	N/A	Conditional*	Conditional*	Conditional*	Optional
Wallet Identifier	DE 48, SE 26, SF 1	Mastercard	327	327	327	327	327	327
Account Number Indicator	DE 48, SE 33, SF 1	Mastercard	H	F	H	H	H	H
Token Requestor ID	DE 48, SE 33, SF 6	Mastercard	50123197928	50127530524	50123197928	Merchant specific SCOF	Merchant specific SCOF	Merchant specific SCOF
Storage Technology	DE 48, SE 33, SF 8	Mastercard	3	3	3	3	3	3

Table 65: e-Commerce purchase submissions: guidance for acquirers - Use case set 2

Transaction identifiers	Data Element	Populated by	Apple Pay eCommerce Checkout Button	Apple Pay Multi Merchant of Record (MMOR)	Apple Pay Recurring Cloud Token	Agentic Payment with Cloud Token & DSRP	Agentic Payment with Cloud Token & DTVC	MVCT (Static Token)
PAN	DE 2	Acquirer	Token	Token	Token	Token	Token	Token
Date, Expiration	DE 14	Acquirer	Token Expiration Date	Token Expiration Date	Token Expiration Date	Token Expiration Date	Token Expiration Date	Token Expiration Date
POS Entry Method	DE 22, SF 1	Acquirer	10, 81	81	10, 81	10, 81	10, 81	10, 81, 82
CIT/MIT indicator	DE 48, SE 22, SF 5	Acquirer	As needed	As needed	Mandatory	As needed	As needed	As needed
Electronic Commerce Security Level Indicator**	DE 48, SE 42, SF 1	Acquirer	242	242	242/247	242/246	210	210/247
Digital Commerce Domain Identifier	DE 48, SE 48, SF 001	Acquirer/Mastercard	N/A	N/A	N/A	1	1	N/A
Digital Commerce Program Type	DE 48, SE 48, SF 002	Acquirer/Mastercard	N/A	N/A	N/A	8	8	N/A

Transaction identifiers	Data Element	Populated by	Apple Pay eCommerce Checkout Button	Apple Pay Multi Merchant of Record (MMOR)	Apple Pay Recurring Cloud Token	Agentic Payment with Cloud Token & DSRP	Agentic Payment with Cloud Token & DTVC	MVCT (Static Token)
Digital Commerce Check-In Entity Type and Name	DE 48, SE 48, SF 003	Acquirer/Mastercard	N/A	N/A	N/A	80	80	N/A
Digital Commerce Check-In Method	DE 48, SE 48, SF 004	Acquirer/Mastercard	N/A	N/A	N/A	Order intent ID	Order intent ID	N/A
CVC 2	DE 48, SE 92	Acquirer	Not required	Not required		Optional	Required	Optional
POS Terminal Attendance	DE 61, SF 1	Acquirer	1	1	1	1	1	1
POS Terminal Location	DE 61, SF 3	Acquirer	2	2	2	2	2	2
POS Cardholder Presence	DE 61, SF 4	Acquirer	5	5	5	5	5	5
POS Card Presence	DE 61, SF 5	Acquirer	1	1	1	1	1	1
Cardholder-Activated Terminal Level	DE 61, SF 10	Acquirer	6	6	6	6	6	6
Digital Payment Cryptogram	DE 104, SE 001	Acquirer	Conditional*	Conditional*	Conditional*	Conditional*	N/A	N/A

Transaction identifiers	Data Element	Populated by	Apple Pay eCommerce Checkout Button	Apple Pay Multi Merchant of Record (MMOR)	Apple Pay Recurring Cloud Token	Agentic Payment with Cloud Token & DSRP	Agentic Payment with Cloud Token & DTVC	MVCT (Static Token)
Wallet Identifier	DE 48, SE 26, SF 1	Mastercard	103	327	327	327	327	327
Account Number Indicator	DE 48, SE 33, SF 1	Mastercard	H	H	H	H	H	F
Token Requestor ID	DE 48, SE 33, SF 6	Mastercard	50110030273	50121755512	Merchant specific	Agent specific	Agent specific	Program specific
Storage Technology	DE 48, SE 33, SF 8	Mastercard	not equal to 3	3	3	3	3	3

Appendix B Guidance for acquirers on typical e-Commerce MITs and CITs transactions

This appendix contains the transaction coding matrix which indicates typical scenarios for processing MDES e-commerce transactions, with the POS Entry Mode of 81 or 10, giving guidance on Mastercard's expectations for valid message values in a subset of relevant key data elements.

This guidance is not exhaustive and other use cases may exist.

Industry Practice MIT does not consider incremental authorizations, so they do not require a CIT or MIT indicator, unless initiated from a credential stored on file.

DSRP cryptograms are optional for MITs, but if provided they must be unique for each transaction request. Mastercard will decline transactions that have replayed DSRP cryptograms

Merchants that tokenize a funding PAN stored on file in the middle of a series of recurring payments, should submit the first tokenized transaction with a DSRP cryptogram and SLI 246 to re-establish the recurring payment series with the Token. In SCA countries, merchants should use the Trace ID from the original CIT performed with the funding PAN for subsequent MITs, rather than the one from the first tokenized transaction.

* Coded with a value of 4 if Trace ID is present, value of 5 if Trace ID is not present

To comply with regulatory requirements in India, the merchants Payment Aggregator or Payment Gateway (domestic or international) supporting Recurring Payments for cards issued in India need to send Recurring Payment Information details, for example, Mandate ID, in DE 48, subelement 41, subfield 1. The Merchants Payment Aggregator or Payment Gateway (domestic or international) will need to connect directly with any one of the prevalent Mandate management systems in India, such as SI hub, Mandate HQ, and ZionHub.

Table 66: Cardholder Initiated Transactions (All Markets except India)

Use case	Description	CIT/MIT Indicator DE 48, SE 22, SF 5	3DS AAV DE 48, SE 43	CVC 2 DE 48, SE 92	DSRP Crypto DE 104, SE 1	Trace ID DE 48, SF 63	Cardholder Presence DE 61, SF 4	eCom SLI DE 48, SE 42	Additional Recurring Payments Info DE 48, SE 41, SF 1
1	Device wallet original transaction with no 3DS AAV, with DSRP crypto	Appropriate CIT indicator	No	Optional	Yes	No	5 or 4 (for 1st recurring)	242	Applicable if 1st recurring transaction for India-issued cards
2	Device wallet original transaction with 3DS AAV, with DSRP crypto	Appropriate CIT indicator	Yes	Optional	Yes	No	5 or 4 (for 1st recurring)	Appropriate 3DS SLI (e.g. 211, 212, 216)	Applicable if 1st recurring transaction for India-issued cards
3	Remote Commerce DSRP original transaction with no 3DS AAV, with DSRP crypto	Appropriate CIT indicator	No	Optional	Yes	No	5 or 4 (for 1st recurring)	242 or 246	Applicable if 1st recurring transaction for India-issued cards

Use case	Description	CIT/MIT Indicator DE 48, SE 22, SF 5	3DS AAV DE 48, SE 43	CVC 2 DE 48, SE 92	DSRP Crypto DE 104, SE 1	Trace ID DE 48, SF 63	Cardholder Presence DE 61, SF 4	eCom SLI DE 48, SE 42	Additional Recurring Payments Info DE 48, SE 41, SF 1
4	Remote Commerce DSRP original transaction with 3DS AAV, with DSRP crypto	Appropriate CIT indicator	Yes	Optional	Yes	No	5 or 4 (for 1st recurring)	Appropriate 3DS SLI (e.g. 211, 212, 216)	Applicable if 1st recurring transaction for India-issued cards
5	Remote Commerce static token original transaction with no 3DS AAV	Appropriate CIT indicator	No	Optional	No	No	5 or 4 (for 1st recurring)	210	Applicable if 1st recurring transaction for India-issued cards
6	Remote Commerce static token original transaction with 3DS AAV	Appropriate CIT indicator	Yes	Optional	No	No	5 or 4 (for 1st recurring)	Appropriate 3DS SLI (e.g. 211, 212, 216)	Applicable if 1st recurring transaction for India-issued cards

Table 67: Merchant Initiated Transactions (Non-SCA Countries except India)

Use case	Description	CIT/MIT Indicator DE 48, SE 22, SF 5	3DS AAV DE 48, SE 43	CVC 2 DE 48, SE 92	DSRP Crypto DE 104, SE 1	Trace ID DE 48, SF 63	Cardholder Presence DE 61, SF 4	eCom SLI DE 48, SE 42	Additional Recurring Payments Info DE 48, SE 41, SF 1
7	Unscheduled Credential On File (Ad hoc) with no 3DS AAV	M101	No	No	Optional	No	4	247	Yes, only cards issued in India & only for auto top-up
8	Unscheduled Credential On File (Ad hoc) with 3DS AAV	M101	Yes	No	Optional	No	4	Appropriate 3DS SLI (e.g. 214, 217)	Yes, only cards issued in India & only for auto top-up
9	Recurring payment transaction with no 3DS AAV	M102, M103	No	No	Optional	No	4	247	Yes, only cards issued in India
10	Recurring payment transaction with 3DS AAV	M102, M103	Yes	No	Optional	No	4	Appropriate 3DS SLI (e.g. 214, 217)	Yes, only cards issued in India
11	Installment transaction with no 3DS AAV	M104	No	No	Optional	No	5	247	Yes, only cards issued in India

Use case	Description	CIT/MIT Indicator DE 48, SE 22, SF 5	3DS AAV DE 48, SE 43	CVC 2 DE 48, SE 92	DSRP Crypto DE 104, SE 1	Trace ID DE 48, SF 63	Cardholder Presence DE 61, SF 4	eCom SLI DE 48, SE 42	Additional Recurring Payments Info DE 48, SE 41, SF 1
12	Installment transaction with 3DS AAV	M104	Yes	No	Optional	No	5	Appropriate 3DS SLI (e.g. 214, 217)	Yes, only cards issued in India
13	Partial shipment transaction with no 3DS AAV	M205	No	No	Optional	No	5	247	N/A
14	Partial shipment transaction with 3DS AAV	M205	Yes	No	Optional	No	5	Appropriate 3DS SLI (e.g. 214, 217)	N/A
15	Related/delayed charge transaction with no 3DS AAV	M206	No	No	Optional	No	5	247	N/A
16	Related/delayed charge transaction with 3DS AAV	M206	Yes	No	Optional	No	5	Appropriate 3DS SLI (e.g. 214, 217)	N/A

Use case	Description	CIT/MIT Indicator DE 48, SE 22, SF 5	3DS AAV DE 48, SE 43	CVC 2 DE 48, SE 92	DSRP Crypto DE 104, SE 1	Trace ID DE 48, SF 63	Cardholder Presence DE 61, SF 4	eCom SLI DE 48, SE 42	Additional Recurring Payments Info DE 48, SE 41, SF 1
17	No show charge transaction with no 3DS AAV	M207	No	No	Optional	No	5	247	N/A
18	No show charge transaction with 3DS AAV	M207	Yes	No	Optional	No	5	Appropriate 3DS SLI (e.g. 214, 217)	N/A
19	Resubmission transaction with no 3DS AAV	M208	No	No	Optional	No	5	247	N/A
20	Resubmission transaction with 3DS AAV	M208	Yes	No	Optional	No	5	Appropriate 3DS SLI (e.g. 214, 217)	N/A

Table 68: Merchant Initiated Transactions (SCA Countries)

Use case	Description	CIT/MIT Indicator DE 48, SE 22, SF 5	3DS AAV DE 48, SE 43	CVC 2 DE 48, SE 92	DSRP Crypto DE 104, SE 1	Trace ID DE 48, SF 63	Cardholder Presence DE 61, SF 4	eCom SLI DE 48, SE 42	Additional Recurring Payments Info DE 48, SE 41, SF 1
7EEA	Unscheduled Credential On File (Ad hoc) with no 3DS AAV	M101	No	No	Optional	Yes	4	247	Yes, only cards issued in India & only for auto top-up
8EEA	Unscheduled Credential On File (Ad hoc) with 3DS AAV	M101	Yes	No	Optional	Yes	4	Appropriate 3DS SLI (e.g. 214, 217)	Yes, only cards issued in India & only for auto top-up
9EEA	Recurring payment transaction with no 3DS AAV	M102, M103	No	No	Optional	Yes	4	247	Yes, only cards issued in India
10EE A	Recurring payment transaction with 3DS AAV	M102, M103	Yes	No	Optional	Yes	4	Appropriate 3DS SLI (e.g. 214, 217)	Yes, only cards issued in India
11EE A	Installment transaction with no 3DS AAV	M104	No	No	Optional	Yes	4	247	Yes, only cards issued in India

Use case	Description	CIT/MIT Indicator DE 48, SE 22, SF 5	3DS AAV DE 48, SE 43	CVC 2 DE 48, SE 92	DSRP Crypto DE 104, SE 1	Trace ID DE 48, SF 63	Cardholder Presence DE 61, SF 4	eCom SLI DE 48, SE 42	Additional Recurring Payments Info DE 48, SE 41, SF 1
12EE A	Installment transaction with 3DS AAV	M104	Yes	No	Optional	Yes	4	Appropriate 3DS SLI (e.g. 214, 217)	Yes, only cards issued in India
13EE A	Partial shipment transaction with no 3DS AAV	M205	No	No	Optional	Yes	4	247	N/A
14EE A	Partial shipment transaction with 3DS AAV	M205	Yes	No	Optional	Yes	4	Appropriate 3DS SLI (e.g. 214, 217)	N/A
15EE A	Related/delayed charge transaction with no 3DS AAV	M206	No	No	Optional	Yes	4	247	N/A
16EE A	Related/delayed charge transaction with 3DS AAV	M206	Yes	No	Optional	Yes	4	Appropriate 3DS SLI (e.g. 214, 217)	N/A

Use case	Description	CIT/MIT Indicator DE 48, SE 22, SF 5	3DS AAV DE 48, SE 43	CVC 2 DE 48, SE 92	DSRP Crypto DE 104, SE 1	Trace ID DE 48, SF 63	Cardholder Presence DE 61, SF 4	eCom SLI DE 48, SE 42	Additional Recurring Payments Info DE 48, SE 41, SF 1
17EE A	No show charge transaction with no 3DS AAV	M207	No	No	Optional	Yes	4	247	N/A
18EE A	No show charge transaction with 3DS AAV	M207	Yes	No	Optional	Yes	4	Appropriate 3DS SLI (e.g. 214, 217)	N/A
19EE A	Resubmission transaction with no 3DS AAV	M208	No	No	Optional	If applicable	4 or 5*	247	N/A
20EE A	Resubmission transaction with 3DS AAV	M208	Yes	No	Optional	If applicable	4 or 5*	Appropriate 3DS SLI (e.g. 214, 217)	N/A

Table 69: Cardholder Initiated Transactions (India)

Use case	Description	CIT/MIT Indicator DE 48, SE 22, SF 5	3DS AAV DE 48, SE 43	CVC 2 DE 48, SE 92	DSRP Crypto DE 104, SE 1	Trace ID DE 48, SF 63	Cardholder Presence DE 61, SF 4	eCom SLI DE 48, SE 42	Additional Recurring Payments Info DE 48, SE 41, SF 1
1IND	Device wallet original transaction with no 3DS AAV, with DSRP crypto	Appropriate CIT indicator	No	Optional	Yes	No	5 or 4 (for 1st recurring)	242	Yes if 1st recurring transaction
2IND	Device wallet original transaction with 3DS AAV, with DSRP crypto	Appropriate CIT indicator	Yes	Optional	Yes	No	5 or 4 (for 1st recurring)	Appropriate 3DS SLI (e.g. 211, 212, 216)	Yes if 1st recurring transaction
3IND	Remote Commerce DSRP original transaction with no 3DS AAV, with DSRP crypto	Appropriate CIT indicator	No	Optional	Yes	No	5 or 4 (for 1st recurring)	242 or 246	Yes if 1st recurring transaction

Use case	Description	CIT/MIT Indicator DE 48, SE 22, SF 5	3DS AAV DE 48, SE 43	CVC 2 DE 48, SE 92	DSRP Crypto DE 104, SE 1	Trace ID DE 48, SF 63	Cardholder Presence DE 61, SF 4	eCom SLI DE 48, SE 42	Additional Recurring Payments Info DE 48, SE 41, SF 1
4IND	Remote Commerce DSRP original transaction with 3DS AAV, with DSRP crypto	Appropriate CIT indicator	Yes	Optional	Yes	No	5 or 4 (for 1st recurring)	Appropriate 3DS SLI (e.g. 211, 212, 216)	Yes if 1st recurring transaction
5IND	Remote Commerce static token original transaction with no 3DS AAV	Appropriate CIT indicator	No	Optional	No	No	5 or 4 (for 1st recurring)	210	Yes if 1st recurring transaction
6IND	Remote Commerce static token original transaction with 3DS AAV	Appropriate CIT indicator	Yes	Optional	No	No	5 or 4 (for 1st recurring)	Appropriate 3DS SLI (e.g. 211, 212, 216)	Yes if 1st recurring transaction

Table 70: Merchant Initiated Transactions (India)

Use case	Description	CIT/MIT Indicator DE 48, SE 22, SF 5	3DS AAV DE 48, SE 43	CVC 2 DE 48, SE 92	DSRP Crypto DE 104, SE 1	Trace ID DE 48, SF 63	Cardholder Presence DE 61, SF 4	eCom SLI DE 48, SE 42	Additional Recurring Payments Info DE 48, SE 41, SF 1
7IND	Unscheduled Credential On File (Ad hoc) with no 3DS AAV	M101	No	No	Optional	No	4	247	Yes, only for auto top-up
8IND	Unscheduled Credential On File (Ad hoc) with 3DS AAV	M101	Yes	No	Optional	No	4	Appropriate 3DS SLI (e.g. 214, 217)	Yes, only for auto top-up
9IND	Recurring payment transaction with no 3DS AAV	M102, M103	No	No	Optional	No	4	247	Yes
10IND	Recurring payment transaction with 3DS AAV	M102, M103	Yes	No	Optional	No	4	Appropriate 3DS SLI (e.g. 214, 217)	Yes
11IND	Installment transaction with no 3DS AAV	M104	No	No	Optional	No	5	247	Yes

Use case	Description	CIT/MIT Indicator DE 48, SE 22, SF 5	3DS AAV DE 48, SE 43	CVC 2 DE 48, SE 92	DSRP Crypto DE 104, SE 1	Trace ID DE 48, SF 63	Cardholder Presence DE 61, SF 4	eCom SLI DE 48, SE 42	Additional Recurring Payments Info DE 48, SE 41, SF 1
12IND	Installment transaction with 3DS AAV	M104	Yes	No	Optional	No	5	Appropriate 3DS SLI (e.g. 214, 217)	Yes
13IND	Partial shipment transaction with no 3DS AAV	M205	No	No	Optional	No	5	247	N/A
14IND	Partial shipment transaction with 3DS AAV	M205	Yes	No	Optional	No	5	Appropriate 3DS SLI (e.g. 214, 217)	N/A
15IND	Related/delayed charge transaction with no 3DS AAV	M206	No	No	Optional	No	5	247	N/A
16IND	Related/delayed charge transaction with 3DS AAV	M206	Yes	No	Optional	No	5	Appropriate 3DS SLI (e.g. 214, 217)	N/A

Use case	Description	CIT/MIT Indicator DE 48, SE 22, SF 5	3DS AAV DE 48, SE 43	CVC 2 DE 48, SE 92	DSRP Crypto DE 104, SE 1	Trace ID DE 48, SF 63	Cardholder Presence DE 61, SF 4	eCom SLI DE 48, SE 42	Additional Recurring Payments Info DE 48, SE 41, SF 1
17IND	No show charge transaction with no 3DS AAV	M207	No	No	Optional	No	5	247	N/A
18IND	No show charge transaction with 3DS AAV	M207	Yes	No	Optional	No	5	Appropriate 3DS SLI (e.g. 214, 217)	N/A
19IND	Resubmission transaction with no 3DS AAV	M208	No	No	Optional	No	5	247	N/A
20IND	Resubmission transaction with 3DS AAV	M208	Yes	No	Optional	No	5	Appropriate 3DS SLI (e.g. 214, 217)	N/A

Appendix C MDES service indicators

MDES Service Indicators provide acquirers and issuers with real-time visibility into the MDES services or programs which apply to a transaction. MDES performed various types of authentication, controls, and other value-added services, which these indicators help to identify.

MDES sends the MDES Service Indicators as a concatenated string when multiple indicators apply to a transaction. Each indicator is a two-byte alphanumeric value. For example, "X1X2X3". A transaction can include up to 15 indicators for issuers and 15 for acquirers.

Issuers and acquirers should ensure that new indicator values do not result in parsing or processing failures.

Issuers

MDES populates Service Indicators for issuers in DE 119 SE 003 within the 0100 Authorization Request and 0200 Financial Transaction Request messages.

Issuers who want to receive this element can opt in using the **Authorization Profiles** in the MDES Manager application.

Table 71: Service Indicators

Indicator	Definition
M1	MVCT Browser Autofill
M2	MVCT for issuers without controls
M3	MCVT for issuers with controls

Acquirers

MDES populates Service Indicators for acquirers in DE 119 SE 004 within the 0110 Authorization Request Response and 0210 Financial Transaction Request Response messages.

Table 72: Service Indicator

Indicator	Definition
A1	Token Multi-Factor Authentication

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